Copyless Cost Register Automata and Algebraic Groups: Exact Characterization for Invertible Series, Efficient Decision, and Non-Closure under Reversal

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Copyless cost register automata (CCRA) compute functions from words to a field with deterministic finite control and polynomial register updates that never duplicate a register value. Over the rationals they form a natural resource-linear subclass of weighted automata, but no characterization of the functions they compute is known, and recent pumping results of Mazowiecki, Puch and Smertnig led them to conjecture that pumping-type characterizations fail. We prove two of their conjectures: their uniform pumping condition is not sufficient for CCRA-computability, and CCRA over $\mathbb{Q}$ are not closed under reversal.

We characterize CCRA-computability exactly for all rational series whose minimal linear representation consists of invertible matrices. For such a series, let $\mathcal{G}$ be the Zariski closure of the group generated by the minimal letter matrices. The series is computed by a CCRA if and only if the identity component of $\mathcal{G}$ is triangularizable over $\mathbb{Q}$ and any two unipotent elements of $\mathcal{G}$ commute; equivalently, the generated group has a finite-index subgroup that is triangular over $\mathbb{Q}$ and in which unipotent Jordan parts commute. Equivalently again, it is computed by a CCRA whose control is a permutation automaton and whose registers are each updated in isolation, $x := \alpha x + \beta$. In the invertible sector, copyless registers never need to interact: they meet only in the output. The competing machines may use arbitrary finite control and singular, resetting or nonlinear updates.

The necessity proof compresses copyless dynamics to affine coordinate permutations acting on invariant subquotients that recover the minimal representation, and transfers the resulting algebraic-group constraints through finite control by return blocks and a power-map argument. Consequences: in the invertible sector a series is polynomially ambiguous iff the identity component is triangularizable over $\mathbb{Q}$ (an intrinsic form of a characterization of Puch and Smertnig), so the CCRA series are exactly the polynomially ambiguous series with commutative unipotent radical, and they are closed under reversal; polynomial ambiguity and CCRA membership of invertible series are decidable in deterministic polynomial time; and for quasi-unipotent series the CCRA series are exactly the polynomials in letter counts twisted by a finite group automaton. The criterion gives a short proof that a polynomially ambiguous example of Mazowiecki, Puch and Smertnig is not CCRA-computable, and proves non-computability of their position-sum candidate, which satisfies their uniform pumping condition.

Beyond the invertible sector, the witness for non-closure under reversal is $\sum_i \prod_{j<i} |u_j|$ over blocks $u_0\#u_1\#\cdots$: it is not CCRA-computable, while its reversal is. The obstruction is arithmetic: a compressed register must carry the inverse of a weight of the unipotent part, and translations must be rational as well. We characterize CCRA-computability exactly when all singular minimal letter matrices have rank at most one, and for one invertible letter with a rank-two reset such that every word containing the reset has rank two. A characterization for arbitrary singular series remains open.

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1 Introduction

Copyless registers. A finite automaton reads its input once, from left to right, with constant memory. Equipping it with finitely many registers that hold values from a semiring, and updating them at each step by expressions in the old values, yields the cost register automata (CRA) of Alur, D'Antoni, Deshmukh, Raghothaman and Yuan [ADD+13]. Over a field, CRA with linear updates are exactly weighted automata, whereas unrestricted polynomial updates such as $x := x^2$ leave the realm of rational series. The *copyless* restriction requires that every old register is used at most once in each update and at most once in the output. It originates in streaming string transducers, where copyless register updates capture exactly the MSO-definable, or regular, string-to-string functions [AČ10, AČ11]. Copylessness is a resource-linearity discipline: a stored value may be moved, combined with others, or discarded, but never duplicated.

Over a field, copyless polynomial CRA (CCRA) compute only rational series [MR16, MR19, MPS26a]: substituting copyless updates into squarefree monomials produces squarefree polynomials, so the induced action on multiaffine polynomials is linear (Section 2). As noted in [MPS26a], CCRA and polynomially ambiguous weighted automata are the richest studied classes known to lie strictly between deterministic and general weighted automata. Over $\mathbb{Q}$ and a one-letter alphabet both classes coincide with linear recurrence sequences whose characteristic roots are roots of rationals [BFLM22]; over larger alphabets they are incomparable [MPS26a].

Which rational series are computed by CCRA? Mazowiecki, Puch and Smertnig [MPS26a] proved a uniform pumping condition necessary for CCRA, used it to separate CCRA from polynomially ambiguous automata, and conjectured that this condition is not sufficient. They proposed a position-sum candidate whose nonrecognizability they could not prove, and suggested that in some sense a characterization should not exist, partly because CCRA are expected not to be closed under reversal (as is the case over the tropical semiring [MR16]); we confirm this over $\mathbb{Q}$ in Section 10. The unipotent case of our results (Theorem 3(1)) already settles the candidate.

Main result. Every rational series $f : \Sigma^* \to \mathbb{Q}$ has a minimal linear representation $f(w) = \lambda M_w \gamma$, unique up to simultaneous conjugation, where $M_{a_1 \cdots a_n} = M_{a_1} \cdots M_{a_n}$. We call f *invertible* if its minimal letter matrices are invertible. Let $\Gamma_f \le \mathrm{GL}_d(\mathbb{Q})$ be the group they generate and $\mathcal{G}_f$ its Zariski closure, a linear algebraic group with identity component $\mathcal{G}_f^\circ$. A group of matrices is *triangularizable over* $\mathbb{Q}$ if some basis of $\mathbb{Q}^d$ makes all of its elements upper triangular.

Theorem 1 (Main theorem). *For an invertible rational series $f : \Sigma^* \to \mathbb{Q}$ the following are equivalent.*

1. *f is computed by a deterministic finite-state polynomial CCRA over $\mathbb{Q}$.*
2. *f is computed by a CCRA whose control is a permutation automaton, in which every register is updated in isolation by a map $x \mapsto \alpha x + \beta$ with $\alpha \in \mathbb{Q}^\times$, $\beta \in \mathbb{Q}$, and whose outputs are linear combinations of products of distinct registers.*
3. *$\mathcal{G}_f^\circ$ is triangularizable over $\mathbb{Q}$, and any two unipotent elements of $\mathcal{G}_f$ commute.*
4. *Γ_f has a subgroup of finite index that is triangularizable over $\mathbb{Q}$ and in which any two elements have commuting unipotent Jordan parts.*

Moreover, every function computed by a machine as in (2) is invertible.

Condition (3) is intrinsic, because minimal representations are unique up to conjugation. In (1) the competing machines are unrestricted CCRA: arbitrary finite control, any number of registers, arbitrary rational constants, and singular, resetting, merging or multiplicative updates. Condition (2) is a strong normal form: *in the invertible sector, copyless registers never need to interact*; they are combined only by the output polynomial. The class is rich. It contains exponential growth, finite-state phenomena such as

alternating sums, and the function $\sum_i 2^i b_i$ on bit strings, for which Mazowiecki, Puch and Smertnig gave a dedicated construction [MPS26b, Example 27]; its group is the Baumslag–Solitar group $BS(1,2)$, which is not virtually nilpotent. The Zariski closure in (3) cannot be replaced by Γ_f itself: for $f(w) = 2^{|w|_a} 3^{|w|_b} [w : ab]$, where $[w : ab]$ counts scattered occurrences of ab, all unipotent elements of Γ_f are central, yet f is not CCRA-computable (Example 19).

The invertible CCRA series are exactly the series computed by reversible CCRA, while the necessity direction (1)⇒(3) holds against arbitrary machines.

Corollary 2 (Reversible machines). *Call a CCRA* reversible *if its control is a permutation automaton and every update $P_{s,a}$ is a bijection of $\mathbb{Q}^r$. A rational series is invertible and CCRA-computable if and only if it is computed by a reversible CCRA.*

Proof. We use the notation of Section 2. A copyless polynomial bijection P of $\mathbb{Q}^r$ is an affine coordinate permutation: no coordinate of P is constant, since otherwise its image lies in a hyperplane; distinct coordinates have disjoint variable supports (Section 2), so each of the r coordinates depends on exactly one of the r variables, and a multiaffine polynomial in one variable has the form $\alpha x + \beta$, with $\alpha \neq 0$ since the coordinate is nonconstant. Hence P^{-1} is again copyless and B_P is invertible on V_r with inverse $B_{P^{-1}}$. As the control permutes S, the letter operators $(B_a h)_s = h_{\delta(s,a)} \circ P_{s,a}$ of the linearization in Section 2 are invertible. The minimal representation is an invariant subquotient of the linearization, so its letter matrices are invertible. Conversely, the machines in Theorem 1(2) are reversible. This also proves the last sentence of Theorem 1. □

Why this matters. Copyless computation is one-pass computation that never duplicates a stored value. Over a field this discipline has sharp algebraic consequences. In the invertible sector, whose CCRA series are exactly those of reversible machines (Corollary 2), it is captured exactly by a condition on the Zariski closure of the transition group that is decidable in polynomial time, and it collapses to machines whose registers never interact (Theorem 1). Outside this sector the reading direction matters: CCRA over $\mathbb{Q}$ are not closed under reversal (Corollary 57), whereas polynomially ambiguous weighted automata are, as one sees by reversing the automaton; this is an obstruction of a different kind from pumping. Deciding whether a rational series lies in a subclass, such as the series of deterministic, unambiguous or polynomially ambiguous automata, is an active line of work [BS21, BS23, JMP24, PS24], and our results add CCRA to this list in the invertible sector. For the general question, Mazowiecki, Puch and Smertnig suggested that a characterization might not exist [MPS26a]; our results show that one exists in the invertible sector, while non-closure under reversal shows that no characterization of all CCRA series can be invariant under transposition of the minimal representation (Remark 62). The proof transfers structure in both directions between a syntactic resource restriction and the Zariski closure of a matrix group, in the spirit of algebraic approaches to decision problems for weighted and quantum automata [DJK05, NPS+22, PS24]; we expect this bridge to be useful for other copyless and resource-linear models.

Sectors and consequences. Two sectors give especially concrete forms. Call f *unipotent* (resp. *quasi-unipotent*) if all eigenvalues of all word matrices M_w of its minimal representation equal 1 (resp. are roots of unity). For a morphism $\varphi : \Sigma^* \to F$ to a finite group, let $|w|_{h,a}$ count the positions i of w with $w_i = a$ and $\varphi(w_1 \cdots w_{i-1}) = h$.

Theorem 3. (1) *A unipotent f is CCRA-computable iff its minimal letter matrices commute, iff f is invariant under permutations of input positions, iff $f(w)$ is a rational polynomial in the letter counts $(|w|_a)_a$.*

(2) *A quasi-unipotent f is CCRA-computable iff any two word matrices M_x, M_y that are unipotent commute, iff Γ_f is virtually abelian, iff there are a finite group F, a morphism $\varphi : \Sigma^* \to F$ and polynomials $P_g \in \mathbb{Q}[n_{h,a} : h \in F, a \in \Sigma]$ with $f(w) = P_{\varphi(w)}\big((|w|_{h,a})_{h,a}\big)$ for all words w.*

Corollary 4. *Let f be invertible and CCRA-computable. Then f is computed by a polynomially ambiguous weighted automaton, and its reversal $w \mapsto f(w^{\mathrm{rev}})$ is CCRA-computable. Consequently any CCRA series whose reversal is not CCRA-computable has a minimal letter matrix that is singular of rank at least two (Theorem 50); such series exist (Corollary 57). The polynomially ambiguous series of* [MPS26b, Example 17] *is invertible but not CCRA-computable.*

Conversely, an invertible series is polynomially ambiguous iff $\mathcal{G}_f^\circ$ is triangularizable over $\mathbb{Q}$ (Theorem 22); this is an intrinsic form of the characterization of polynomial ambiguity by Puch and Smertnig [PS24], and what is new is the condition on $\mathcal{G}_f^\circ$, a direct proof and a deterministic polynomial-time test. Thus in the invertible sector CCRA-computability is exactly polynomial ambiguity plus commutation of unipotent elements, and polynomial ambiguity is decidable in deterministic polynomial time (Corollary 23).

Theorem 5. *From a rational weighted representation one can compute its minimal representation and decide whether it is invertible in polynomial bit time. For invertible series, CCRA-computability is decidable by a Zariski-closure computation (Section 8) and, more efficiently, as stated below. Unipotence of a series is decidable in polynomial bit time, and for unipotent series CCRA-computability is decidable in polynomial bit time. More generally, quasi-unipotence of an invertible series and, for quasi-unipotent series, CCRA-computability are decidable in deterministic polynomial bit time, and CCRA-computability of invertible series is decidable in deterministic polynomial time (Theorems 28 and 40).*

Finally, the position-sum function of [MPS26b, Example 26],

$$f(b_1 \cdots b_j) = \sum_{i=1}^{j} i\, b_i, \qquad b_i \in \{0, 1\}, \tag{1}$$

is unipotent and not permutation-invariant. By Theorem 3(1) it is not CCRA-computable, although, as Mazowiecki, Puch and Smertnig computed [MPS26b, Example 26], it satisfies their uniform pumping condition, with ring $\mathbb{Z}[1/2]$ and exponent $m = 1$ (Theorem 21 records the computation). This disproves sufficiency of that condition, as conjectured in [MPS26a, Section 3.2]. Our characterization is global rather than of pumping type; Theorem 1 shows that in the invertible sector a clean characterization nevertheless exists.

Techniques. The necessity direction (1)⇒(3) has three steps. *Compression.* A copyless map acts linearly, by pullback, on the 2^r-dimensional space of multiaffine polynomials in r registers. A word whose register map has the fewest nonconstant coordinates yields a copyless retraction whose pullback is a polynomial in a word operator. It therefore acts as the identity on the minimal subquotient, and, for a one-state machine, it compresses all letters simultaneously to *affine coordinate permutations* $x_i \mapsto \alpha_i x_{\pi(i)} + \beta_i$ acting on one common invariant subquotient that is isomorphic to the minimal representation (Section 3). This step has no counterpart in pumping arguments, which constrain iterated factors one at a time: compression extracts, from an arbitrary machine with resets and products, a normal form for the action of the whole monoid at once. *Algebraic groups.* Pullbacks of affine coordinate permutations form a linear algebraic group whose identity component is triangular over $\mathbb{Q}$ and whose unipotent elements are pullbacks of translations, hence commute. Both properties pass to Zariski closures, images and invariant subquotients. *Finite control.* Return blocks of a minimum-rank control word form a monoid of words that fixes a set of control states pointwise. The Zariski closure of its matrices contains all N-th powers of elements of $\mathcal{G}_f^\circ$, hence all of $\mathcal{G}_f^\circ$,

because a connected group of finite exponent is trivial. Local quotients over all prefixes jointly embed the minimal representation (Section 4). The sufficiency direction (3)⇒(2) uses the decomposition $\mathcal{G}_f^\circ = T \ltimes U$ of a $\mathbb{Q}$-split solvable group. A permutation automaton tracks the coset of the prefix in $\Gamma_f/(\Gamma_f \cap \mathcal{G}_f^\circ)$; a cocycle moves the remaining dynamics into $\mathcal{G}_f^\circ$; the torus contributes products of rational scalars; and because U is commutative, its logarithm accumulates in isolated affine registers $z := \chi(t)^{-1}z + \beta$ (Section 5).

Related work. Mazowiecki and Riveros [MR16, MR19] developed the structure theory of CCRA over semirings, including a normal form with permutations in the finite control and reset analysis under nonzero-semiring hypotheses; those hypotheses are unavailable for signed rational computation with cancellation, and their reversal counterexample is tropical. Our normal form (2) is much stronger in the invertible sector. The pumping theory of [MPS26a] is the direct starting point. The squarefree linearization is folklore [MPS26b, Lemma 50]. Puch and Smertnig [PS24] characterize, for invertible weighted automata, finite and polynomial ambiguity by arithmetic conditions, using unit equations and monomial covers, and obtain decidability via Zariski closures [DJK05, NPS+22]; our decision procedure of Section 8 follows the same pattern, while Section 9 gives a closure-free procedure; the property and the proof are different. Theorem 22 recasts their characterization of polynomial ambiguity intrinsically, in terms of $\mathcal{G}_f^\circ$, as triangularizability of $\mathcal{G}_f^\circ$ over $\mathbb{Q}$, with a direct proof and a deterministic polynomial-time test, and shows that the invertible CCRA series are exactly the polynomially ambiguous ones with commutative unipotent radical. Bell and Smertnig characterized unambiguous series [BS21] and decided whether a series is computed by a deterministic, respectively unambiguous, weighted automaton [BS23], and Jecker, Mazowiecki and Purser [JMP24] study determinisation and unambiguisation of polynomially ambiguous rational weighted automata. Weber and Seidl [WS91] characterize polynomial ambiguity of an automaton, as opposed to a series. Kiefer, Murawski, Ouaknine, Wachter and Worrell [KMO+13] analyse equivalence and minimisation of $\mathbb{Q}$-weighted automata, and Zariski closures of matrix monoids also underlie the computation of polynomial invariants of affine programs [HOPW18]. Pumping lemmas for weighted automata were developed by Chattopadhyay, Mazowiecki, Muscholl and Riveros [CMMR21], and Daviaud, Reynier and Talbot [DRT16] minimise cost register automata. Clemente [Cle25, Cle26] studies commutativity problems and commutative products of series, and Benalioua, Lhote and Reynier [BLR24] minimize linear and affine CRA; neither addresses the existence of copyless realizations. Douéneau-Tabot [DT22] characterizes the unary-output functions of pebble and blind transducers through $\mathbb{N}$-rational series and their Cauchy and Hadamard products, and decides membership in the blind class; there outputs are natural numbers and the machines are two-way.

Organization. The proof of Theorem 1 uses only Sections 2–6; decidability, the polynomial-time algorithm in the quasi-unipotent sector and the Monte Carlo algorithm do not depend on the height bound of Section 9.5. Section 2 fixes models and algebraic-group tools. Section 3 proves the compression theorem for monoids of copyless maps, Section 4 handles finite control and proves (1)⇒(3), and Section 5 proves (3)⇒(2). Section 6 proves (3)⇔(4) and Theorem 3. Section 7 proves Corollary 4, discusses examples, establishes the pumping property of (1), and characterizes polynomial ambiguity in the invertible sector (Section 7.2). Section 8 proves the first claims of Theorem 5 and gives polynomial-size synthesis and short certificates for explicit pattern sums. Section 9 gives the polynomial-time decision procedures, proves that they run in deterministic polynomial time, and proves the remaining claims. Section 10 derives a local necessary condition from Theorem 1 (Corollary 48) and treats singular series: rank-one resets, the inverse-weight and lifting obstructions, non-closure under reversal, and an exact characterization for one invertible letter and one rank-two reset. Section 11 lists open problems.

2 Models, representations and algebraic groups

2.1 Copyless machines and their linearization

A polynomial expression is *copyless* if each variable occurs at most once in it. A tuple of expressions is copyless if each variable occurs at most once across the entire tuple. Consequently its coordinate polynomials are multiaffine (of degree at most one in each variable) and distinct coordinates have disjoint variable supports; cancellation can remove variables but does not invalidate these conclusions. Compositions of copyless tuples are copyless.

A deterministic copyless polynomial cost register automaton (CCRA) over $\mathbb{Q}$ consists of a finite set S of control states with initial state s_0, a complete deterministic transition function $\delta : S \times \Sigma \to S$, registers $x_1, \dots, x_r$ with an initial valuation $x^{(0)} \in \mathbb{Q}^r$, a copyless polynomial update $P_{s,a} : \mathbb{Q}^r \to \mathbb{Q}^r$ for every transition, and a copyless output polynomial ω_s for every state; all constants are rational. Extend δ to words, and let $P_{s,z}$ be the composite update along the run of z from s, so $P_{s,zz'} = P_{\delta(s,z),z'} \circ P_{s,z}$. The machine computes $f(w) = \omega_{\delta(s_0,w)}\big(P_{s_0,w}(x^{(0)})\big)$. A *permutation automaton* is a control in which every letter permutes S.

Let V_r be the space of multiaffine polynomials in $x_1, \dots, x_r$, with basis the squarefree monomials $x_I = \prod_{i \in I} x_i$, $I \subseteq \{1, \dots, r\}$. For a copyless map P, the pullback $B_P(h) = h \circ P$ preserves V_r, because substituting into a squarefree monomial multiplies polynomials with disjoint supports. Thus $B_{P \circ P'} = B_{P'} B_P$, and every CCRA computes a rational series: on $\bigoplus_{s \in S} V_r$ let $(B_a h)_s = h_{\delta(s,a)} \circ P_{s,a}$, let $\lambda(h) = h_{s_0}(x^{(0)})$ and $o = (\omega_s)_s$; then $f(w) = \lambda B_{a_1} \cdots B_{a_n} o$.

2.2 Weighted representations

A representation of f is a triple of a row vector λ, matrices M_a and a column vector γ with $f(w) = \lambda M_w \gamma$. It is minimal if the row vectors λM_w span the dual space and the columns $M_w \gamma$ span the space. Minimal representations exist, are computable, and are unique up to simultaneous conjugation [Sch61, BR11]. More precisely, for an arbitrary representation on a space X put

$$W = \operatorname{span}\{M_w \gamma : w \in \Sigma^*\}, \qquad N = \{h \in W : \lambda M_v h = 0 \text{ for all } v \in \Sigma^*\}. \tag{2}$$

Both are invariant, and the map $W \to \mathbb{Q}^{\Sigma^*}$, $h \mapsto (v \mapsto \lambda M_v h)$, has kernel N and sends $M_w \gamma$ to the residual $v \mapsto f(vw)$. Hence the induced representation on W/N is isomorphic to every minimal representation. In particular, *the minimal representation is an invariant subquotient of every representation*, and if all M_a are invertible then so are the minimal letter matrices.

2.3 Algebraic-group tools

For a group $\Gamma \leq \mathrm{GL}_d(\mathbb{Q})$ its Zariski closure $\mathcal{G} \subseteq \mathrm{GL}_d(\overline{\mathbb{Q}})$ is the set of common zeros in $\mathrm{GL}_d(\overline{\mathbb{Q}})$ of all polynomials in the entries and $\det^{-1}$ that vanish on Γ. It is a linear algebraic group defined over $\mathbb{Q}$. We use the following standard facts [Bor91, Hum75].

- (A1) $\mathcal{G}$ has a connected normal closed subgroup $\mathcal{G}^\circ$ of finite index, whose cosets are the irreducible components of $\mathcal{G}$; every closed subgroup of finite index contains $\mathcal{G}^\circ$.
- (A2) Every $g \in \mathcal{G}$ has a Jordan decomposition $g = g_s g_u = g_u g_s$ with g_s semisimple, g_u unipotent, and $g_s, g_u \in \mathcal{G}$. For $g \in \mathrm{GL}_d(\mathbb{Q})$ both factors are polynomials in g with rational coefficients. Morphisms of algebraic groups preserve Jordan decompositions. Images of algebraic groups under morphisms are closed, and the image of the identity component is the identity component of the image.
- (A3) In characteristic zero every unipotent element of $\mathcal{G}$ lies in $\mathcal{G}^\circ$: if $u \neq 1$ is unipotent, the closure of $\langle u \rangle$ is the irreducible curve $\{\exp(t \log u)\}$.

(A4) A connected algebraic group all of whose elements are semisimple is a torus [Bor91, 11.5]. A connected group whose maximal tori are trivial is unipotent, since each element lies in a Borel subgroup $B = T_B B_u$ with T_B a maximal torus [Bor91, 10.6, 11.10]. In characteristic zero, a connected group of finite exponent is trivial: unipotent elements of finite order are trivial, so all elements are semisimple, and a torus of finite exponent is trivial.

(A5) Let H be connected, defined over $\mathbb{Q}$, and triangularizable over $\mathbb{Q}$. Then H is solvable, its unipotent elements form a closed connected normal subgroup H_u defined over $\mathbb{Q}$, and there is a torus $T \subseteq H$ defined over $\mathbb{Q}$ such that multiplication $T \times H_u \to H$ is an isomorphism of varieties [Bor91, 10.6, 18.2]. Galois invariance of the unique decomposition $h = tu$ gives $H(\mathbb{Q}) = T(\mathbb{Q})H_u(\mathbb{Q})$. In a rational basis making H upper triangular, $t \mapsto (t_{11}, \ldots, t_{dd})$ is an injective morphism defined over $\mathbb{Q}$ (its kernel lies in $T \cap H_u = 1$), so T is $\mathbb{Q}$-isomorphic to a subtorus of the split diagonal torus and is split over $\mathbb{Q}$; hence its weight spaces on $\mathbb{Q}^d$ and on matrices (adjoint action) are defined over $\mathbb{Q}$, that is, T and its adjoint action are diagonalizable over $\mathbb{Q}$ [Bor91, §8].

(A6) For connected H, a subspace is H-invariant iff it is invariant under the Lie algebra of H [Hum75, 13.2].

For a unipotent rational matrix u, $\log u = \sum_{j\geq 1}(-1)^{j+1}(u-I)^j/j$ is a rational nilpotent matrix; exp and log are inverse bijections between unipotent and nilpotent matrices, and $\exp(X+Y) = \exp X \exp Y$ when X, Y commute.

Lemma 6. *Let $S \subseteq \mathrm{GL}_d(\mathbb{Q})$ be a submonoid generating the group Γ, with Zariski closure $\mathcal{G}$ of Γ. Then S is Zariski dense in $\mathcal{G}$, and $S \cap \mathcal{G}^\circ$ is Zariski dense in $\mathcal{G}^\circ$. The same holds for Γ, and $\Gamma \cap \mathcal{G}^\circ$ is a normal subgroup of finite index in Γ.*

Proof. Let $\bar{S}$ be the closure of S. Right multiplication by $s \in S$ is a homeomorphism mapping S into S, so $\bar{S}s \subseteq \bar{S}$. Hence for $x \in \bar{S}$ the closed set $\{y : xy \in \bar{S}\}$ contains S, hence $\bar{S}$; so $\bar{S}$ is a monoid. For $x \in \bar{S}$ the descending chain $\bar{S} \supseteq x\bar{S} \supseteq x^2\bar{S} \supseteq \cdots$ of closed sets stabilizes, and cancelling a power of x gives $\bar{S} = x\bar{S}$, so $x^{-1} \in \bar{S}$. Thus $\bar{S}$ is a closed group containing Γ, and $\bar{S} = \mathcal{G}$. As $\mathcal{G}$ is the finite disjoint union of the closed cosets $g\mathcal{G}^\circ$, we have $\mathcal{G} = \bigcup_g \overline{S \cap g\mathcal{G}^\circ}$ with $\overline{S \cap g\mathcal{G}^\circ} \subseteq g\mathcal{G}^\circ$; intersecting with $\mathcal{G}^\circ$ gives $\overline{S \cap \mathcal{G}^\circ} = \mathcal{G}^\circ$. The last claim follows from (A1). □

Lemma 7. *If u is a unipotent $d \times d$ rational matrix and $t \geq 1$, then $u = \sum_{j=0}^{d-1} \binom{1/t}{j}(u^t - I)^j$. In particular u commutes with every matrix commuting with u^t, and g_u commutes with every matrix commuting with $(g^t)_u = (g_u)^t$.*

Proof. In $\mathbb{Q}[X]/(X^d)$ substitute $Y = (1+X)^t - 1$ into the truncated series $(1+Y)^{1/t} = 1 + X$, and put $X = u - I$. The second statement applies this to g_u, using $(g^t)_u = (g_u)^t$. □

3 Compression of a monoid of copyless maps

Throughout this section, $\mathcal{Y}$ is a monoid acting on $\mathbb{Q}^r$ by copyless polynomial maps P_y with $P_1 = \mathrm{id}$ and $P_{yz} = P_z \circ P_y$. Its pullbacks $B_y = B_{P_y}$ on V_r satisfy $B_{yz} = B_y B_z$. We fix subspaces $N \subseteq W \subseteq V_r$ invariant under every B_y, write $D = W/N$, and let $\rho(y)$ be the operator induced by B_y on D. In Section 4 the monoid $\mathcal{Y}$ will consist of return words of a CCRA.

For a copyless map $P : \mathbb{Q}^r \to \mathbb{Q}^r$ let $k(P)$ be the number of nonconstant coordinates of P, let L_P be the coordinate affine subspace obtained by fixing each constant coordinate to its value, and let $V(L_P)$ be the multiaffine polynomials in its $k(P)$ free coordinates.

Lemma 8. *P maps $\mathbb{Q}^r$ onto $L_P(\mathbb{Q})$, $\operatorname{rank} B_P = 2^{k(P)}$, and $\ker B_P = \{h \in V_r : h|_{L_P} = 0\}$.*

Proof. A nonconstant multiaffine q over $\mathbb{Q}$ is onto $\mathbb{Q}$: write $q = x_i a + b$ with $a \neq 0$ free of x_i, specialize the other variables to rationals where $a \neq 0$ (possible since a nonzero polynomial over an infinite field does not vanish on all rational points), and solve for x_i. Distinct nonconstant coordinates of P have disjoint supports, so these choices are independent; this gives surjectivity. Restriction $V_r \to V(L_P)$ is onto, and pullback $V(L_P) \to V_r$ through P is injective because P is onto $L_P(\mathbb{Q})$. Factoring B_P through these maps proves the remaining claims. □

Choose $e \in \mathcal{Y}$ minimizing $k(P_e)$ and put $P = P_e$, $k = k(P)$, $L = L_P$. The element ee has at most k nonconstant coordinates, hence exactly k; as P is onto L, all k free coordinates of $A = P|_L : L \to L$ are nonconstant. Restriction preserves disjointness of supports, so each free output of A uses exactly one free variable and all are used. By multiaffinity,

$$A_i(t) = \alpha_i t_{\pi(i)} + \beta_i, \qquad \alpha_i \in \mathbb{Q}^\times, \tag{3}$$

for a permutation π. We call such a map an *affine coordinate permutation*. Define the copyless retraction $Q = A^{-1} \circ P : \mathbb{Q}^r \to L$. It is onto L and $Q|_L = \mathrm{id}_L$, so $E = B_Q$ is idempotent.

Lemma 9. *E is the projection onto* $\operatorname{im} B_e$ *along* $\ker B_e$*, and $E = p(B_e)$ for some $p \in \mathbb{Q}[T]$.*

Proof. Composition with A^{-1} is an automorphism of $V(L)$, so $\operatorname{im} B_Q = \operatorname{im} B_P$; both maps are onto L, so Lemma 8 gives $\ker B_Q = \ker B_P$. Since $k(P_{ee}) = k(P_e)$, $\operatorname{rank} B_e^2 = \operatorname{rank} B_e$, hence $V_r = \ker B_e \oplus \operatorname{im} B_e$ and B_e is invertible on its image. If q is the minimal polynomial of $B_e|_{\operatorname{im} B_e}$, then $q(0) \neq 0$, and the Chinese remainder theorem gives $p \in \mathbb{Q}[T]$ with $p \equiv 0 \bmod T$ and $p \equiv 1 \bmod q$. Then $p(B_e)$ vanishes on $\ker B_e$ and is the identity on $\operatorname{im} B_e$. □

Proposition 10 (Compression). *Assume that $\rho(y)$ is invertible for every $y \in \mathcal{Y}$. Then there are $k \geq 0$, affine coordinate permutations R_y of $\mathbb{Q}^k$ ($y \in \mathcal{Y}$), and subspaces $N'' \subseteq W'' \subseteq V_k$ invariant under every pullback B_{R_y}, together with a linear isomorphism $W''/N'' \cong D$ intertwining the operator induced by B_{R_y} with $\rho(y)$ for every y.*

Proof. By Lemma 9, E preserves W and N; let $\bar{E}$ be its induced operator on D. Since $\operatorname{im} B_e = \operatorname{im} E$, we have $(I - E)B_e = 0$, so $(I - \bar{E})\rho(e) = 0$, and invertibility of $\rho(e)$ gives $\bar{E} = I$, that is, $(I - E)W \subseteq N$. The physical map P_e need not be invertible.

Let $J : V(L) \to \operatorname{im} E$, $J(h) = h \circ Q$, with inverse $g \mapsto g|_L$, and put $R_y = Q \circ P_y|_L : L \to L$ and $C_y = EB_yE$. Direct substitution gives $C_y J(h) = h \circ Q \circ P_y \circ Q = J(B_{R_y} h)$. Each R_y is copyless. The register map of eye is $P_e \circ P_y \circ P_e = A \circ R_y \circ (A \circ Q)$, where $A \circ Q = P_e$ is onto L and A rescales and translates free coordinates. Hence P_{eye} has as many nonconstant coordinates as R_y has nonconstant free outputs, and minimality of e forces k of them; the support count used for (3) shows that R_y is an affine coordinate permutation.

Put $W' = EW \subseteq W$ and $N' = W' \cap N$. Both are invariant under every C_y, because E and B_y preserve W and N and E is the identity on W'. The map $W'/N' \to W/N$, $h + N' \mapsto h + N$, is injective by definition of N' and surjective because $h - Eh \in N$ for $h \in W$. For $h \in W'$ we have $C_y h - B_y h = (E - I)B_y h \in N$, so the map intertwines C_y with $\rho(y)$. Finally set $W'' = J^{-1}W'$ and $N'' = J^{-1}N'$. If $k = 0$, all objects are at most one-dimensional and the statement is trivial. □

The assignment $y \mapsto R_y$ need not be multiplicative; only the induced operators on W''/N'' form a representation of $\mathcal{Y}$, and only this is used below.

Let $\mathfrak{A}_k$ be the algebraic group of affine coordinate permutations of affine k-space over $\overline{\mathbb{Q}}$, that is, $(\overline{\mathbb{Q}}^\times \ltimes \overline{\mathbb{Q}})^k \rtimes S_k$, and let $\mathcal{W}_k = \{B_R : R \in \mathfrak{A}_k\} \subseteq \mathrm{GL}(V_k \otimes \overline{\mathbb{Q}})$.

Lemma 11. *$\mathcal{W}_k$ is a closed subgroup. Its identity component consists of the pullbacks of the maps $x \mapsto (\alpha_i x_i + \beta_i)_i$; it preserves the complete rational flag spanned by the squarefree monomials taken in order of*

nondecreasing degree. The unipotent elements of $\mathcal{W}_k$ are exactly the pullbacks of translations; in particular they commute pairwise.

Proof. $R \mapsto B_R$ is an anti-homomorphism of algebraic groups, so its image is closed (A2), and its identity component is the image of the connected finite-index subgroup $(\overline{\mathbb{Q}}^\times \ltimes \overline{\mathbb{Q}})^k$. For $R(x) = (\alpha_i x_i + \beta_i)_i$ we have $B_R x_I = \prod_{i\in I}(\alpha_i x_i + \beta_i) = (\prod_{i\in I}\alpha_i)x_I + \sum_{I'\subsetneq I} c_{I'} x_{I'}$, which gives the triangular form with diagonal entries $\prod_{i\in I}\alpha_i$. Such B_R is unipotent iff $\alpha_i = 1$ for all i (take $I = \{i\}$), that is, iff R is a translation. Every unipotent element of $\mathcal{W}_k$ lies in $\mathcal{W}_k^\circ$ by (A3). Translations commute, and so do their pullbacks. □

Corollary 12 (One-state constraint). *Under the hypothesis of Proposition 10, let $\mathcal{H}$ be the Zariski closure of $\{\rho(y) : y \in \mathcal{Y}\}$ in $\mathrm{GL}(D)$. Then any two unipotent elements of $\mathcal{H}$ commute, and $\mathcal{H}^\circ$ is triangularizable over $\mathbb{Q}$.*

Proof. By Lemma 6, $\mathcal{H}$ is a group. Let $\mathcal{S} \subseteq \mathrm{GL}(V_k \otimes \overline{\mathbb{Q}})$ be the closed subgroup stabilizing W'' and N'', and let $\pi : \mathcal{S} \to \mathrm{GL}(W''/N'') \cong \mathrm{GL}(D)$ be the induced-action morphism. Let $\mathcal{G}_L$ be the Zariski closure of the group generated by the B_{R_y}; it lies in $\mathcal{W}_k \cap \mathcal{S}$. By (A2), $\pi(\mathcal{G}_L)$ is closed; it contains every $\rho(y)$ and is contained in the closure of the group they generate, so $\pi(\mathcal{G}_L) = \mathcal{H}$. If $h \in \mathcal{H}$ is unipotent, write $h = \pi(g)$ with $g \in \mathcal{G}_L$; then $h = h_u = \pi(g_u)$ with $g_u \in \mathcal{G}_L$ unipotent, hence the pullback of a translation by Lemma 11. Two such elements commute, and so do their images. Moreover $\mathcal{H}^\circ = \pi(\mathcal{G}_L^\circ)$ with $\mathcal{G}_L^\circ \subseteq \mathcal{W}_k^\circ$, which preserves the rational monomial flag of Lemma 11; its traces on W'' modulo N'' form a chain of rational $\mathcal{H}^\circ$-invariant subspaces with successive quotients of dimension at most one. □

4 Finite control: proof of (1)⇒(3)

Fix a CCRA C as in Section 2 computing an invertible f, with minimal representation $(\lambda, (M_a)_a, \gamma)$ of dimension $d \geq 1$ (the case $f = 0$ is trivial). Write $\mathcal{G} = \mathcal{G}_f$.

Words act on the finite set S of control states. Choose a word t for which the image $R = \delta(S, t)$ has minimum size. Since $\delta(S, tt) = \delta(R, t)$ cannot be smaller, t permutes R. Let $N = |R|!$ and put $e = t^N$. Then $\delta(S, e) = R$ and e fixes every state of R.

Lemma 13. *For every word x, the word exe permutes R, and $(exe)^N$ fixes every state of R.*

Proof. $\delta(S, exe) \subseteq \delta(S, e) = R$ and minimality force equality. Since e maps S onto R and fixes R, $\delta(S, exe)$ is the image of R under exe; thus exe maps R onto itself, and its order as a permutation divides $|R|!$, which divides N. □

Let $L_R = \{z \in \Sigma^* : \delta(r, z) = r \text{ for all } r \in R\}$, a monoid containing e and every $(exe)^N$. Let $\mathcal{K}$ be the Zariski closure of $\{M_z : z \in L_R\}$, a closed subgroup of $\mathcal{G}$ by Lemma 6, and let $\mathcal{A}_R = \mathrm{span}\{M_z : z \in L_R\}$, a unital algebra. By Cayley–Hamilton, M_e^{-1} is a polynomial in M_e, so $M_e^{-1} \in \mathcal{A}_R$.

Lemma 14. $\mathcal{K} \supseteq \mathcal{G}^\circ$.

Proof. Let $g = M_e \in \mathcal{G}$ and $\Phi : \mathcal{G} \to \mathcal{G}$, $\Phi(h) = (ghg)^N$. Then $\Phi(M_x) = M_{(exe)^N} \in \mathcal{K}$ for every word x. As Φ is continuous and the word matrices are dense in $\mathcal{G}$ (Lemma 6), $\Phi(\mathcal{G}) \subseteq \mathcal{K}$. For $k \in \mathcal{G}^\circ$ we have $g^{-1}kg^{-1} \in \mathcal{G}$ and $\Phi(g^{-1}kg^{-1}) = k^N$; hence $\mathcal{K}$ contains every N-th power of an element of $\mathcal{G}^\circ$. Let $\mathcal{H}$ be the closure of the group generated by these powers. It is a closed subgroup of $\mathcal{K} \cap \mathcal{G}^\circ$, normal in $\mathcal{G}^\circ$ because its generating set is conjugation invariant. The quotient $\mathcal{G}^\circ/\mathcal{H}$ is a connected algebraic group of exponent dividing N, hence trivial by (A4). Thus $\mathcal{G}^\circ = \mathcal{H} \subseteq \mathcal{K}$. □

Lemma 15 (Local quotients). *Let $u \in \Sigma^*$, $q = \delta(s_0, ue) \in R$, and let $x_u = P_{s_0,ue}(x^{(0)})$ be the valuation after ue. Let L_R act on $\mathbb{Q}^r$ by $P_z = P_{q,z}$, and for $s \in \Sigma^*$ let $o_s = \omega_{\delta(q,s)} \circ P_{q,s} \in V_r$. Put*

$$W = \mathrm{span}\{o_s : s \in \Sigma^*\}, \quad N = \{h \in W : (B_z h)(x_u) = 0 \ \forall z \in L_R\}, \quad K_u = \{v \in \mathbb{Q}^d : \lambda M_{ue} M_z v = 0 \ \forall z \in L_R\}.$$

Then W and N are invariant under every B_z, K_u is $\mathcal{K}$-invariant, and $\sum_s c_s o_s \mapsto \sum_s c_s M_s \gamma + K_u$ induces a linear isomorphism $W/N \cong \mathbb{Q}^d/K_u$ intertwining B_z with the operator induced by M_z, for every $z \in L_R$.

Proof. Since $z \in L_R$ returns to q, $P_{zz'} = P_{z'} \circ P_z$ and $B_z o_s = o_s \circ P_{q,z} = o_{zs}$; so W is invariant, and so is N. Evaluating, $(B_z o_s)(x_u) = o_{zs}(x_u) = f(uezs) = \lambda M_{ue} M_z M_s \gamma$. If $h = \sum c_s o_s$, then $(B_z h)(x_u) = \lambda M_{ue} M_z (\sum c_s M_s \gamma)$ for all z. Hence the map is well defined (if $h = 0$ then $\sum c_s M_s \gamma \in K_u$) and its kernel is exactly N. It is onto because the $M_s\gamma$ span $\mathbb{Q}^d$, and it intertwines because $B_z o_s = o_{zs} \mapsto M_z M_s \gamma$. Finally K_u is invariant under each M_z, $z \in L_R$, hence under the closure $\mathcal{K}$. □

Proof of (1)⇒(3). Fix u. Each M_z is invertible, so its induced operator on $\mathbb{Q}^d/K_u$ is invertible. By Lemma 15, Proposition 10 and Corollary 12 apply to the monoid L_R acting by $P_{q,z}$. Thus the Zariski closure $\mathcal{H}_u$ of the induced operators on $\mathbb{Q}^d/K_u$ has pairwise commuting unipotent elements and $\mathcal{H}_u^\circ$ is triangularizable over $\mathbb{Q}$. The restriction morphism res_u from the stabilizer of K_u to $\mathrm{GL}(\mathbb{Q}^d/K_u)$ maps $\mathcal{K}$ onto $\mathcal{H}_u$ by (A2).

Commutation. Let $g_1, g_2 \in \mathcal{G}$ be unipotent. By (A3) and Lemma 14 they lie in $\mathcal{G}^\circ \subseteq \mathcal{K}$, and $\mathrm{res}_u(g_1), \mathrm{res}_u(g_2)$ are unipotent elements of $\mathcal{H}_u$, hence commute. So $[g_1, g_2] = g_1 g_2 - g_2 g_1$ maps $\mathbb{Q}^d$ into K_u, that is, $\lambda M_{ue} a [g_1, g_2] = 0$ for every $a \in \mathcal{A}_R$. Taking $a = M_e^{-1}$ gives $\lambda M_u [g_1, g_2] = 0$. Since u is arbitrary and the λM_u span the dual space, $[g_1, g_2] = 0$.

Triangularizability. Taking $a = M_e^{-1}$ also shows $K_u \subseteq \ker(\lambda M_u)$, so $\bigcap_u K_u = 0$, and some finite set U_0 of words satisfies $\bigcap_{u \in U_0} K_u = 0$. The group $\mathcal{G}^\circ \subseteq \mathcal{K}$ preserves each K_u, and $\mathrm{res}_u(\mathcal{G}^\circ)$ is connected, hence contained in $\mathcal{H}_u^\circ$, which preserves a complete rational flag of $\mathbb{Q}^d/K_u$. The diagonal map $\mathbb{Q}^d \to \bigoplus_{u \in U_0} \mathbb{Q}^d/K_u$ is injective and $\mathcal{G}^\circ$-equivariant. Concatenating the flags gives a $\mathcal{G}^\circ$-invariant chain of rational subspaces of the direct sum with successive quotients of dimension at most one; intersecting it with the image of $\mathbb{Q}^d$ gives a complete $\mathcal{G}^\circ$-invariant rational flag of $\mathbb{Q}^d$. □

No hypothesis on the physical register maps was used: they may be singular, resetting or multiplicative. Invertibility of the minimal letter matrices enters through $\bar{E} = I$ in Proposition 10, through the invertibility of the induced block operators, through Lemma 6 (the closure $\mathcal{K}$ is a group), and through $M_e^{-1} \in \mathcal{A}_R$.

5 The normal form: proof of (3)⇒(2)

Assume (3) and write $\Gamma = \Gamma_f$, $\mathcal{G} = \mathcal{G}_f$. By Lemma 6, $\Gamma^\circ = \Gamma \cap \mathcal{G}^\circ$ is a normal subgroup of finite index; let $F = \Gamma/\Gamma^\circ$ and $\varphi : \Sigma^* \to F$, $\varphi(w) = M_w \Gamma^\circ$. The image $\varphi(\Sigma^*)$ is a submonoid of a finite group containing the images of the generators of Γ, hence equals F. Choose words r_g with $\varphi(r_g) = g$ and $r_1 = \varepsilon$, and put

$$c(g, a) = M_{r_g} M_a M_{r_{g\varphi(a)}}^{-1} \in \Gamma^\circ \qquad (g \in F,\ a \in \Sigma).$$

For $w = a_1 \cdots a_n$ let $g_0 = 1$ and $g_i = \varphi(a_1 \cdots a_i)$. Telescoping gives

$$M_w = c(g_0, a_1) c(g_1, a_2) \cdots c(g_{n-1}, a_n) M_{r_{g_n}}. \tag{4}$$

By (3) and (A5), $\mathcal{G}^\circ = T \ltimes U$, where T is a torus diagonalizable over $\mathbb{Q}$, U is the group of unipotent elements of $\mathcal{G}^\circ$, which is commutative by (3), and $\Gamma^\circ \subseteq \mathcal{G}^\circ(\mathbb{Q}) = T(\mathbb{Q}) U(\mathbb{Q})$. Since U is commutative, $\mathfrak{u} = \log U$ is a linear space of pairwise commuting nilpotent matrices, defined over $\mathbb{Q}$, with $U(\mathbb{Q}) = \exp(\mathfrak{u} \cap M_d(\mathbb{Q}))$. By (A5) the adjoint action of T on $\mathfrak{u}$ is diagonalizable over $\mathbb{Q}$: fix a rational basis $E_1, \ldots, E_p$ of $\mathfrak{u}$ with $t E_j t^{-1} = \chi_j(t) E_j$ for rational characters χ_j. Likewise decompose $\mathbb{Q}^d = \bigoplus_\psi V_\psi$ into rational weight spaces of T, with projections Π_ψ, so $t = \sum_\psi \psi(t) \Pi_\psi$.

Write $c(g, a) = t_{g,a} \exp(X_{g,a})$ with $t_{g,a} \in T(\mathbb{Q})$ and $X_{g,a} = \sum_j x_j(g, a) E_j \in \mathfrak{u}$, $x_j(g, a) \in \mathbb{Q}$. Along w define $T_0 = I$, $Z_0 = 0$ and

$$T_i = T_{i-1} t_{g_{i-1}, a_i}, \qquad Z_i = t_{g_{i-1}, a_i}^{-1} Z_{i-1} t_{g_{i-1}, a_i} + X_{g_{i-1}, a_i}.$$

Lemma 16. *$c(g_0, a_1) \cdots c(g_{i-1}, a_i) = T_i \exp(Z_i)$ for every i.*

Proof. By induction, with $t = t_{g_{i-1},a_i}$ and $X = X_{g_{i-1},a_i}$: $T_{i-1}\exp(Z_{i-1})\, t \exp(X) = T_{i-1} t\ \exp(t^{-1}Z_{i-1}t)\exp(X) = T_i \exp(t^{-1}Z_{i-1}t + X)$, using that $\mathfrak{u}$ is commutative and normalized by T. □

In coordinates $Z_i = \sum_j z_j^{(i)} E_j$ the recursion reads

$$z_j^{(i)} = \chi_j(t_{g_{i-1},a_i})^{-1} z_j^{(i-1)} + x_j(g_{i-1}, a_i), \qquad y_\psi^{(i)} = \psi(t_{g_{i-1},a_i})\, y_\psi^{(i-1)}, \tag{5}$$

where $y_\psi^{(i)} = \psi(T_i)$. These are isolated affine updates with rational coefficients, $\chi_j(t)^{-1}, \psi(t) \in \mathbb{Q}^\times$, that depend only on the current element of F and the letter read. By (4) and Lemma 16,

$$f(w) = \lambda T_n \exp(Z_n) M_{r_{g_n}} \gamma = \sum_\psi y_\psi^{(n)}\ \lambda \Pi_\psi \Big(\sum_{l=0}^{d-1} \frac{1}{l!} \Big(\sum_j z_j^{(n)} E_j \Big)^l \Big) M_{r_{g_n}} \gamma, \tag{6}$$

since exp of a nilpotent $d\times d$ matrix truncates after degree $d-1$. Expanding, $f(w) = \sum_{\psi,\mu} c_{\psi,\mu,g_n}\, y_\psi^{(n)} \prod_j (z_j^{(n)})^{\mu_j}$ over finitely many exponent vectors μ, with rational coefficients depending on $g_n = \varphi(w)$.

Now build the machine. Its control states are the elements of F, with initial state 1 and transitions $g \xrightarrow{a} g\varphi(a)$; this is a permutation automaton. For every pair (ψ, μ) allocate one register $y_{\psi,\mu}$, initialized to 1, and μ_j registers $z_{\psi,\mu,j,1}, \ldots, z_{\psi,\mu,j,\mu_j}$ for each j, initialized to 0. On letter a in state g, every copy of y_ψ and of z_j is updated in isolation by (5). The output in state g is

$$\sum_{\psi,\mu} c_{\psi,\mu,g}\, y_{\psi,\mu} \prod_j \prod_{l=1}^{\mu_j} z_{\psi,\mu,j,l}.$$

Each register occurs once in its own update and at most once in each output, so the machine is copyless, of the form (2), and computes f by (6). This proves (3)⇒(2). The implication (2)⇒(1) is trivial.

For the final claim of Theorem 1: in a machine as in (2), every letter acts on $\bigoplus_s V_r$ by a block permutation matrix whose blocks are pullbacks of maps $x \mapsto (\alpha_i x_i + \beta_i)_i$ with $\alpha_i \neq 0$, which are invertible by Lemma 11. The minimal representation is an invariant subquotient of this representation (Section 2), so its letter matrices are invertible.

6 The elementary reformulation and the two sectors

Lemma 17. *Let $\mathcal{G}^\circ$ be solvable and let $\Delta \subseteq \mathcal{G}^\circ$ be a Zariski dense subgroup. Then the group U of unipotent elements of $\mathcal{G}^\circ$ is the Zariski closure of the group generated by $\{g_u : g \in \Delta\}$.*

Proof. Let $\mathcal{U}$ be that closure. The generating set is invariant under conjugation by Δ, because $(hgh^{-1})_u = hg_u h^{-1}$, so $\mathcal{U}$ is normalized by Δ and hence by $\mathcal{G}^\circ$. Since $\mathcal{G}^\circ$ is connected and solvable, U is a closed subgroup [Bor91, 10.6] containing the generators, so $\mathcal{U} \subseteq U$. In the connected solvable group $\mathcal{G}^\circ/\mathcal{U}$, the image of Δ consists of semisimple elements by (A2). Commutators in a connected solvable group are unipotent (Lie–Kolchin), so the image of Δ is commutative; its closure is the whole quotient, which is therefore commutative and, since the semisimple elements of a commutative algebraic group form a closed subgroup [Bor91, 4.7], consists of semisimple elements; hence it is a torus. Its unipotent subgroup $U/\mathcal{U}$ is therefore trivial. □

Proof of (3)⇔(4). If (3) holds, $\Gamma \cap \mathcal{G}^\circ$ has finite index, is triangularizable over $\mathbb{Q}$, and the unipotent parts of its elements lie in $\mathcal{G}^\circ$ by (A2), so they commute. Conversely let Γ_1 be as in (4). Its closure has finite index in $\mathcal{G}$, so contains $\mathcal{G}^\circ$ by (A1), and fixes the rational flag fixed by Γ_1; hence $\mathcal{G}^\circ$ is triangularizable over $\mathbb{Q}$ and solvable. Every $g \in \Gamma$ has a power $g^k \in \Gamma_1$, $k \leq [\Gamma : \Gamma_1]$, so by Lemma 7 the unipotent parts of all elements

of Γ commute pairwise. By Lemma 17 applied to $\Delta = \Gamma \cap \mathcal{G}^\circ$ (dense by Lemma 6), the group U of unipotent elements of $\mathcal{G}^\circ$ is the closure of a commutative group, hence commutative. Every unipotent element of $\mathcal{G}$ lies in $\mathcal{G}^\circ$ (A3), hence in U. □

This completes the proof of Theorem 1.

Lemma 18. *If f is quasi-unipotent, then $\mathcal{G}_f^\circ$ is unipotent and triangularizable over $\mathbb{Q}$, and the unipotent word matrices are exactly the word matrices in $\mathcal{G}_f^\circ$; they are Zariski dense in $\mathcal{G}_f^\circ$.*

Proof. Eigenvalues of word matrices are roots of unity of degree at most d over $\mathbb{Q}$; Euler's totient of their order is at most d, so these orders divide a fixed m. Thus $(M_w^m - I)^d = 0$ for all words, and by Lemma 6 this polynomial identity holds on $\mathcal{G}_f$. A torus in $\mathcal{G}_f^\circ$ would then consist of elements of order dividing m, hence be trivial, and $\mathcal{G}_f^\circ$ is unipotent by (A4). The elements of $\Gamma_f \cap \mathcal{G}_f^\circ$ are unipotent rational matrices, so by Kolchin's theorem they fix a complete rational flag, and so does their closure $\mathcal{G}_f^\circ$. Unipotent elements of $\mathcal{G}_f$ lie in $\mathcal{G}_f^\circ$ by (A3), and density follows from Lemma 6. □

Proof of Theorem 3. (2) Let f be quasi-unipotent. By Lemma 18, condition (3) of Theorem 1 says exactly that $\mathcal{G}_f^\circ$ is commutative. Since commutation is a closed condition and unipotent word matrices are dense in $\mathcal{G}_f^\circ$, this is equivalent to pairwise commutation of unipotent word matrices. If $\mathcal{G}_f^\circ$ is commutative, $\Gamma_f \cap \mathcal{G}_f^\circ$ is abelian of finite index; conversely, the closure of a finite-index abelian subgroup is commutative and contains $\mathcal{G}_f^\circ$. In the proof of (3)⇒(2) the torus $T \subseteq \mathcal{G}_f^\circ$ is now trivial, so (5) reads $z_j := z_j + x_j(g, a)$, whence $z_j^{(n)} = \sum_{h,a} x_j(h, a)|w|_{h,a}$, and (6) expresses $f(w)$ as a polynomial in the counts $|w|_{h,a}$ with coefficients depending on $\varphi(w)$. Conversely such a function is computed by the CCRA with control F that keeps one counter register per occurrence of a variable in each monomial of each P_g; counters of $n_{h,a}$ increment on a in state h.

(1) If f is unipotent, the identity $(g - I)^d = 0$ holds on M_{Σ^*} and hence on $\mathcal{G}_f$, so $\mathcal{G}_f = \mathcal{G}_f^\circ$ is unipotent and F is trivial. By part (2), CCRA-computability is equivalent to commutation of all word matrices, that is, of the letter matrices, and to f being a polynomial in the letter counts. If f is permutation invariant, then $\lambda M_u(M_aM_b - M_bM_a)M_v\gamma = 0$ for all u, v, and minimality gives $M_aM_b = M_bM_a$; the converse implications are clear. □

7 Consequences and examples

Proof of Corollary 4. *Polynomial ambiguity.* Take a machine as in Theorem 1(2), with control states S, and its linearization on $\bigoplus_{s\in S} V_r$, viewed as a weighted automaton with states (s, I), $I \subseteq \{1, \dots, r\}$, and initial vector supported on the states (s_0, I). The pullback of $x_{I'}$ under $x \mapsto (\alpha_i x_i + \beta_i)_i$ contains only monomials x_I with $I \subseteq I'$, so a transition from (s, I) to (s', I') on a requires $I \subseteq I'$ and $s' = \delta(s, a)$. On a word of length n the control component of a run with nonzero initial weight is determined, and the monomial component is a chain $I_0 \subseteq \cdots \subseteq I_n$ determined by the time at which each register enters it, so there are at most $(n + 2)^r$ such runs.

Reversal. The reversal has minimal representation $(\gamma^\mathsf{T}, (M_a^\mathsf{T})_a, \lambda^\mathsf{T})$. The automorphism $g \mapsto (g^\mathsf{T})^{-1}$ of GL_d maps Γ_f onto the group generated by the M_a^T, hence $\mathcal{G}_f$ onto its closure, and it preserves condition (3) of Theorem 1: it maps upper triangular matrices to lower triangular ones and unipotent elements to unipotent elements.

Example 17 of [MPS26b]. Its automaton has letter matrices

$$M_a = \begin{pmatrix} 2 & 2 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad M_b = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix},$$

initial vector e_1^T and final vector e_3. This representation is minimal: the rows $e_1^\mathsf{T}, e_1^\mathsf{T}M_a, e_1^\mathsf{T}M_aM_b$ and the columns $e_3, M_be_3, M_aM_be_3$ are bases. The unipotent Jordan parts $(M_a)_u = I + E_{12}$ and $(M_b)_u = I + E_{23}$ do not commute, and both lie in $\mathcal{G}_f$ by (A2). By Theorem 1(3) the series is not CCRA-computable. This recovers [MPS26b, Theorem 18], whose proof uses a Fermat-quotient argument on pumping sequences. □

Example 19 (Zariski closure is needed)**.** Let $M_a = 2(I + E_{12})$, $M_b = 3(I + E_{23})$, $\lambda = e_1^\mathsf{T}$, $\gamma = e_3$, so $f(w) = 2^{|w|_a}3^{|w|_b}[w : ab]$. The representation is minimal. An element of Γ_f is $2^\alpha 3^\beta h$ with h in the integral Heisenberg group and α, β the exponent sums of the two generators; it is unipotent iff $\alpha = \beta = 0$, in which case h is central. So all unipotent elements of Γ_f commute. Nevertheless $(M_a)_u = I + E_{12}$ and $(M_b)_u = I + E_{23}$ do not commute, so f is not CCRA-computable.

Example 20 (positive cases)**.** (i) The alternating sum $\sum_i (-1)^i b_i$ has Γ_f infinite dihedral and $\mathcal{G}_f^\circ \cong \mathbb{G}_a$; the normal form uses a two-state parity control and one counter. (ii) For $g(b_1 \cdots b_j) = \sum_i 2^i b_i$, the closure is the affine group of the line; the normal form uses one state, registers $h := h/2 + b$ and $p := 2p$ with $h = 0$, $p = 1$ initially, and output hp. (iii) The alternating position sum $\sum_i (-1)^i i b_i$ is quasi-unipotent but not CCRA-computable: here F has order two, and two of the unipotent cocycles $c(g, a) \in \Gamma_f \cap \mathcal{G}_f^\circ$ of Section 5 do not commute. Explicitly, on coordinates (v, t, s) with $t = n(-1)^n$ and $s = (-1)^n$, appending b acts by $B_b = \left(\begin{smallmatrix} 1 & -b & -b \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{smallmatrix}\right)$; this three-dimensional representation (transposed, with initial row $(0, 0, 1)$ and final column $(1, 0, 0)^\mathsf{T}$) is minimal, and the unipotent products $U = B_0^2 = I + 2E_{23}$ and $V = B_1B_0 = I + E_{12} + 2E_{13} + 2E_{23}$ satisfy $UV - VU = -2E_{13} \neq 0$.

7.1 The uniform pumping property of the position sum

Write $\ell(w) = |w|$, $c(w) = \sum_i w_i$, and let f be (1). Then $f(uv) = f(u) + f(v) + \ell(u)c(v)$. On column coordinates $(f, \ell, 1)^\mathsf{T}$, appending a bit b acts by the unitriangular matrix A_b with rows $(1, b, b)$, $(0, 1, 1)$, $(0, 0, 1)$, so the row representation with initial row $(0, 0, 1)$, matrices A_b^T and final column $(1, 0, 0)^\mathsf{T}$ recognizes f; it is unipotent. Since $f(01) = 2 \neq 1 = f(10)$, Theorem 3(1) shows that no CCRA computes f.

For a subsemiring R of a field K, an *R-generable exponential polynomial sequence* [MPS26a, Definition 11] is obtained by pointwise sums and products from constant sequences in R, the sequence $(n1_K)_n$, the sequences $(\alpha^n)_n$ with $\alpha \in R$, and the geometric sums $(\sum_{i<n} \alpha^i)_n$ with $\alpha \in R \setminus \{1\}$. In characteristic zero, the degree-j coefficient sum of its eventual exponential-polynomial normal form $\sum_{\lambda,j} a_{\lambda,j} n^j \lambda^n$ is $\sum_\lambda a_{\lambda,j}$. The uniform condition of [MPS26a] asks for one R and one $m \geq 1$ such that $(f(uw^{m(n+1)}v))_n$ is R-generable for all words u, w, v, with all degree-coefficient sums in R.

Theorem 21. *The function* (1) *satisfies the uniform condition with the finitely generated subsemiring* $R = \mathbb{Z}[1/2]$ *and* $m = 1$*, but is not computed by any CCRA over* $\mathbb{Q}$*.*

Proof. For words u, w, v and $n \geq 0$, the j-th copy of w ($0 \leq j < n$) contributes $f(w) + (\ell(u) + j\ell(w))c(w)$ and the final v contributes $f(v) + (\ell(u) + n\ell(w))c(v)$. Hence

$$f(uw^nv) = C + An + B\tfrac{n(n-1)}{2}, \quad C = f(u) + f(v) + \ell(u)c(v),\ A = f(w) + \ell(u)c(w) + \ell(w)c(v),\ B = \ell(w)c(w),$$

and $f(uw^{n+1}v) = (C + A) + (A + B/2)n + (B/2)n^2$. All coefficients lie in $R = \mathbb{Z}[1/2]$, which is generated as a semiring by -1 and $1/2$. The sequence is obtained from constants and $(n)_n$ by sums and products, the only

exponential base is 1, and the degree-coefficient sums are the displayed coefficients. Nonrecognizability was shown above. This calculation is the baseline of [MPS26a, Example 26] and [MPS26b, Appendix B.4]. □

7.2 Polynomial ambiguity in the invertible sector

A weighted automaton $\mathcal{P}$ over $\mathbb{Q}$ consists of a finite state set S, an initial row $\alpha \in \mathbb{Q}^{1\times S}$, a final column $\beta \in \mathbb{Q}^{S\times 1}$ and letter matrices $A_a \in \mathbb{Q}^{S\times S}$, where $(A_a)_{pq}$ is the weight of the transition $p \xrightarrow{a} q$ (0 if there is none); it computes $f(w) = \alpha A_w \beta$, $A_{a_1\cdots a_n} = A_{a_1}\cdots A_{a_n}$. A *path* labelled $w = a_1\cdots a_n$ from p to q is a sequence $p = s_0, \ldots, s_n = q$ with $(A_{a_i})_{s_{i-1}s_i} \neq 0$ for all i; it is *accepting* if moreover $\alpha_{s_0} \neq 0 \neq \beta_{s_n}$. The automaton is *polynomially ambiguous* if the number of accepting paths labelled w is bounded by a polynomial in $|w|$ [MPS26a, PS24]. A series is polynomially ambiguous if some polynomially ambiguous weighted automaton over $\mathbb{Q}$ computes it.

Theorem 22. *For an invertible rational series $f : \Sigma^* \to \mathbb{Q}$ the following are equivalent.*

(1) *f is polynomially ambiguous.*
(2) *$\mathcal{G}_f^\circ$ is triangularizable over $\mathbb{Q}$.*
(3) *Γ_f has a subgroup of finite index that is triangularizable over $\mathbb{Q}$.*

Together with Theorem 1 and Corollary 4 this settles the relation between the two classes in the invertible sector.

Corollary 23. *An invertible series is CCRA-computable if and only if it is polynomially ambiguous and any two unipotent elements of $\mathcal{G}_f$ commute; equivalently, iff it is polynomially ambiguous and the unipotent radical of $\mathcal{G}_f^\circ$ is commutative. Given a rational weighted automaton, one can decide in deterministic polynomial time whether its series is invertible and polynomially ambiguous.*

Proof. The first claim follows from Theorem 22 and the equivalence (1)⇔(3) of Theorem 1; if $\mathcal{G}_f^\circ$ is triangularizable over $\mathbb{Q}$, every unipotent element of $\mathcal{G}_f$ lies in $\mathcal{G}_f^\circ$ (A3), and the unipotent elements of $\mathcal{G}_f^\circ$ form its unipotent radical (A5). For the second claim, minimize and test invertibility (Theorem 5) and compute the block form of Section 9.1 (step 1 of the algorithm there, in polynomial time); condition (2) is condition (a) of Section 9, which is decidable in deterministic polynomial time by Lemma 32. □

Thus in the invertible sector the CCRA series are exactly the polynomially ambiguous series for which the unipotent radical of $\mathcal{G}_f^\circ$ is commutative; the series of [MPS26b, Example 17] and of Example 19 are polynomially ambiguous, with $\mathcal{G}_f^\circ$ triangular over $\mathbb{Q}$ and noncommutative unipotent radical. In particular, all eigenvalues of all word matrices of an invertible polynomially ambiguous series are roots of rationals (elements of $\Gamma_f \cap \mathcal{G}_f^\circ$ are triangular over $\mathbb{Q}$, and $g^k \in \Gamma_f \cap \mathcal{G}_f^\circ$ for $g \in \Gamma_f$ and $k = [\Gamma_f : \Gamma_f \cap \mathcal{G}_f^\circ]$), and $\mathcal{G}_f^\circ$ modulo its unipotent radical is a $\mathbb{Q}$-split torus (A5); a non-split torus, as for a rotation matrix of infinite order, never occurs. This is in line with the one-letter statement of [BFLM22].

Remark 24. Puch and Smertnig [PS24, Theorem 1.6] characterize polynomial ambiguity of an invertible series by a spectral condition on the word matrices (finitely generated spectrum and a power-splitting property); with [PS24, Theorem 1.3, Proposition 9.4] this is equivalent to the representation being an epimorphic image of a block-triangular representation with monomial diagonal blocks. That structural form is closely related to condition (3) of Theorem 22, since a block-triangular group with monomial diagonal blocks has a triangularizable subgroup of finite index. The sufficiency direction there uses unit equations, and decidability is obtained through Zariski-closure computations. Theorem 22 gives an intrinsic condition on the Zariski closure of the minimal group, its proof below is direct (it uses neither their result nor unit

equations), and it yields the polynomial-time bound of Corollary 23. The proof of Theorem 22 uses only that $\mathbb{Q}$ has characteristic zero and holds verbatim over any field of characteristic zero, with automata and triangularizability taken over that field; the complexity claim is specific to $\mathbb{Q}$.

Proof of Theorem 22: (2)⇔(3) and (2)⇒(1). Write $\Gamma = \Gamma_f$, $\mathcal{G} = \mathcal{G}_f$ and $\Gamma^\circ = \Gamma \cap \mathcal{G}^\circ$, a normal subgroup of finite index (Lemma 6). If (2) holds, Γ° fixes the rational flag fixed by $\mathcal{G}^\circ$. If Γ_1 is as in (3), its closure has finite index in $\mathcal{G}$, hence contains $\mathcal{G}^\circ$ (A1), and fixes the rational flag fixed by Γ_1.

Assume (2) and choose $h_0 \in \mathrm{GL}_d(\mathbb{Q})$ such that $h_0^{-1} g h_0$ is upper triangular for all $g \in \mathcal{G}^\circ$. With $F = \Gamma/\Gamma^\circ$, φ, the words r_g and the cocycle $c(g,a) = M_{r_g} M_a M_{r_{g\varphi(a)}}^{-1} \in \Gamma^\circ$ as in Section 5 (their construction and (4) use only Lemma 6, not condition (3) of Theorem 1), the matrices $c'(g,a) = h_0^{-1} c(g,a) h_0$ are rational and upper triangular, and by (4)

$$f(w) = (\lambda h_0)\, c'(g_0, a_1) \cdots c'(g_{n-1}, a_n)\, (h_0^{-1} M_{r_{g_n}} \gamma).$$

Let $\mathcal{P}$ have states $F \times \{1, \dots, d\}$, a transition $(g,i) \xrightarrow{a} (g\varphi(a), j)$ of weight $c'(g,a)_{ij}$ whenever this entry is nonzero, initial weights $(\lambda h_0)_i$ on $(1,i)$ and final weights $(h_0^{-1} M_{r_g} \gamma)_j$ on (g,j). Its letter matrices are block permutation matrices with blocks $c'(g,a)$, so $\mathcal{P}$ computes f by the displayed formula. On a word of length n the first components along an accepting path are $g_0, \dots, g_n$, and the second components form a nondecreasing sequence $i_0 \le \cdots \le i_n$ in $\{1, \dots, d\}$, because each $c'(g,a)$ is upper triangular. Such a sequence is determined by the number of occurrences of each index, so there are at most $(n+2)^d$ accepting paths, and $\mathcal{P}$ is polynomially ambiguous. □

For (1)⇒(2), fix a polynomially ambiguous automaton $\mathcal{P}$ computing f, which we may assume *trim*: every state lies on an accepting path (removing the other states changes neither f nor the accepting paths).

Lemma 25. *Let C be a strongly connected component of the transition graph of $\mathcal{P}$. For every word w and all $p, q \in C$ there is at most one path labelled w from p to q, and its states lie in C.*

Proof. Every state of such a path is reachable from p and reaches q, so it lies in C. Suppose that $\pi_1 \neq \pi_2$ are two paths labelled w from p to q; then $|w| \ge 1$. Let τ be a path from q to p, labelled v (empty if $p = q$). Then $\pi_1\tau$ and $\pi_2\tau$ are distinct paths labelled wv from p to p. As $\mathcal{P}$ is trim, there are paths labelled u from a state with nonzero initial weight to p and labelled u' from p to a state with nonzero final weight. Choosing one of the two cycles for each of the k blocks gives 2^k accepting paths labelled $u(wv)^k u'$, pairwise distinct because the blocks have the same length and are aligned; as $|u(wv)^k u'|$ grows linearly in k, this contradicts polynomial ambiguity. □

Lemma 26 (Component lemma). *Let C be a finite set and $\rho(a) \in \mathbb{Q}^{C \times C}$ for $a \in \Sigma$, with $\rho(a_1 \cdots a_n) = \rho(a_1) \cdots \rho(a_n)$. Define paths as above and assume:*

(U) *for every word w and all $p, q \in C$ there is at most one path labelled w from p to q.*

Let $Y \subseteq X \subseteq \mathbb{Q}^C$ be subspaces invariant under every $\rho(a)$ (acting on columns), such that every $\rho(a)$ induces an invertible operator $\sigma(a)$ on $D = X/Y$, and let $\mathcal{H}$ be the Zariski closure of $\{\sigma(w) : w \in \Sigma^*\}$ in $\mathrm{GL}(D)$. Then $\mathcal{H}^\circ$ is a torus split over $\mathbb{Q}$; in particular $\mathcal{H}^\circ$ is diagonalizable over $\mathbb{Q}$.

Proof. *Supports.* By (U), $\rho(w)_{pq}$ is the product of the weights along the unique path labelled w from p to q if it exists, and 0 otherwise. Let $B_w \subseteq C \times C$ be the support of $\rho(w)$. Then $B_{uv} = B_u \circ B_v$ (composition of relations), so $w \mapsto B_w$ is a morphism onto a finite monoid $\mathfrak{B}$ of relations. For words $z_1, \dots, z_k$ the sequences $p = s_0, \dots, s_k = q$ with $(s_{i-1}, s_i) \in B_{z_i}$ for all i correspond injectively to paths labelled $z_1 \cdots z_k$ from p to q, so by (U) there is at most one such sequence, and

$$\rho(z_1 \cdots z_k)_{pq} = \textstyle\prod_{i=1}^{k} \rho(z_i)_{s_{i-1} s_i} \tag{7}$$

for that sequence if it exists, and 0 otherwise.

An idempotent with a group around it. The intersection $\mathfrak{J}$ of all two-sided ideals of $\mathfrak{B}$ is nonempty (it contains the product of all of them), hence the minimum ideal, and $\mathfrak{B}^1 x \mathfrak{B}^1 = \mathfrak{J}$ for every $x \in \mathfrak{J}$. Some power of an element of $\mathfrak{J}$ is an idempotent $E \in \mathfrak{J}$; write $E = B_e$ for a word e. The finite monoid $\mathfrak{M} = E\mathfrak{B}E$ with identity E is a group: for $x \in \mathfrak{M}$ we have $E = sxt$ with $s, t \in \mathfrak{B}^1$, hence $E = (EsE)x(EtE)$; so $t'(s'x) = E$ for $s' = EsE$, $t' = EtE \in \mathfrak{M}$, because in a finite monoid $yz = 1$ implies $zy = 1$; thus x has a left inverse, hence is a unit. Let $n = |\mathfrak{M}|$. Then $B_{(exe)^n} = (EB_xE)^n = E$ for every word x. Put $L = \{z \in \Sigma^* : B_z = E\}$, a semigroup containing e and all $(exe)^n$. If $D \neq 0$ then $\sigma(e)$ is invertible, so $\rho(e) \neq 0$ and $E \neq \emptyset$; if $D = 0$ there is nothing to prove.

Fixed points of E. Let $\Theta = \{\theta \in C : (\theta, \theta) \in E\}$. By the uniqueness statement above with all $B_{z_i} = E$, for every $k \geq 1$ and all p, q there is at most one sequence of length k from p to q with consecutive pairs in E (an *E-sequence*). Hence: (i) if $\theta \neq \theta'$ are in Θ then $(\theta, \theta') \notin E$, since otherwise θ, θ, θ' and θ, θ', θ' are two E-sequences; (ii) for $(p, q) \in E$ there is exactly one $\theta \in \Theta$ with $(p, \theta), (\theta, q) \in E$. Uniqueness is clear. For existence, $E = E^k$ for $k = |C| + 1$ yields an E-sequence $s_0 = p, \ldots, s_k = q$, in which some state repeats, $s_i = s_j$ with $i < j$; then $(s_i, s_i) \in E^{j-i} = E$, so $\theta = s_i \in \Theta$, and $(p, \theta) \in E$ (if $i = 0$ then $p = \theta$) and $(\theta, q) \in E$ (if $j = k$ then $\theta = q$).

Factorization through a diagonal character. Let $\iota \in \mathbb{Q}^{C \times \Theta}$ be the inclusion of coordinates. For $z \in L$ the matrix $\chi(z) = \iota^{\mathsf{T}} \rho(z) \iota$ is diagonal by (i), with nonzero diagonal entries. For $z, z' \in L$, (7) and the uniqueness of E-sequences give $\chi(zz') = \chi(z)\chi(z')$: the only E-sequence θ, r, θ is $r = \theta$. Moreover, for $z, y, z' \in L$,

$$\rho(zyz') = \rho(z)\, \iota\, \chi(y)\, \iota^{\mathsf{T}} \rho(z'). \tag{8}$$

Indeed, the (p, q) entry of the right side is $\sum_{\theta \in \Theta} \rho(z)_{p\theta} \chi(y)_{\theta\theta} \rho(z')_{\theta q}$, whose nonzero terms correspond to E-sequences p, θ, θ, q; the left side is given by (7) with E-sequences p, r, s, q of length three. If $(p, q) \notin E = E^3$, both sides vanish. If $(p, q) \in E$, the unique θ of (ii) gives an E-sequence p, θ, θ, q, which is therefore the only E-sequence of length three from p to q, and both sides equal $\rho(z)_{p\theta} \rho(y)_{\theta\theta} \rho(z')_{\theta q}$.

A morphism from a $\mathbb{Q}$-split diagonalizable group onto the closure. Let $\mathcal{D} \subseteq \mathbb{Q}^{\Theta \times \Theta}$ be the linear span of $\chi(L)$, a space of diagonal matrices. By (8) with $z = z' = e$, the linear map $\delta \mapsto \rho(e) \iota \delta \iota^{\mathsf{T}} \rho(e)$ sends $\chi(y)$ to $\rho(eye)$ for $y \in L$; hence on $\mathcal{D}$ it takes values preserving X and Y and induces a linear map $\Lambda : \mathcal{D} \to \mathrm{End}(D)$ with $\Lambda(\chi(y)) = \sigma(eye) = \sigma(e)\sigma(y)\sigma(e)$. The linear map $\Psi(\delta) = \sigma(e)^{-1} \Lambda(\delta) \sigma(e)^{-1}$, defined over $\mathbb{Q}$ and extended to $\mathcal{D} \otimes \overline{\mathbb{Q}}$, satisfies $\Psi(\chi(y)) = \sigma(y)$ for $y \in L$. Let $\mathcal{T}$ be the Zariski closure of $\chi(L)$ in the diagonal torus of GL_Θ; it lies in $\mathcal{D} \otimes \overline{\mathbb{Q}}$ and is a closed subgroup by the proof of Lemma 6, which applies to subsemigroups of GL with one more line: from $\bar{S} = x\bar{S}$ and $x \in \bar{S}$ we get $x = xy$ with $y \in \bar{S}$, so $1 = y \in \bar{S}$ and then $x^{-1} \in \bar{S}$. The polynomial identity $\Psi(ab) = \Psi(a)\Psi(b)$ holds on $\chi(L) \times \chi(L)$, hence on its closure $\mathcal{T} \times \mathcal{T}$; and $\Psi(1) = 1$, because $\Psi(1)\Psi(\chi(e)) = \Psi(\chi(e)) = \sigma(e)$ is invertible; then $\Psi(a)\Psi(a^{-1}) = 1$. So $\Psi : \mathcal{T} \to \mathrm{GL}(D)$ is a morphism of algebraic groups defined over $\mathbb{Q}$. Its image is closed (A2), contains $\sigma(L)$, and lies in the closure $\mathcal{K}$ of $\sigma(L)$; hence $\mathcal{K} = \Psi(\mathcal{T})$ and $\mathcal{K}^\circ = \Psi(\mathcal{T}^\circ)$. The torus $\mathcal{T}^\circ$ is a subtorus of a diagonal torus defined over $\mathbb{Q}$, hence split over $\mathbb{Q}$, and so is its image $\mathcal{K}^\circ$ under the $\mathbb{Q}$-morphism Ψ (characters of $\mathcal{K}^\circ$ pull back injectively and Galois-equivariantly to characters of $\mathcal{T}^\circ$).

The closure contains $\mathcal{H}^\circ$. This is the argument of Lemma 14. The group $\mathcal{H}$ is the closure of $\sigma(\Sigma^*)$ (Lemma 6), and $\mathcal{K} \subseteq \mathcal{H}$ is a closed subgroup. Let $g = \sigma(e)$ and $\Xi(h) = (ghg)^n$. For every word x, $\Xi(\sigma(x)) = \sigma((exe)^n) \in \mathcal{K}$, so $\Xi(\mathcal{H}) \subseteq \mathcal{K}$ by density; for $k \in \mathcal{H}^\circ$, $\Xi(g^{-1}kg^{-1}) = k^n \in \mathcal{K}$. The closure of the group generated by all k^n, $k \in \mathcal{H}^\circ$, is normal in $\mathcal{H}^\circ$ and contained in $\mathcal{K}$, and the quotient of $\mathcal{H}^\circ$ by it is connected of finite exponent, hence trivial (A4). Thus $\mathcal{H}^\circ \subseteq \mathcal{K} \subseteq \mathcal{H}$, so $\mathcal{H}^\circ = \mathcal{K}^\circ$ is a $\mathbb{Q}$-split torus, and a $\mathbb{Q}$-split torus acting on a rational vector space is diagonalizable over $\mathbb{Q}$ (A5). □

Proof of Theorem 22: (1)⇒(2). Let $\mathcal{P} = (\alpha, (A_a)_a, \beta)$ be trim and polynomially ambiguous, computing f. Number its strongly connected components $C_1, \dots, C_r$ so that every transition goes from C_i to C_j with $i \le j$. Then $V_j = \operatorname{span}\{e_q : q \in C_1 \cup \cdots \cup C_j\}$ is invariant under every A_a acting on columns, and A_w induces on $V_j/V_{j-1} \cong \mathbb{Q}^{C_j}$ the diagonal block $\rho_j(w) = (A_w)_{C_jC_j}$; the ρ_j satisfy (U) by Lemma 25.

By Section 2, with $W = \operatorname{span}\{A_w\beta\}$ and $N = \{h \in W : \alpha A_v h = 0 \ \forall v\}$, the operators induced on W/N form a representation isomorphic to the minimal one; identify W/N with $\mathbb{Q}^d$. Put $W_j = W \cap V_j$ and $\Phi_j = (W_j + N)/N$, so $0 = \Phi_0 \subseteq \cdots \subseteq \Phi_r = \mathbb{Q}^d$ are rational subspaces invariant under every M_a, hence under Γ_f and $\mathcal{G}_f$. Restricting coordinates to C_j maps W_j equivariantly onto $X_j \subseteq \mathbb{Q}^{C_j}$ with kernel W_{j-1}, and the surjection $W_j \to \Phi_j/\Phi_{j-1}$ has kernel $W_j \cap (W_{j-1} + N) \supseteq W_{j-1}$. Hence $\Phi_j/\Phi_{j-1} \cong X_j/Y_j$, where Y_j is the image of $W_j \cap (W_{j-1} + N)$, and this isomorphism intertwines the operator induced by M_w with the operator $\sigma_j(w)$ induced by $\rho_j(w)$. As M_w is invertible, so is $\sigma_j(w)$. By Lemma 26 the closure $\mathcal{H}_j$ of $\{\sigma_j(w)\}$ has an identity component diagonalizable over $\mathbb{Q}$. The induced-action morphism ϖ_j from $\mathcal{G}_f$ to $\mathrm{GL}(\Phi_j/\Phi_{j-1})$ maps $\mathcal{G}_f$ onto $\mathcal{H}_j$ and $\mathcal{G}_f^\circ$ onto $\mathcal{H}_j^\circ$ (A2). Choosing in each Φ_j/Φ_{j-1} a rational basis diagonalizing $\mathcal{H}_j^\circ$ and lifting, we obtain a complete rational flag of $\mathbb{Q}^d$ fixed by $\mathcal{G}_f^\circ$. □

The proof shows more: $\mathcal{G}_f^\circ$ acts on each graded piece Φ_j/Φ_{j-1} through a $\mathbb{Q}$-split torus, so the unipotent radical of $\mathcal{G}_f^\circ$ is exactly the subgroup acting trivially on all of them (that subgroup is normal and consists of unipotent elements, and unipotent elements act trivially through a torus). Polynomial ambiguity thus controls the reductive part of $\mathcal{G}_f^\circ$, whereas copylessness additionally forces the unipotent radical to be commutative.

8 Algorithms

8.1 Deciding the invertible sector

Proof of Theorem 5, first two claims. The input is a weighted automaton with binary-encoded rational entries, and the question decided is whether the series it computes is computed by some CCRA over $\mathbb{Q}$; we work in the Turing model with exact rational arithmetic. Minimization is a polynomial-bit-time computation [Sch61, BR11, Tze92], and invertibility of the minimal letter matrices is a determinant test. Now let f be invertible, with minimal letter matrices M_a. Derksen, Jeandel and Koiran [DJK05] give an algorithm that, from finitely many invertible matrices with algebraic entries, computes finitely many polynomials defining the Zariski closure of the group they generate; we apply it to the inverse-closed generating set $\{M_a, M_a^{-1}\}$ and do not claim any complexity bound for this step (see [NPS+22] for a complexity analysis of Zariski-closure computation). From these polynomials one computes the vanishing ideal of $\mathcal{G}_f$ (the radical of the ideal they generate, by standard Gröbner-basis methods) and hence the Lie algebra $\mathfrak{g} \subseteq \mathfrak{gl}_d(\mathbb{Q})$ of $\mathcal{G}_f^\circ$ as the tangent space at the identity, that is, the kernel of the Jacobian at I of generators of the ideal. By (A6), $\mathcal{G}_f^\circ$ is triangularizable over $\mathbb{Q}$ iff $\mathfrak{g}$ stabilizes a complete rational flag. Let $\mathcal{A}$ be the unital associative algebra generated by $\mathfrak{g}$ and J its radical, computed by Dickson's trace criterion in characteristic zero. Then $\mathfrak{g}$ stabilizes a complete rational flag iff $\mathcal{A}/J \cong \mathbb{Q}^s$ as algebras: if $\mathfrak{g}$ is triangular, the strictly triangular part of $\mathcal{A}$ is a nilpotent ideal and $\mathcal{A}/J$ is a quotient of a subalgebra of the diagonal algebra; conversely, if $\mathcal{A}/J \cong \mathbb{Q}^s$ then all simple $\mathcal{A}$-modules are one-dimensional over $\mathbb{Q}$ and a composition series of $\mathbb{Q}^d$ is a complete invariant flag. The condition $\mathcal{A}/J \cong \mathbb{Q}^s$ is checked on a basis of $\mathcal{A}/J$: the basis elements must commute and their minimal polynomials must split into linear factors over $\mathbb{Q}$, which is tested by polynomial factorization over $\mathbb{Q}$. If it holds, the unipotent elements of $\mathcal{G}_f$ form the connected group $U = R_u(\mathcal{G}_f^\circ)$ with Lie algebra $\mathfrak{g} \cap J$ (the nilpotent elements of $\mathfrak{g}$), and U is commutative iff $\mathfrak{g} \cap J$ is abelian. All of this is exact linear algebra once the closure is known. □

Section 9 replaces the Zariski-closure step by a direct procedure that uses only linear algebra over $\mathbb{Q}$, finiteness tests and, for the torus part, the algorithm of Friedl and Rónyai and rational root finding; it runs in deterministic polynomial time (Theorem 40).

8.2 The unipotent sector in polynomial time

For $d \geq 1$, all word matrices M_w are unipotent iff $\mathrm{tr}(M_w) = d$ for all words: for the reverse direction, apply the trace condition to w^j, $1 \leq j \leq d$; Newton's identities give characteristic polynomial $(T-1)^d$. Build the invariant span of $(\mathrm{vec}(M_w), 1)$ in dimension d^2+1, starting with $(\mathrm{vec}(I), 1)$ and closing under $(\mathrm{vec}(X), c) \mapsto (\mathrm{vec}(XM_a), c)$. Breadth-first closure decides whether the functional $\mathrm{tr}(X) - dc$ vanishes on the span. Retained basis vectors are word vectors of length at most d^2. First minimize the input representation: an invisible eigenvalue must not reject a property of the function. If the test passes, Theorem 3(1) reduces CCRA-computability to pairwise commutator tests.

All steps have polynomial bit complexity. With supplied dimension n and encoding length L, a common denominator has $O(L)$ bits, entries of length-k products have $O(k(L+\log(n+1))+L)$ bits, and basis words in minimization have length at most n; Hadamard's bound and fraction-free elimination bound minors. The trace span uses words of length at most d^2+1 and at most d^2+1 retained vectors. This proves the unipotent claims of Theorem 5; the remaining claims are Theorems 28 and 40.

8.3 Explicit pattern sums

For words w, p let $[w : p]$ count scattered embeddings of p into w, with $[w : \varepsilon] = 1$.

Proposition 27. *Given explicitly listed words p over an explicitly given alphabet and binary-encoded rational coefficients c_p, let $f(w) = \sum_p c_p[w : p]$ and $D = \sum_p (|p|+1)$. Then f is unipotent; CCRA-computability is equivalent to permutation invariance and decidable in polynomial bit time. A YES instance has a one-state copyless realization of polynomial encoding size with at most $\sum_p |p|$ registers. A NO instance has computable words $uabv$ and $ubav$ of length at most $2D$ with different values.*

Proof. For $p = p_1 \cdots p_\ell$ take states $0, \ldots, \ell$, $T_a = I + \sum_{i:p_i=a} E_{i-1,i}$, $\lambda = e_0^{\mathsf{T}}$ and $\gamma = e_\ell$; a run stays or advances one edge per letter, so it counts $[w : p]$. Direct sums scaled by c_p give a unitriangular representation of dimension D, and unipotence descends to the minimal subquotient. Build bases of reachable rows λT_u and observable columns $T_v\gamma$ with $|u|, |v| \leq D-1$, and test $r(T_aT_b - T_bT_a)c$ for basis rows r, basis columns c and letters a, b. A nonzero value yields the words $uabv$, $ubav$; otherwise bilinearity gives invariance under adjacent transpositions in every context, hence permutation invariance. Entries of length-k products count embeddings and are at most 2^k, so all quantities have polynomially many bits.

For synthesis, order the alphabet. On a YES instance $f(w) = f(\mathrm{sort}(w))$, and a nondecreasing pattern p has $\prod_a \binom{|w|_a}{m_{p,a}}$ embeddings into the sorted word, $m_{p,a} = |p|_a$, while other patterns have none. For each nondecreasing listed p, letter a and $0 \leq j < m_{p,a}$, allocate a register $x_{p,a,j}$, initially $-j$, incremented on a. The output $\sum_p \frac{c_p}{\prod_a m_{p,a}!} \prod_{a,j} x_{p,a,j}$ is correct by the falling-factorial identity and is syntactically copyless. □

9 Polynomial-time decision procedures

Throughout this section $\Gamma = \langle M_1, \ldots, M_k \rangle \leq \mathrm{GL}_d(\mathbb{Q})$, where $M_1, \ldots, M_k$ are the minimal letter matrices of an invertible series (computed in polynomial time by Theorem 5), $\mathcal{G}$ is the Zariski closure of Γ, and σ is the total bit length of the entries of the $M_a^{\pm 1}$. We work in the Turing model with exact rational arithmetic. The *height* $\mathrm{ht}(V)$ of a subspace $V \subseteq \mathbb{Q}^N$ is the largest bit length of an entry of its reduced row echelon basis. Condition (3) of Theorem 1 is the conjunction of

(a) $\mathcal{G}^\circ$ is triangularizable over $\mathbb{Q}$, and

(b) any two unipotent elements of $\mathcal{G}$ commute.

Theorem 28. (1) (Quasi-unipotent sector.) *Given a rational weighted automaton, one can decide in deterministic polynomial time whether its series f is invertible and quasi-unipotent, and if so whether f is CCRA-computable, equivalently whether Γ_f is virtually abelian. The algorithm uses only exact linear algebra over $\mathbb{Q}$ and Hermite normal forms of integer matrices.*

(2) (Invertible sector.) *CCRA-computability of an invertible series is decided by a deterministic algorithm whose running time is polynomial in the input length and in the number B_f of Definition 36 (the largest height of the Lie algebras of the unipotent radicals of those truncations of $\mathcal{G}_f$ that the algorithm visits), and is $2^{O(m\log(dk))}$ times a polynomial in the input length, where $m \le d$ is the nilpotency index of the radical of* span Γ_f. *In particular it runs in polynomial time when m is bounded, and in the quasi-unipotent sector. By Theorem 40, B_f is always polynomially bounded, so this algorithm runs in deterministic polynomial time on all invertible series.*

(3) (Randomized.) *CCRA-computability of invertible series is decidable in Monte Carlo polynomial time (bounded two-sided error).*

The algorithms in (2) and (3) use the deterministic polynomial-time algorithm of Friedl and Rónyai [FR85] *(see also* [Bea99, Theorem 2.6]*) for the simple components of a semisimple algebra over $\mathbb{Q}$, and rational root finding for polynomials over $\mathbb{Q}$.*

Theorem 40 (Section 9.5) shows that B_f is always bounded by a polynomial in the input length, extending Lemma 37 from the quasi-unipotent sector. With Theorem 28(2) this gives a deterministic polynomial-time algorithm for the whole invertible sector, so part (3) is superseded; we keep it because its proof is short.

The proof has three ingredients: a block-triangular form whose diagonal part is completely reducible (Section 9.1); a test of (a) by finiteness testing (Section 9.2); and a layer-by-layer computation of the Lie algebra of the unipotent radical in which the possibly exponential finite quotient $\mathcal{G}/\mathcal{G}^\circ$ is handled implicitly, through commutants and a cohomological minimality argument, and never enumerated (Section 9.3). Section 9.5 then bounds the heights of the computed Lie algebras by a polynomial. In outline, the deterministic polynomial-time claim for the invertible sector is proved as follows: Lemma 32 tests (a); Proposition 35 computes the Lie algebra of U one distance layer at a time; the size analysis (Lemma 38 and the proof of Theorem 28(2)) bounds the running time polynomially in the input length and in B_f; and Theorem 40, via Lemmas 41–45 and Proposition 46, bounds B_f by a polynomial.

9.1 Block form and the reductive quotient

Let $A = \operatorname{span}\Gamma$, computed by closing $\{I\}$ under right multiplication by the M_a (Section 8); it is the span of the words of length less than d^2, and $M_a^{-1} \in A$. In characteristic zero its radical is $J = \{x \in A : \operatorname{tr}(xy) = 0 \text{ for all } y \in A\}$: this is an ideal consisting of elements x with $\operatorname{tr}(x^i) = 0$ for all $i \ge 1$, hence a nil ideal, and every element of the radical satisfies the trace condition. Put $F_i = J^i\mathbb{Q}^d$ and let m be the least index with $F_m = 0$. The F_i are A-submodules, and F_i/F_{i+1} is annihilated by J, hence a module over the semisimple algebra A/J and therefore semisimple.

Choose subspaces E_p with $F_{m-1-p} = E_p \oplus F_{m-p}$ $(0 \le p < m)$ and a basis of $\mathbb{Q}^d$ adapted to $\mathbb{Q}^d = E_0 \oplus \cdots \oplus E_{m-1}$ (computable in polynomial time, with polynomial bit size). In this basis Γ lies in the group P of invertible block upper triangular matrices. Let L_P be the block diagonal matrices, U_P the block unipotent ones, and $b : P \to L_P$ the homomorphism that keeps the diagonal blocks. For $t \ge 0$ let $\mathfrak{v}_t$ be the space of matrices supported on the blocks (q, p) with $p - q = t$, the *distance-t part*; $\mathfrak{u}_P = \bigoplus_{t\ge1} \mathfrak{v}_t$ is the Lie algebra of U_P, and $X \mapsto X_t$ denotes the projection onto $\mathfrak{v}_t$. Matrices supported at distance $> j$ form an ideal $\mathfrak{I}_{j+1}$ of the algebra of block upper triangular matrices; let π_j be reduction modulo $\mathfrak{I}_{j+1}$ (delete all blocks at distance $> j$), so that $P_j = \pi_j(P)$ is again a linear algebraic group and $\pi_j : P \to P_j$ a morphism defined over $\mathbb{Q}$. We

write $\mathfrak{u}_{P,j} = \pi_j(\mathfrak{u}_P) = \bigoplus_{1\le t\le j} \mathfrak{v}_t$. Note that $\mathfrak{v}_{j+1}$ is central in the Lie algebra $\mathfrak{u}_{P,j+1}$, and that $\mathrm{Ad}(u)$ acts trivially on $\mathfrak{v}_{j+1}$ for $u \in \pi_{j+1}(U_P)$.

Put $\mathcal{G}_j = \pi_j(\mathcal{G})$, the closure of $\Gamma_j = \pi_j(\Gamma)$ (A2), let H be the closure of $b(\Gamma)$, and write $b_a = b(M_a)$. Since b factors through every π_j, $b(\mathcal{G}_j) = H$.

Lemma 29. *$U := \mathcal{G} \cap U_P$ is the unipotent radical $R_u(\mathcal{G})$, and for every j, $U_j := \pi_j(U) = \mathcal{G}_j \cap \pi_j(U_P) = R_u(\mathcal{G}_j)$. In particular $U = U_{m-1}$.*

Proof. A subspace of E_p is H-stable iff it is $b(\Gamma)$-stable, and $E_p \cong F_{m-1-p}/F_{m-p}$ as $b(\Gamma)$-modules, so H acts completely reducibly on each E_p. Let $N = b(R_u(\mathcal{G}_j))$, a connected unipotent normal subgroup of H (A2). For an irreducible H-submodule W of some E_p, the fixed space of N on $W \otimes \overline{\mathbb{Q}}$ is nonzero (Kolchin) and, as N is defined over $\mathbb{Q}$, Galois-stable, hence spanned by its rational points; so $W^N \neq 0$, and it is H-stable, so $W^N = W$; hence N acts trivially on $\bigoplus_p E_p$ and $N = 1$, i.e. $R_u(\mathcal{G}_j) \subseteq \ker(b|_{\mathcal{G}_j}) = \mathcal{G}_j \cap \pi_j(U_P)$. Conversely $\mathcal{G}_j \cap \pi_j(U_P)$ is a closed normal unipotent subgroup, connected in characteristic zero, hence contained in $R_u(\mathcal{G}_j)$. Finally, if $x = \pi_j(g) \in \mathcal{G}_j \cap \pi_j(U_P)$ with $g \in \mathcal{G}$, then $b(g) = b(x) = 1$, so $g \in U$; thus $\pi_j(U) = \mathcal{G}_j \cap \pi_j(U_P)$. For $j = m-1$, π_j is the identity. □

Lemma 30. *Condition (a) holds iff H° is a torus that is split over $\mathbb{Q}$. If (a) holds, then (b) holds iff U is commutative; thus condition (3) is equivalent to: H° is a $\mathbb{Q}$-split torus and U is commutative.*

Proof. If H° is a $\mathbb{Q}$-split torus, it is diagonalizable over $\mathbb{Q}$ on each E_p (A5); refining the block basis by rational weight bases shows that $\mathcal{G}^\circ \subseteq P$, whose diagonal blocks lie in $b(\mathcal{G}^\circ) = H^\circ$ (A2), is upper triangular. Conversely, if $\mathcal{G}^\circ$ fixes a complete rational flag, then $H^\circ = b(\mathcal{G}^\circ)$ fixes the induced flags of the $E_p \cong F_{m-1-p}/F_{m-p}$, so H° is connected, solvable and, by the proof of Lemma 29, has trivial unipotent radical; it is therefore a torus (A5), and it is split over $\mathbb{Q}$ because it is triangularizable over $\mathbb{Q}$ (A5). If (a) holds, every unipotent element of $\mathcal{G}$ lies in $\mathcal{G}^\circ$ (A3), and the unipotent elements of the connected solvable group $\mathcal{G}^\circ$ form the connected normal subgroup $(\mathcal{G}^\circ)_u$ (A5), which equals $R_u(\mathcal{G}^\circ) = R_u(\mathcal{G}) = U$ by Lemma 29. □

9.2 Testing condition (a)

Lemma 31 (Finiteness). *Let $\Delta = \langle g_1, \dots, g_k\rangle \le \mathrm{GL}_n(\mathbb{Q})$ be given by its generators. Then:*

1. *finiteness of Δ is decidable in deterministic polynomial time;*
2. *if Δ is finite, then $|\Delta| \le (2n+1)^{n^2}$, and every element of Δ has entries of bit length polynomial in n and the input length; more precisely, if $w_1, \dots, w_r$ is a basis of $\operatorname{span}\Delta$ consisting of positive words of length $< r$ in the generators, and $w_1^\#, \dots, w_r^\#$ is the dual basis for the trace form, then $h = \sum_i \operatorname{tr}(hw_i)\, w_i^\#$ with $\operatorname{tr}(hw_i) \in \{-n, \dots, n\}$ for every $h \in \Delta$.*

Deterministic polynomial-time finiteness testing over $\mathbb{Q}$ is due to Babai, Beals and Rockmore [BBR93]; we include a short proof of the version we need.

Proof. Compute words $w_1 = I, \dots, w_r$ forming a basis of $\mathcal{A} = \operatorname{span}\Delta$ by span closure and the Gram matrix $T = (\operatorname{tr}(w_i w_j))_{ij}$. If Δ is finite, then: every element has finite order, so its trace is a sum of n roots of unity lying in $\mathbb{Q}$, hence an integer of absolute value $\le n$; $\mathcal{A}$ is semisimple (Maschke), so the trace form is nondegenerate (its radical is a nil ideal, as in Section 9.1) and T is invertible. Therefore (2) follows by expanding h in the dual basis, since $hw_i \in \Delta$; the entries of h are integer combinations, with coefficients of absolute value at most n, of the entries of the $w_i^\# = \sum_j (T^{-1})_{ij} w_j$, which have polynomial bit length, and $h \mapsto (\operatorname{tr}(hw_i))_i$ is injective with values in $\{-n, \dots, n\}^r$, $r \le n^2$.

The test rejects (declares Δ infinite) if T is singular or has an entry outside $\{-n, \dots, n\}$. Otherwise let $\Lambda^\# = \{y \in \mathcal{A} : \operatorname{tr}(yw_i) \in \mathbb{Z}\ \forall i\}$, a lattice containing $\Lambda_0 = \bigoplus_i \mathbb{Z} w_i$ with index $|\det T|$. Compute

$\Lambda_{s+1} = \Lambda_s + \sum_a \Lambda_s g_a$ (Hermite normal forms of integer coordinate matrices with respect to the dual basis [KB79]); reject as soon as some $\Lambda_s \not\subseteq \Lambda^\#$. If Δ is finite, every Λ_s is spanned by positive words, i.e. elements of Δ, so $\Lambda_s \subseteq \Lambda^\#$. Each strict inclusion at least doubles the index over Λ_0, so after at most $\log_2 |\det T| + 1$ steps either we have rejected or $\Lambda := \Lambda_s$ satisfies $\Lambda g_a \subseteq \Lambda$ for all a. Then Λ contains all positive words, and it is spanned by positive words, so it is the ring $\mathcal{O}$ spanned by them, a finitely generated $\mathbb{Z}$-module; $\mathcal{O}\mathbb{Z}^n$ is a lattice in $\mathbb{Q}^n$ with $g_a \mathcal{O}\mathbb{Z}^n \subseteq \mathcal{O}\mathbb{Z}^n$.

Finally let $\rho(g)X = g^\mathsf{T} X g$ on symmetric matrices $\mathrm{Sym}_n(\mathbb{Q})$, let $S^\Delta = \bigcap_a \ker(\rho(g_a) - 1)$ and $R = \sum_a \mathrm{im}(\rho(g_a) - 1)$. Reject unless $\mathrm{Sym}_n = S^\Delta \oplus R$; then write $I = I_0 + I_1$ with $I_0 \in S^\Delta$, $I_1 \in R$, and reject unless I_0 is positive definite (leading principal minors). Note that $\rho(gh) = \rho(h)\rho(g)$, so $\rho(gh) - 1 = (\rho(h) - 1)\rho(g) + (\rho(g) - 1)$ and $\rho(g^{-1}) - 1 = -(\rho(g) - 1)\rho(g^{-1})$; hence $R = \sum_{h\in\Delta} \mathrm{im}(\rho(h) - 1)$. If Δ is finite, the averaging operator $\alpha = \frac{1}{|\Delta|}\sum_{h\in\Delta} \rho(h)$ satisfies $\alpha(\rho(g) - 1) = 0$, fixes S^Δ pointwise and has image in S^Δ, and $X - \alpha X = \frac{1}{|\Delta|}\sum_h (1 - \rho(h))X \in R$; so $\mathrm{Sym}_n = S^\Delta \oplus R$, α is the projection onto S^Δ along R, and $I_0 = \alpha(I) = \frac{1}{|\Delta|}\sum_h h^\mathsf{T} h$ is positive definite. Hence finite groups are accepted. Conversely, if all tests pass, Δ preserves the positive definite form I_0, so $|\det g_a| = 1$ and therefore $g_a \mathcal{O}\mathbb{Z}^n = \mathcal{O}\mathbb{Z}^n$; thus Δ lies in the intersection of the discrete group $\mathrm{GL}(\mathcal{O}\mathbb{Z}^n)$ with the compact group $O(I_0)(\mathbb{R})$, which is finite. All steps are polynomial: the words have length $< n^2$, the Hermite normal forms involve integers bounded by $|\det T| \le (n\sqrt{r})^r$, and the last step is linear algebra in dimension $n(n+1)/2$. □

Lemma 32. *Let $\theta(g)X = gXg^{-1}$ be the conjugation action of $b(\Gamma)$ on $M_d(\mathbb{Q})$ and $A_\theta = \mathrm{span}\,\theta(b(\Gamma))$. Condition (a) holds iff A_θ is semisimple and the following algebra C is commutative and the minimal polynomial of every element of some (equivalently, every) basis of C splits into linear factors over $\mathbb{Q}$: C is the centralizer in $M_d(\mathbb{Q})$ of $\mathcal{B}$, the sum of those isotypic components $eM_d(\mathbb{Q})$ (e a primitive central idempotent of A_θ) on which $\theta(b(\Gamma))$ acts as a finite group. Consequently (a) is decidable in deterministic polynomial time.*

Proof. Let $\mathbb{Q}[H^\circ]$ denote the span of $b(\Gamma) \cap H^\circ$, which is Zariski dense in H° (Lemma 6); its $\overline{\mathbb{Q}}$-span is the span of H°. If C is commutative and a basis $c_1, \ldots, c_s$ has split minimal polynomials, then every element of C has only rational eigenvalues: modulo the radical, C is a product of number fields, in each of which the images of the c_i are roots of their minimal polynomials, hence rational, and they span; so each field is $\mathbb{Q}$.

Suppose the conditions hold. Each $eM_d(\mathbb{Q})$ on which the action is finite is fixed pointwise by H°, because the closure of a finite group is finite and H° is connected; hence $\mathcal{B} \subseteq M_d(\mathbb{Q})^{H^\circ}$, which is the centralizer of $\mathbb{Q}[H^\circ]$, and so $\mathbb{Q}[H^\circ] \subseteq C$. Thus H° is commutative. Its unipotent elements form a closed connected subgroup [Bor91, 4.7], normal in H, which is trivial by the argument in the proof of Lemma 29; so all elements of H° are semisimple and H° is a torus (A4). The elements of $b(\Gamma) \cap H^\circ \subseteq C$ are therefore commuting semisimple matrices with rational eigenvalues, hence simultaneously diagonalizable over $\mathbb{Q}$, and so is their closure H°. Thus H° is a $\mathbb{Q}$-split torus, and (a) holds by Lemma 30.

Conversely suppose (a), so H° is a $\mathbb{Q}$-split torus and H/H° is finite. Every rational H-module is then completely reducible (decompose under H° into weight spaces, take an H°-stable complement and average the projection over H/H°), so $M_d(\mathbb{Q})$ is a semisimple A_θ-module, A_θ is semisimple, and $M_d(\mathbb{Q}) = \bigoplus_e eM_d(\mathbb{Q})$ is the isotypic decomposition for $b(\Gamma)$, equivalently for H. The subspace $M_d(\mathbb{Q})^{H^\circ}$ is H-stable; on an irreducible H-submodule V the fixed space V^{H° is 0 or V, so each isotypic component lies in $M_d(\mathbb{Q})^{H^\circ}$ or meets it trivially, and $M_d(\mathbb{Q})^{H^\circ}$ is the sum of the components it contains. On these H acts through the finite group H/H°, while a component on which $\theta(b(\Gamma))$ is finite is fixed by H° as above. Hence $\mathcal{B} = M_d(\mathbb{Q})^{H^\circ}$, the centralizer of the commutative split semisimple algebra $\mathbb{Q}[H^\circ] = \bigoplus_\chi \mathbb{Q}\varepsilon_\chi$ (weight idempotents). This centralizer is $\bigoplus_\chi \mathrm{End}(\varepsilon_\chi \mathbb{Q}^d)$, whose centralizer is $\bigoplus_\chi \mathbb{Q}\varepsilon_\chi$. So $C = \mathbb{Q}[H^\circ]$, which is commutative and diagonal over $\mathbb{Q}$.

Algorithm: compute $A_\theta \subseteq \mathrm{End}(M_d(\mathbb{Q}))$ by span closure, test semisimplicity (radical by the trace criterion), compute its primitive central idempotents by the algorithm of Friedl and Rónyai [FR85], test finiteness on each component by Lemma 31, compute C by linear algebra, test commutativity, and test that the minimal polynomial of each basis element of C splits into linear factors over $\mathbb{Q}$ (find its rational roots, e.g. by factorization over $\mathbb{Q}$, and compare degrees). All matrices involved have size at most $d^2 \times d^2$ and polynomial bit length. The semisimplicity test in fact always succeeds. Indeed, H acts completely reducibly on $\bigoplus_p E_p$ over $\mathbb{Q}$, hence over $\overline{\mathbb{Q}}$ (characteristic zero); the argument in the proof of Lemma 29, applied to the unipotent radical $R_u(H)$ itself, shows that $R_u(H)$ acts trivially on every irreducible H-submodule, so $R_u(H) = 1$ and H is reductive; and for a reductive group in characteristic zero (averaging over the finite group H/H° when H is disconnected) the module $M_d(\overline{\mathbb{Q}}) \cong \overline{\mathbb{Q}}^d \otimes (\overline{\mathbb{Q}}^d)^*$ is completely reducible, so A_θ is semisimple. Thus (a) holds iff C is commutative and split; we keep the semisimplicity condition in the statement because the primitive central idempotents of A_θ are needed to define $\mathcal{B}$. □

In the quasi-unipotent sector no idempotents are needed:

Lemma 33. *An invertible series is quasi-unipotent iff $b(\Gamma)$ is finite. In this case (a) holds and $U = \mathcal{G}^\circ$, so f is CCRA-computable iff U is commutative.*

Proof. Word matrices and their block diagonal parts have the same eigenvalues. If $b(\Gamma)$ is finite, all eigenvalues of word matrices are roots of unity. Conversely, if f is quasi-unipotent then by the proof of Lemma 18 all elements of $\mathcal{G}$, hence of $H = b(\mathcal{G})$, satisfy $(g^N - I)^d = 0$ for some N; so H° is unipotent (A4), and it is trivial because H acts completely reducibly (proof of Lemma 29); thus H and $b(\Gamma)$ are finite. In that case $H^\circ = 1$ is a split torus, (a) holds, $\mathcal{G}^\circ$ is unipotent (Lemma 18) and equals $R_u(\mathcal{G}) = U$; conclude by Lemma 30. □

9.3 The unipotent radical, layer by layer

From now on assume (a), so that H° is a $\mathbb{Q}$-split torus and every rational H-module, over $\mathbb{Q}$ or over $\overline{\mathbb{Q}}$, is completely reducible (proof of Lemma 32; over $\mathbb{Q}$ the coset representatives of H/H° can be taken in $b(\Gamma)$, which meets every coset).

Lemma 34 (Abelian extensions). *Let $E = L \ltimes \mathcal{M}$ be a linear algebraic group over $\mathbb{Q}$, where $\mathcal{M}$ is a vector group, identified with its Lie algebra and acted on by L through a rational representation* Ad*, and every rational L-module, over $\mathbb{Q}$ or over $\overline{\mathbb{Q}}$, is completely reducible. Let $g_a = \ell_a \exp(c_a) \in E(\mathbb{Q})$ $(1 \le a \le k)$ with $\ell_a \in L(\mathbb{Q})$ generating a Zariski dense subgroup of L and $c_a \in \mathcal{M}(\mathbb{Q})$, and let $\mathcal{D}$ be the closure of $\langle g_1, \dots, g_k \rangle$. Let $\mathcal{E}$ be the commutant of $\{\mathrm{Ad}(\ell_a)|_{\mathcal{M}(\mathbb{Q})}\}_a$ and*

$$\Phi = \left\{ \varphi \in \mathcal{E} : \ \exists y \in \mathcal{M}(\mathbb{Q})\ \forall a \quad \varphi(c_a) = (1 - \mathrm{Ad}(\ell_a)^{-1})\, y \right\}.$$

Then $\mathcal{D} \cap \mathcal{M} = \exp W$ for the subspace $W = \bigcap_{\varphi \in \Phi} \ker \varphi$ of $\mathcal{M}(\mathbb{Q})$, and there is $y \in \mathcal{M}(\mathbb{Q})$ with $c_a - (1 - \mathrm{Ad}(\ell_a)^{-1})y \in W$ for all a.

Proof. For $g \in \Gamma' = \langle g_a \rangle$ write $g = \ell(g) \exp(c(g))$; then ℓ is a homomorphism and $c(gh) = \mathrm{Ad}(\ell(h))^{-1} c(g) + c(h)$, so c is a 1-cocycle for the right action $X \cdot g = \mathrm{Ad}(\ell(g))^{-1} X$, determined by its values c_a on generators (and $c(g^{-1}) = -\mathrm{Ad}(\ell(g))c(g)$). Call a subspace $W' \subseteq \mathcal{M}(\mathbb{Q})$ *working* if it is $\mathrm{Ad}(L)$-stable and some y satisfies $c_a \equiv (1 - \mathrm{Ad}(\ell_a)^{-1})y \pmod{W'}$ for all a; since both sides are cocycles, the congruence then holds for all $g \in \Gamma'$.

$\mathcal{D} \cap \mathcal{M}$ is a closed subgroup of a vector group, hence a subspace $\exp W_0$ defined over $\mathbb{Q}$. It is normalized by $\mathcal{D}$, and since $\mathcal{M}$ is commutative, conjugation by $\ell \exp(X)$ acts on $\mathcal{M}$ as $\mathrm{Ad}(\ell)$; the projection of $\mathcal{D}$ to L is closed and contains $\langle \ell_a \rangle$, hence is L, so W_0 is $\mathrm{Ad}(L)$-stable.

W_0 is working. The image of $\mathcal{D}$ in $E/\exp W_0 = L \ltimes (\mathcal{M}/W_0)$ meets $\mathcal{M}/W_0$ trivially and maps onto L bijectively, hence isomorphically (characteristic zero); so it is the graph $\{\ell \exp s(\ell)\}$ of a morphism $s : L \to \mathcal{M}/W_0$ satisfying $s(\ell\ell') = \mathrm{Ad}(\ell')^{-1}s(\ell) + s(\ell')$. On $\mathcal{M}/W_0 \oplus \overline{\mathbb{Q}}$ let ℓ act by $(x, \tau) \mapsto (\mathrm{Ad}(\ell)x - \tau\,\mathrm{Ad}(\ell)s(\ell), \tau)$, a rational representation (the cocycle identity gives the homomorphism property). The submodule $\mathcal{M}/W_0 \oplus 0$ has an L-stable complement, spanned by some $(y_0, 1)$; then $\mathrm{Ad}(\ell)y_0 - \mathrm{Ad}(\ell)s(\ell) = y_0$, i.e. $s(\ell) = (1 - \mathrm{Ad}(\ell)^{-1})y_0$. Hence $c_a \equiv s(\ell_a) = (1 - \mathrm{Ad}(\ell_a)^{-1})y_0$ modulo W_0. This is a linear system with rational coefficients that has a solution over $\overline{\mathbb{Q}}$, so it has one over $\mathbb{Q}$.

Every working W' contains W_0. If y witnesses that W' is working, then, because $\mathcal{M}$ is commutative, $\exp(y)\, g \exp(-y) = \ell(g) \exp\big(c(g) - (1 - \mathrm{Ad}(\ell(g))^{-1})y\big) \in L \ltimes \exp W'$ for all $g \in \Gamma'$. The latter is a closed subgroup, so it contains $\exp(y)\mathcal{D}\exp(-y)$, and intersecting with $\mathcal{M}$ (on which conjugation by $\exp(y)$ is trivial) gives $W_0 \subseteq W'$.

$W = W_0$. The commutant $\mathcal{E}$ is the algebra of L-equivariant endomorphisms of the completely reducible module $\mathcal{M}(\mathbb{Q})$, a semisimple algebra, and Φ is a left ideal of $\mathcal{E}$: if $\varphi(c_a) = (1 - \mathrm{Ad}(\ell_a)^{-1})y$ then $(\psi\varphi)(c_a) = (1 - \mathrm{Ad}(\ell_a)^{-1})\psi y$ for $\psi \in \mathcal{E}$. Hence $\Phi = \mathcal{E}\pi_0$ for an idempotent $\pi_0 \in \Phi$, and $W = \ker \pi_0$. Since $\mathcal{M}(\mathbb{Q}) = \ker \pi_0 \oplus \mathrm{im}\, \pi_0$ as L-modules, $\pi_0 \in \Phi$ says exactly that $\ker \pi_0$ is working; so $W_0 \subseteq W$. Conversely let π be the L-equivariant projection onto an L-stable complement of W_0 along W_0; applying π to the congruence that makes W_0 working shows $\pi \in \Phi$, so $W \subseteq \ker \pi = W_0$. □

Call $v \in U_P(\mathbb{Q})$ a *Levi conjugator at level j* if $\pi_j(vb_av^{-1}) \in \mathcal{G}_j$ for all a, equivalently $\pi_j(vHv^{-1}) \subseteq \mathcal{G}_j$, equivalently $\pi_j\big((vb_av^{-1})^{-1}M_a\big) \in U_j$ for all a. The identity is a Levi conjugator at level 0. Let $\mathfrak{u}_j \subseteq \mathfrak{u}_{P,j}$ be the Lie algebra of U_j; when U_j is commutative it is a vector group and $U_j = \exp \mathfrak{u}_j$ (this is the only case used).

Proposition 35 (Layer step). *Assume (a). Let $0 \le j \le m-2$, assume U_j is commutative, and let v be a Levi conjugator at level j. View elements of $\mathfrak{u}_{P,j}$ as elements of $\mathfrak{u}_{P,j+1}$ with zero distance-$(j+1)$ part.*

1. *U_{j+1} is commutative iff $[X, Y]_{j+1} = 0$ for all $X, Y \in \mathfrak{u}_j$.*
2. *Suppose it is. Put $\mathcal{M} = \mathfrak{u}_j \oplus \mathfrak{v}_{j+1}$, $\ell_a = \pi_{j+1}(vb_av^{-1})$ and $c_a = \log\big(\ell_a^{-1}\pi_{j+1}(M_a)\big)$. Then $\mathcal{M}$ is a commutative Lie algebra stable under every $\mathrm{Ad}(\ell_a)$, each c_a lies in $\mathcal{M}$, and Lemma 34, applied with $L = \pi_{j+1}(vHv^{-1})$, these ℓ_a, c_a, yields $W = \mathfrak{u}_{j+1}$.*
3. *There is $y \in \mathfrak{v}_{j+1}(\mathbb{Q})$ with $c_a - (1 - \mathrm{Ad}(b_a)^{-1})y \in \mathfrak{u}_{j+1}$ for all a, and for every such y the element $\exp(-y)v$ is a Levi conjugator at level $j+1$.*

Proof. (1) $\mathcal{M}$ is the preimage of $\mathfrak{u}_j$ in $\mathfrak{u}_{P,j+1}$, and $\mathfrak{u}_{j+1} \subseteq \mathcal{M}$ maps onto $\mathfrak{u}_j$. Since $\mathfrak{v}_{j+1}$ is central and $[X, Y] \in \mathfrak{v}_{j+1}$ for $X, Y \in \mathfrak{u}_j$ (the image of $[X, Y]$ in $\mathfrak{u}_{P,j}$ vanishes), $[\mathcal{M}, \mathcal{M}]$ is spanned by the $[X, Y]_{j+1}$. If these vanish, $\mathfrak{u}_{j+1} \subseteq \mathcal{M}$ is commutative. Conversely, lifts $\tilde{X}, \tilde{Y} \in \mathfrak{u}_{j+1}$ of X, Y differ from X, Y by central elements, so $[X, Y]_{j+1} = [\tilde{X}, \tilde{Y}] = 0$ if $\mathfrak{u}_{j+1}$ is commutative.

(2) Let $\mathcal{E}_j \subseteq P_{j+1}$ be the preimage of $\mathcal{G}_j$; then $\tilde{U} := \mathcal{E}_j \cap \ker b$ is the preimage of U_j, with Lie algebra $\mathcal{M}$, a vector group by (1). As v is a Levi conjugator, $L = \pi_{j+1}(vHv^{-1}) \subseteq \mathcal{E}_j$; b maps L isomorphically onto H and kills $\tilde{U}$, so $\mathcal{E}_j = L \ltimes \tilde{U}$, and rational L-modules are completely reducible. We have $\ell_a \in L$, $b(\ell_a) = b_a$, so the ℓ_a generate a dense subgroup of L; $\ell_a^{-1}\pi_{j+1}(M_a) \in \mathcal{E}_j \cap \ker b = \tilde{U}$ gives $c_a \in \mathcal{M}$. Now $\mathcal{G}_{j+1} \subseteq \mathcal{E}_j$ is the closure of $\langle \pi_{j+1}(M_a) \rangle$ and $\mathcal{G}_{j+1} \cap \tilde{U} = \mathcal{G}_{j+1} \cap \ker b = U_{j+1}$ (Lemma 29). Lemma 34 gives $\exp W = U_{j+1}$, i.e. $W = \mathfrak{u}_{j+1}$.

(3) Lemma 34 gives $y_0 \in \mathcal{M}(\mathbb{Q})$ with $c_a - (1 - \mathrm{Ad}(\ell_a)^{-1})y_0 \in \mathfrak{u}_{j+1}$. Write $y_0 = w + y$ with $w \in \mathfrak{u}_{j+1}$, $y \in \mathfrak{v}_{j+1}$ (possible as $\mathfrak{u}_{j+1} + \mathfrak{v}_{j+1} = \mathcal{M}$). Conjugation by any element of $\mathcal{E}_j$ acts on $\mathcal{M}$ as conjugation by an element of $\mathcal{G}_{j+1}$ with the same image under b, so it preserves $\mathfrak{u}_{j+1}$; hence y also works, and on $\mathfrak{v}_{j+1}$, $\mathrm{Ad}(\ell_a) = \mathrm{Ad}(b_a)$

because $\mathrm{Ad}(\pi_{j+1}(v))$ is trivial there. Then, $\mathcal{M}$ being commutative, $\exp(y)\pi_{j+1}(M_a)\exp(-y) = \ell_a \exp\big(c_a - (1-\mathrm{Ad}(\ell_a)^{-1})y\big) \in \ell_a U_{j+1}$, i.e. $\pi_{j+1}\big((\exp(-y)vb_av^{-1}\exp(y))^{-1}M_a\big) \in \exp(-y)U_{j+1}\exp(y) = U_{j+1}$. □

Definition 36. For an invertible series f satisfying (a), let j^* be the largest $j \le m-1$ such that $U_0, \dots, U_{j^*}$ are commutative, and $B_f = \max_{j \le j^*} \mathrm{ht}(\mathfrak{u}_j)$, heights taken in the adapted basis with the matrix coordinates ordered by increasing distance. (Put $B_f = 0$ if (a) fails.) B_f depends on the choice of the adapted basis; the algorithm uses the basis computed in step 1.

The algorithm.

(1) Minimize, test invertibility (Theorem 5); compute A, J, the flag, the adapted basis, b_a.
(2) Test (a) by Lemma 32; if (a) fails, answer NO. (For part (1) of Theorem 28 one instead tests finiteness of $b(\Gamma)$ by Lemma 31; if $b(\Gamma)$ is infinite, f is not quasi-unipotent (Lemma 33) and we report this; otherwise (a) holds and we proceed.)
(3) Put $\mathfrak{u}_0 = 0$, $v = I$. For $j = 0, \dots, m-2$: if $[X,Y]_{j+1} \neq 0$ for two basis elements of $\mathfrak{u}_j$, answer NO. Otherwise compute ℓ_a, c_a, the matrices of $\mathrm{Ad}(\ell_a)$ on $\mathcal{M}$, the commutant $\mathcal{E}$, a basis $\varphi_1, \dots, \varphi_s$ of Φ (projection of the solution space of a homogeneous linear system in (φ, y)), and $\mathfrak{u}_{j+1} = \bigcap_i \ker \varphi_i$, stored by its reduced echelon basis with respect to coordinates ordered by increasing distance. Then compute the new Levi conjugator: let $u_a \in \mathfrak{u}_{j+1}$ be the combination of the echelon basis vectors whose pivots have distance $\le j$, with coefficients the entries of c_a at these pivots (so $u_a \equiv c_a$ modulo $\mathfrak{v}_{j+1}$), let $r_a = (c_a - u_a)_{j+1}$, solve
$$(1-\mathrm{Ad}(b_a)^{-1})\, y + w_a = r_a \qquad (1 \le a \le k), \quad y \in \mathfrak{v}_{j+1},\ w_a \in \mathfrak{u}_{j+1} \cap \mathfrak{v}_{j+1},$$
with free variables set to zero, and replace v by $\pi_{j+1}(\exp(-y)v)$. (The basis vectors with pivot at distance $j+1$ are supported at distance $j+1$ and span $\mathfrak{u}_{j+1} \cap \mathfrak{v}_{j+1}$; the solutions y of this system are exactly the $y \in \mathfrak{v}_{j+1}$ of Proposition 35(3).)
(4) Answer YES.

By Lemmas 29, 30, 33 and Proposition 35, the answer is YES iff condition (3) holds (note $U_0 = 1$ and $U = U_{m-1}$), i.e. iff f is CCRA-computable. The number of arithmetic operations is polynomial in d and k: $\dim \mathcal{M} \le d^2$, $\mathcal{E}$ is the kernel of a linear map on d^4 unknowns, and $m \le d$.

9.4 Bit complexity

Lemma 37 (A priori heights in the quasi-unipotent sector). *If $b(\Gamma)$ is finite, then $\mathrm{ht}(\mathfrak{u}_j)$ is bounded by a polynomial in d and σ for every j (in the adapted basis, whose bit size is polynomial).*

Proof. Let $K = \Gamma \cap U_P$, the kernel of $b|_\Gamma$, of index $|b(\Gamma)| \le (2d+1)^{d^2}$ (Lemma 31). By Schreier's lemma K is generated by elements $r_h M_a r_{hb_a}^{-1}$, where r_h is a shortest positive word with $b(r_h) = h \in b(\Gamma)$; these are words of length $< 2|b(\Gamma)|$ in the $M_a^{\pm 1}$. A word $g = M_{a_1}^{\varepsilon_1} \cdots M_{a_s}^{\varepsilon_s}$ factors as $g = b(g)\prod_{i=1}^{s}(1+x_i)$ with $1 + x_i = \beta_i^{-1}\, b(M_{a_i}^{\varepsilon_i})^{-1} M_{a_i}^{\varepsilon_i}\, \beta_i$, $\beta_i = b(M_{a_{i+1}}^{\varepsilon_{i+1}} \cdots M_{a_s}^{\varepsilon_s}) \in b(\Gamma)$. By Lemma 31(2) all elements of $b(\Gamma)$ have polynomial bit length and a common denominator of polynomial bit length, so the x_i are strictly block upper triangular with entries in $D^{-1}\mathbb{Z}$ of absolute value $\le 2^p$, for some D and p polynomial in d, σ. Since products of m of the x_i vanish, $\prod_i(1+x_i)$ is a sum of at most ms^{m-1} products of fewer than m factors x_i; hence the entries of g have bit length $O(m \log s + mp + m \log D)$, polynomial for $s < 2|b(\Gamma)|$. The same holds for the logarithms of the Schreier generators. In the quasi-unipotent sector $K = \Gamma \cap \mathcal{G}^\circ$ (Lemma 33) is dense in $U = \mathcal{G}^\circ$ (Lemma 6). The Lie algebra $\mathfrak{l}$ generated by the logarithms of the Schreier generators equals $\mathfrak{u}$: $\exp \mathfrak{l}$ is a closed subgroup (Baker–Campbell–Hausdorff; exp and log are inverse polynomial bijections between $\mathfrak{u}_P$ and U_P) containing K, so $\mathfrak{u} \subseteq \mathfrak{l}$; and $\mathfrak{u}$ contains the logarithm of every unipotent element of U (A3) and is a Lie algebra, so $\mathfrak{l} \subseteq \mathfrak{u}$. As $\mathfrak{l}$ is spanned by left-normed brackets of fewer than m such logarithms,

it, and every $\mathfrak{u}_j = \pi_j(\mathfrak{u})$, is spanned by vectors of polynomial bit length, and the reduced echelon basis of such a span has polynomial height. □

Lemma 38 (Size of the Levi conjugators). *There is a polynomial p such that the following holds in step 3. If the entries of the $M_a^{\pm 1}$ and $b_a^{\pm 1}$ (in the adapted basis) and of the echelon bases of $\mathfrak{u}_1, \dots, \mathfrak{u}_j$ have bit length at most $\beta \geq 1$, then every number computed at level $j+1$, including the echelon basis of $\mathfrak{u}_{j+1}$, has bit length at most $p(d,k)\,\beta$.*

Proof. Write $y_t \in \mathfrak{v}_t$ for the correction computed at level t. Since y_t^2 and $y_t(v-1)$ are supported at distances $> t$, the update $v \mapsto \pi_t(\exp(-y_t)v)$ is $v \mapsto v - y_t$; hence $v = I - \sum_{t \leq j} y_t$ at level j. The coefficient matrix C_t of the system for y_t involves only the $b_a^{\pm 1}$ and the echelon basis of $\mathfrak{u}_t \cap \mathfrak{v}_t$; the solution with free variables zero is $y_t = \operatorname{adj}(C_t'')\,\gamma_t r' / \det(C_t'')$, where $C_t'' = \gamma_t C_t'$ is an integer matrix, C_t' is a nonsingular square submatrix of C_t determined by C_t alone, γ_t is a common denominator of C_t', and r' collects entries of the r_a. Thus the earlier corrections enter only through the right-hand sides.

Let δ be the product of $m!$, of 2^β, of the denominators of all entries of the $b_a^{\pm 1}$, $M_a^{\pm 1}$ and of the echelon bases of $\mathfrak{u}_1, \dots, \mathfrak{u}_j$, and of the integers $\gamma_t |\det C_t''|$ for $t \leq j$; its bit length is at most $p_1(d,k)\beta$ for a polynomial p_1. For $c, \kappa \geq 1$ call a matrix X supported at distances ≥ 1 (c,κ)-*tame* if, for every $t \geq 1$, $\delta^{ct} X_t$ is an integer matrix and every entry of X_t has absolute value at most κ^t. A (c,κ)-tame matrix is (c',κ')-tame for $c' \geq c$, $\kappa' \geq \kappa$. Moreover:

- if X is (c,κ)-tame and Y is (c,κ')-tame, then XY and $X+Y$ are $(c, dm\max(\kappa,\kappa'))$-tame, since an entry of $(XY)_t$ is a sum of at most dm products of entries of X_{t_1} and Y_{t_2} with $t_1 + t_2 = t$;
- $b_a^{\pm 1} X b_a^{\mp 1}$ is $(c+2, d^2\delta^2\kappa)$-tame, since $\delta b_a^{\pm 1}$ is an integer matrix, the entries of $b_a^{\pm 1}$ have absolute value at most $2^\beta \leq \delta$, and $\delta^2 \leq \delta^{2t}$ for $t \geq 1$;
- $X b_a^{\pm 1}$ and $b_a^{\pm 1} X$ are $(c+1, d\delta\kappa)$-tame, for the same reasons;
- $\exp(X) - 1$, $\log(1+X)$ and $(1+X)^{-1} - 1$ are $(c+1, (dm)^{m+1}\kappa)$-tame, because they are combinations of $X, \dots, X^{m-1}$ with coefficients of absolute value at most 1 of the form $\pm 1/i$ or $1/i!$ with $i < m$, and $i! \mid \delta$;
- if X is (c,κ)-tame and $u = \sum_i X_{p_i} e_i$, where the e_i are echelon basis vectors of some $\mathfrak{u}_t$ with pivot p_i at distance s_i and X_{p_i} is the entry of X at p_i, then u is $(c+1, d^2\delta\kappa)$-tame: the entries of e_i vanish at distances $< s_i$ (echelon form with coordinates ordered by distance), have denominators dividing δ and absolute value at most $2^\beta \leq \delta$, and $\delta^{cs_i+1} \mid \delta^{(c+1)s}$ for $s \geq s_i \geq 1$.

Also $v_a - 1$, where $v_a = b_a^{-1} M_a$, is $(2, d\delta^2)$-tame. Suppose $y_1, \dots, y_{t-1}$ are (c,κ)-tame. Then $v-1$ is $(c, dm\kappa)$-tame, and $\ell_a^{-1}\pi_t(M_a) = \pi_t\big(v \cdot b_a^{-1} v^{-1} b_a \cdot v_a\big)$, c_a, u_a and r_a are obtained from $v-1$ and $v_a - 1$ by a bounded number of the operations above; hence r_a is $(c + c_1, F\kappa)$-tame for an absolute constant c_1 and an F with $\log F \leq p_2(d,k)\beta$, independent of t and of c. Since $|\det C_t''| \geq 1$ and $\gamma_t \det C_t'' \mid \delta$, y_t is then $(c + c_1 + 1, F'\kappa)$-tame with $F' = F \cdot \max_{t' \leq m}\big(\dim(\mathfrak{v}_{t'})\,(1 + k\dim\mathfrak{v}_{t'}) \cdot \gamma_{t'} \max |\operatorname{adj} C_{t'}''|\big)$ (maximum over the levels at which these systems are defined), again with $\log F' \leq p_3(d,k)\beta$ independent of t and c. By induction on t, using monotonicity, every y_t $(t \leq j)$ is $(c_0 + (c_1+1)t, (F')^t\kappa_0)$-tame for absolute c_0 and $\kappa_0 = d\delta^2$, and the same holds, with t replaced by m, for $v - 1$, $\ell_a^{\pm 1} - b_a^{\pm 1}$ and c_a at level $j+1$. An entry at distance $s \leq m$ of a (c,κ)-tame matrix has denominator at most δ^{cs} and numerator at most $\kappa^s \delta^{cs}$, hence bit length at most $2cs\log\delta + s\log\kappa + 2$; with $c = O(m)$, $\log\kappa = O(m\,p_3(d,k)\beta)$ and $m \leq d$ this is at most $p_4(d,k)\beta$. The matrices of the $\operatorname{Ad}(\ell_a)$ on $\mathcal{M}$ (whose basis consists of the echelon basis of $\mathfrak{u}_j$ and elementary matrices), the commutant $\mathcal{E}$, the system defining Φ and the kernel $\bigcap_i \ker\varphi_i$ are then linear-algebra problems of dimension at most $\operatorname{poly}(d,k)$ with coefficients of bit length $O(p_4(d,k)\beta)$, so their reduced echelon solutions, in particular the echelon basis of $\mathfrak{u}_{j+1}$, have bit length at most $p_5(d,k)\beta$ (Cramer's rule and Hadamard's bound).

It remains to bound the correction y_{j+1}, whose system involves $\mathfrak{u}_{j+1}$. Repeat the argument with β replaced by $\beta' = p_5(d,k)\beta$, which bounds the bit lengths of the echelon bases of $\mathfrak{u}_1, \dots, \mathfrak{u}_{j+1}$; this adds the denominators of the echelon basis of $\mathfrak{u}_{j+1}$ and $\gamma_{j+1}|\det C''_{j+1}|$ to δ and shows that y_{j+1}, and the new v, have bit length at most $p_4(d,k)\beta'$. Taking $p = p_4 p_5$ proves the lemma. □

Proof of Theorem 28. (1) Minimization and invertibility are polynomial (Theorem 5); quasi-unipotence is decided by Lemma 33 and Lemma 31(1), which uses linear algebra and Hermite normal forms only. In this sector condition (a) holds and CCRA-computability is equivalent to commutativity of U (Lemma 33) and to virtual abelianness of Γ_f (Theorem 3(2)). Step 3 decides commutativity of U. By Lemma 37 every $\mathfrak{u}_j$ has polynomial height β; since step 3 stores $\mathfrak{u}_{j+1}$ in reduced echelon form, Lemma 38 applies at every level with the same polynomial β, so every level runs in polynomial time, and there are fewer than d levels.

(2) Step 2 is polynomial by Lemma 32 and Lemma 31. In the adapted basis the entries of the $M_a^{\pm 1}$ and $b_a^{\pm 1}$ have bit length at most σ', a polynomial in d and σ; below, σ stands for σ' (the difference is absorbed in the polynomials). Step 3 performs the bracket test only on $\mathfrak{u}_j$ with $j \le j^*$ and computes $\mathfrak{u}_{j+1}$ only if $j+1 \le j^*$, so all heights involved are at most B_f, and so by Lemma 38 its running time is polynomial in the input length and B_f. For the second bound, Lemma 38 and the bit-size bounds of Gaussian elimination show that the reduced echelon basis of $\mathfrak{u}_{j+1}$ computed at level $j+1$ has height at most $p(d,k)\,(\sigma + \max_{t \le j} \operatorname{ht} \mathfrak{u}_t)$ for a fixed polynomial p. Iterating over at most $m-1$ levels gives $B_f \le (p(d,k)+1)^m(\sigma+1)$, and hence the time bound $2^{O(m\log(dk))}\operatorname{poly}(\sigma)$.

(3) Run steps 1–2 exactly. By the proof of (2), every rational number occurring in step 3 is a quotient of integers of bit length at most $\Sigma = 2^{O(m\log(dk))}\operatorname{poly}(\sigma)$, and the step consists of polynomially many arithmetic operations and reduced echelon computations with a fixed pivoting rule, together with zero tests. All polynomials in Lemma 38 and in the proof of (2) are explicit (they come from Cramer's rule and Hadamard's bound), so Σ, and the range X below, can be computed from d, k, m and σ. Let $\mathcal{Z}$ be the set of nonzero integers formed by the denominators of all rational numbers occurring, the pivot minors of all echelon computations, and the numerators of all brackets tested in step 3; their product has bit length at most $\operatorname{poly}(d,k)\,\Sigma$, so at most that many primes divide it. Perform step 3 modulo a prime $\mathfrak{q} > m$ chosen uniformly among the primes in $[X, 2X]$ with X a suitable polynomial in Σ, so that $\mathfrak{q}$ has $O(m\log(dk) + \log\sigma)$ bits. Here $\mathcal{Z}$ includes, for every echelon computation (also the projection computing Φ and the particular solutions), a nonzero minor certifying its rank profile (nonzero rows on the pivot columns); a prime not dividing it nor any denominator yields the same rank profile, and the reduced echelon form modulo $\mathfrak{q}$ is the reduction of the rational one. If $\mathfrak{q}$ divides no element of $\mathcal{Z}$, every echelon form, solution and zero test computed modulo $\mathfrak{q}$ is the reduction of the rational one, and the answer is correct; otherwise (for instance if an inconsistent system is met) the output is arbitrary, which is accounted for in the failure probability. The random prime is obtained by sampling integers in $[X, 2X]$ and testing primality deterministically. By the prime number theorem the failure probability is at most $1/3$ for a suitable choice of X, and it can be reduced by independent repetition and majority vote. □

Remark 39. (i) The finite quotient $\mathcal{G}/\mathcal{G}^\circ$ may have order exponential in d (for instance the hyperoctahedral group of order $2^d d!$), and generating sets of $\Gamma \cap \mathcal{G}^\circ$ obtained from a transversal can be exponentially long. The algorithm never enumerates it: the finite part enters only through the commutant $\mathcal{E}$ and the minimality argument of Lemma 34, which replaces averaging over the finite group. The finiteness test of Lemma 31(1) is applied only to the given generators; the bounds of Lemma 31(2) and the Schreier generators are used only in the analysis (Lemmas 37 and 45), not in the algorithm.

(ii) Unipotent parts of short words do not suffice to generate the Lie algebra of U. Let $\Gamma \le \mathrm{GL}_{n+2}(\mathbb{Q})$ be the affine Weyl group of type $\tilde{A}_n$, acting on $\mathbb{Q}^{n+1}$ by the affine maps s_i (transposition of coordinates $i, i+1$, $1 \le i \le n$) and $s_0 : x \mapsto (1\ n{+}1)x + e_1 - e_{n+1}$. Any word of length at most n in $s_0, \dots, s_n$ involves at most n

of the $n+1$ Coxeter generators, hence lies in a proper parabolic subgroup, which is finite; so it has finite order and trivial unipotent part. Yet Γ is quasi-unipotent and virtually abelian with U the n-dimensional group of translations by vectors with coordinate sum zero, and $|\mathcal{G}/\mathcal{G}^\circ| = (n+1)!$. (Coxeter elements, of length $n+1$, do have nontrivial unipotent part, so the example only shows that the necessary word length grows with d; we know no polynomial bound in general, and the algorithm avoids the question by working with Levi conjugators rather than with group elements. Replacing translations by $\mathrm{diag}(2, 1, \ldots, 1, \frac{1}{2})$ gives the same phenomenon for the torus.)

(iii) Testing (a) is at least as hard as testing finiteness: for Γ inside the orthogonal group of the anisotropic form $x_1^2 + x_2^2 + x_3^2$ over $\mathbb{Q}$, (a) holds iff Γ is finite, since the identity component of the closure is trivial, a torus without nontrivial rational characters (a split torus would have rational isotropic weight vectors), or SO_3.

(iv) Beals [Bea99] decides virtual solvability of finitely generated linear groups over number fields in polynomial time by the same centralizer construction as in Lemma 32 (his Lemma 3.1 and Section 5), and states that virtual abelianness is decidable in polynomial time; we have not relied on the latter statement, whose proof there is only sketched. Lemma 32 adds the splitting test, which is what distinguishes CCRA-computability: the series with $M = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ has a virtually abelian group but a non-split torus, so it is not CCRA-computable.

9.5 Polynomial heights: a deterministic polynomial-time algorithm

We now prove that B_f is bounded by a polynomial in the input length. Together with Theorem 28(2) this shows that the algorithm of Section 9.3, unchanged, runs in deterministic polynomial time on all invertible series. Only the analysis is new.

Theorem 40. *There is a polynomial p such that, for every invertible series f satisfying (a), the Lie algebra $\mathfrak{u}$ of $U = R_u(\mathcal{G})$ is spanned by matrices whose entries, in the adapted basis of step 1, have bit length at most $p(d, k, \sigma)$. Consequently $\mathrm{ht}(\mathfrak{u}_j) \le p'(d, k, \sigma)$ for all j, for a polynomial p'; in particular $B_f \le p'(d, k, \sigma)$, and CCRA-computability of invertible series is decidable in deterministic polynomial time.*

Conventions. Throughout, (a) holds and $S := H^\circ$, a $\mathbb{Q}$-split torus contained in L_P. A rational matrix or vector has *polynomial size* if the bit lengths of its entries are bounded by a fixed polynomial in d, k, σ (independent of f); the polynomial may change from line to line, finitely many times. We use three standard facts. (F1) A subspace of $\mathbb{Q}^N$ spanned by vectors of bit length $\le \beta$ has height at most $\mathrm{poly}(N)\,(\beta+1)$ (Cramer's rule and Hadamard's bound). (F2) If $V_0 \subseteq \mathbb{Q}^N$ is spanned by vectors of bit length $\le \beta$ and $A_1, \ldots, A_s$ are $N \times N$ matrices with entries of bit length $\le \beta$, then the smallest subspace containing V_0 and stable under all A_i is spanned by the vectors $A_{i_1} \cdots A_{i_r} v$ with $r < N$ and v from the spanning set; a product of r such matrices has a common denominator of at most $rN^2\beta$ bits, so these vectors have bit length $O(N^3(\beta + \log N))$. (F3) If $V \subseteq \mathfrak{u}_P$ is spanned by matrices of bit length $\le \beta$, the Lie algebra generated by V is spanned by the left-normed brackets of fewer than m of them (brackets of length m vanish), which have bit length $O(md^2(\beta + \log d))$; if V is stable under a family of Lie algebra automorphisms or derivations of $\mathfrak{u}_P$, so is the Lie algebra it generates.

Weight basis. The algebra $C = \mathbb{Q}[H^\circ] = \mathbb{Q}[S]$ is computed in Lemma 32 (by the proof of that lemma, under (a) the computed algebra equals $\mathbb{Q}[H^\circ]$, which lies in the span of $H \subseteq L_P$); it is commutative, split and block diagonal, and its primitive idempotents ε_χ (one for each weight χ of S on $\mathbb{Q}^d$) have polynomial size: C has a basis of polynomial size (it is a centralizer computed by linear algebra), a combination of the basis elements with coefficients in $\{0, \ldots, d^2\}$ separates the at most d weights (by the box argument used in Lemma 43), and each ε_χ is a Lagrange interpolation polynomial of degree $< d$ in that element, whose rational eigenvalues are found by rational root finding. Refining the adapted basis on each E_p by bases of

the $\varepsilon_\chi E_p$ changes the basis by a block diagonal matrix of polynomial size; we work in this refined basis and transform back at the end, which preserves polynomial size of spanning sets. Let X_S be the set of weights, $V_\chi = \varepsilon_\chi \mathbb{Q}^d$ and $n_\chi = \dim V_\chi$; each basis vector lies in some V_χ, and S is contained in the diagonal torus $D = \{\sum_\chi x_\chi \varepsilon_\chi\} \cong \mathbb{G}_m^{X_S}$. Write $\mathfrak{z} \subseteq \mathfrak{u}_P$ for the span of the elementary matrices $E_{rc} \in \mathfrak{u}_P$ with r, c in the same weight space, and $\mathfrak{u}_P^{\neq 0}$ for the span of the other elementary matrices in $\mathfrak{u}_P$. These are the sums of the zero and of the nonzero weight spaces of $\mathrm{Ad}(S)$ on $\mathfrak{u}_P$; both are graded by distance, $\mathfrak{z}$ is an associative subalgebra, $Z := C_{U_P}(S) = I + \mathfrak{z} = \exp \mathfrak{z}$, and $\mathrm{Ad}(h)$ preserves $\mathfrak{z}$ and $\mathfrak{u}_P^{\neq 0}$ for $h \in H$, because H normalizes S and hence permutes the V_χ. For $h \in H$ write π_h for the induced permutation of X_S, $hV_\chi = V_{\pi_h \chi}$. Every element of H is semisimple, since a power lies in the torus S.

An element $Y = \sum_\chi y_\chi \varepsilon_\chi \in \mathrm{Lie}(S)$ is *regular* if $y_\chi \neq y_\psi$ whenever $\chi \neq \psi$ and $\mathfrak{u}_P$ contains an elementary matrix E_{rc} with $r \in V_\chi$, $c \in V_\psi$. Then $\mathrm{ad}(Y)$ acts on E_{rc} by $y_\chi - y_\psi$, so it is invertible on $\mathfrak{u}_P^{\neq 0}$ and zero on $\mathfrak{z}$, and a matrix $v \in U_P$ commutes with Y iff $v \in Z$, that is,

$$C_{U_P}(Y) = Z \qquad \text{for regular } Y. \tag{9}$$

If Y is regular, so is $\mathrm{Ad}(h)Y$ for $h \in H$. Since $\chi - \psi$ is a nontrivial character of S for $\chi \neq \psi$, its differential is a nonzero linear form on $\mathrm{Lie}(S)$; so the regular elements form the complement of at most d^2 hyperplanes of $\mathrm{Lie}(S)$.

Lemma 41 (Tame iterations). *Let $\mathfrak{u}_P^{(t)}$ denote the matrices supported at distances $\geq t$.*

1. *Let $h \in L_P(\mathbb{Q})$ be semisimple and $g \in hU_P(\mathbb{Q})$, both of polynomial size. There is $u \in U_P(\mathbb{Q})$ of polynomial size with $u^{-1}gu \in h\, C_{U_P}(h)$. Consequently $g_s = uhu^{-1}$ and the closure of $\langle g_s \rangle$ is $u\, \overline{\langle h \rangle}\, u^{-1}$.*
2. *Let $Y \in \mathrm{Lie}(S)$ be regular and $X \in Y + \mathfrak{u}_P$, both of polynomial size. There is $u \in U_P(\mathbb{Q})$ of polynomial size with $\mathrm{Ad}(u^{-1})X \in Y + \mathfrak{z}$. Consequently $X_s = \mathrm{Ad}(u)Y$.*
3. *The map $\mu : \exp(\mathfrak{u}_P^{\neq 0}) \times Z \to U_P$, $(e^y, z) \mapsto e^y z$, is an isomorphism of varieties defined over $\mathbb{Q}$ (here $\exp(\mathfrak{u}_P^{\neq 0})$ is a closed subvariety of U_P, in general not a subgroup), and if $w \in U_P(\mathbb{Q})$ has polynomial size, so does the y with $w \in e^y Z$.*

Proof. All three are proved by an iteration over $t = 1, \ldots, m-1$ that, at step t, removes an unwanted distance-t component without changing components at smaller distances. We use repeatedly that for $A \in \mathfrak{u}_P^{(t)}$ and $B \in \mathfrak{u}_P$ every Lie bracket involving A and at least one further factor from $\mathfrak{u}_P$ lies in $\mathfrak{u}_P^{(t+1)}$, and the Baker–Campbell–Hausdorff formula, which is finite on $\mathfrak{u}_P$.

(1) Since h is semisimple and preserves each $\mathfrak{v}_t$, we have $\mathfrak{v}_t = \ker(1 - \mathrm{Ad}(h^{-1})) \oplus \mathrm{im}(1 - \mathrm{Ad}(h^{-1}))$ on $\mathfrak{v}_t$. Start with $u = I$. At step t let $R = \log(h^{-1}u^{-1}gu) \in \mathfrak{u}_P$ and suppose that its components at distances $< t$ lie in the kernels. Let Q_t be the component of R_t in the image, choose $Z_t \in \mathfrak{v}_t$ with $(1 - \mathrm{Ad}(h^{-1}))Z_t = -Q_t$, and replace u by ue^{Z_t}. The new element is $h^{-1}e^{-Z_t}u^{-1}gue^{Z_t} = e^{-\mathrm{Ad}(h^{-1})Z_t} e^R e^{Z_t}$, whose logarithm is $R + (1 - \mathrm{Ad}(h^{-1}))Z_t$ modulo $\mathfrak{u}_P^{(t+1)}$. Thus the components at distances $< t$ are unchanged and the distance-t component now lies in the kernel. After step $m-1$ we have $u^{-1}gu = he^R$ with $\mathrm{Ad}(h)R = R$, so h and e^R commute; this is the Jordan decomposition of $u^{-1}gu$, whence $g_s = uhu^{-1}$.

(2) Start with $u = I$. At step t let $R = \mathrm{Ad}(u^{-1})X - Y \in \mathfrak{u}_P$, suppose that its components at distances $< t$ lie in $\mathfrak{z}$, let Q_t be the component of R_t in $\mathfrak{u}_P^{\neq 0}$, let $Z_t \in \mathfrak{v}_t \cap \mathfrak{u}_P^{\neq 0}$ be the solution of $[Z_t, Y] = Q_t$ (entrywise division by the nonzero numbers $y_\psi - y_\chi$), and replace u by ue^{Z_t}. Then $\mathrm{Ad}(e^{-Z_t})(Y + R) = Y + R - [Z_t, Y] - [Z_t, R] + \frac{1}{2}[Z_t, [Z_t, Y + R]] - \cdots$, which is $Y + R - Q_t$ modulo $\mathfrak{u}_P^{(t+1)}$. After step $m-1$, $\mathrm{Ad}(u^{-1})X = Y + N$ with $N \in \mathfrak{z}$ nilpotent and $[Y, N] = 0$, so $X_s = \mathrm{Ad}(u)Y$.

(3) Start with $y = 0$. At step t let $R = \log(e^{-y}w)$, suppose that its components at distances $< t$ lie in $\mathfrak{z}$, and replace y by $y + Q_t$, where Q_t is the component of R_t in $\mathfrak{u}_P^{\neq 0}$. Since $\log(e^{-y-Q_t}e^y) \equiv -Q_t$ modulo $\mathfrak{u}_P^{(t+1)}$,

the new logarithm is $R - Q_t$ modulo $\mathfrak{u}_P^{(t+1)}$. At the end $\log(e^{-y}w) \in \mathfrak{z}$, i.e. $e^{-y}w \in Z$; so μ is onto, with an inverse given by polynomials. It is injective: if $e^y z = e^{y'} z'$ with $y \neq y'$, let t be the least distance with $y_t \neq y'_t$; then $\log(e^{-y'} e^y) \equiv y - y'$ modulo $\mathfrak{u}_P^{(t+1)}$ has the nonzero distance-t component $y_t - y'_t \in \mathfrak{u}_P^{\neq 0}$, but it lies in $\log Z = \mathfrak{z}$.

Sizes. We use the tame calculus of the proof of Lemma 38. Let δ be the product of $m!$, of 2^β where β bounds the bit lengths of the input data ($h^{\pm 1}$ and $h^{-1}g$ in (1), Y and $X - Y$ in (2), w in (3)), of their denominators, and of the numerators and denominators of the following fixed linear maps of $\mathfrak{v}_t$ ($1 \leq t < m$): in (1) the map A_t that projects onto $\mathrm{im}(1 - \mathrm{Ad}(h^{-1}))$ along the kernel and then inverts $1 - \mathrm{Ad}(h^{-1})$ there; in (2) the entrywise division by the numbers $y_\psi - y_\chi$; in (1)–(3) the projections onto $\mathfrak{u}_P^{\neq 0}$ and onto a distance component. These maps have entries of polynomial bit length (for A_t by Cramer's rule applied to the matrix of $\mathrm{Ad}(h^{-1})$ on $\mathfrak{v}_t$), so $\log \delta$ is polynomial. In each step the current u (resp. y) is updated incrementally ($u \mapsto ue^{Z_t}$, $y \mapsto y + Q_t$), and the quantities R, Q_t, Z_t, $e^{\pm Z_t}$ and the updated u (in (1), $h^{-1}u^{-1}gu = (h^{-1}u^{-1}h)(h^{-1}g)u$; in (2), $\mathrm{Ad}(u^{-1})X - Y = (u^{-1} - 1)Y + Y(u - 1) + (u^{-1} - 1)Y(u - 1) + u^{-1}(X - Y)u$) are obtained from the previous ones and the input by a bounded number of the operations of that calculus: sums and products, $\exp(X) - 1$, $\log(1 + X)$, $(1 + X)^{-1} - 1$, conjugation by $h^{\pm 1}$, products with Y, projections, and application of one of the fixed maps above to a single distance component. As in the proof of Lemma 38 (where the same structure occurs: each new quantity is obtained from the previous iterate and the fixed input by $O(1)$ operations, so every quantity is the result of a chain of $O(m)$ operations, and two-argument operations are applied after raising both arguments to the larger (c, κ) by monotonicity), each such operation maps (c, κ)-tame matrices to $(c + 2, F\kappa)$-tame ones with $\log F$ polynomial (for a fixed map with entries of absolute value $\leq \delta$ and denominators dividing δ, applied to a distance-t component, one may take $F = \dim(\mathfrak{v}_t)\,\delta$). Starting from $(2, d\delta^2)$-tameness of the input, after the $O(m)$ operations of the whole iteration all matrices are $(O(m), F^{O(m)} d\delta^2)$-tame, and an entry at distance $s \leq m$ of a (c, κ)-tame matrix has bit length at most $2cs \log \delta + s \log \kappa + 2$, which is polynomial. The block diagonal parts (h, Y, and u having identity diagonal blocks) are fixed throughout. □

The torus from short words. For a word w in the letters $M_a^{\pm 1}$ write M_w for its matrix, $|w|$ for its length, and S_w for the identity component of the closure of $\langle b(M_w) \rangle$; it is a subtorus of S, since $b(M_w)^e \in S$ for $e = |H/S|$.

For a subgroup Δ of $D(\mathbb{Q})$ let $\Lambda(\Delta) \subseteq \mathbb{Q}^{X_S}$ be the span of the vectors $(v_p(t_\chi))_{\chi \in X_S}$ over all $t = \sum_\chi t_\chi \varepsilon_\chi \in \Delta$ and all primes p, where v_p is the p-adic valuation. The identity component of the closure of Δ is cut out in D by the characters $x \mapsto \prod_\chi x_\chi^{a_\chi}$ ($a \in \mathbb{Z}^{X_S}$) that take root-of-unity values, i.e. the values ± 1, on Δ [Bor91, 8.2, 8.12], and $\prod_\chi t_\chi^{a_\chi} = \pm 1$ iff $\sum_\chi a_\chi v_p(t_\chi) = 0$ for all p. Hence, identifying $y \in \mathbb{Q}^{X_S}$ with $\sum_\chi y_\chi \varepsilon_\chi$,

$$\mathrm{Lie}\big(\overline{\Delta}^\circ\big) = \Lambda(\Delta). \tag{10}$$

Lemma 42 (Short words generate the torus). *$\mathrm{Lie}(S) = \sum_w \mathrm{Lie}(S_w)$, the sum over the words of length at most $2d^2 + 2d$. For each such word, $\mathrm{Lie}(S_w)$ is spanned by vectors of polynomial size.*

Proof. For $g \in b(\Gamma)$ and $\chi \in X_S$ let $g_\chi : V_{\pi_g^{-1}\chi} \to V_\chi$ be the restriction of g, a square matrix in the weight basis. Let $\mathcal{P}$ be the finite set of primes dividing a numerator or denominator of some $\det((b_a^{\pm 1})_\chi)$. For $g \in b(\Gamma)$ define $c(g) \in \mathcal{Q} := \mathbb{Q}^{X_S} \otimes \mathbb{Q}^{\mathcal{P}}$ by $c(g)_{\chi,p} = v_p(\det g_\chi)$. Since $(gh)_\chi = g_\chi h_{\pi_g^{-1}\chi}$, we have the cocycle identity $c(gh) = c(g) + \pi_g c(h)$, where $(\pi_g u)_\chi = u_{\pi_g^{-1}\chi}$; in particular every $\det g_\chi$ is a product of the generator determinants, so its valuations outside $\mathcal{P}$ vanish, and $|c(b(M_w))_{\chi,p}| \leq |w|\beta_0$, where β_0 bounds the bit lengths of the determinants $\det((b_a^{\pm 1})_\chi)$ in the weight basis; it is polynomial in d and σ, by Hadamard's

bound. For $t \in S$ we have $\det t_\chi = t_\chi^{n_\chi}$, so with $\mathsf{N} = \operatorname{diag}(n_\chi)$,

$$\big(v_p(t_\chi)\big)_\chi = \mathsf{N}^{-1} c(t)_{\cdot,p} \qquad (t \in S \cap b(\Gamma)). \tag{11}$$

Let $K_\pi = \{g \in \Gamma : \pi_{b(g)} = 1\}$, a normal subgroup of finite index on which $g \mapsto c(b(g))$ is a homomorphism, and $\mathsf{W} = \operatorname{span}\{c(b(g)) : g \in K_\pi\}$. For $g \in K_\pi$ some power g^e has $b(g^e) \in S$ and $c(b(g^e)) = e\,c(b(g))$; conversely $S \cap b(\Gamma) \subseteq b(K_\pi)$. For $u \in \mathcal{Q}$ let $\operatorname{col}(u) \subseteq \mathbb{Q}^{X_S}$ be the span of the vectors $u_{\cdot,p}$ ($p \in \mathcal{P}$), and for a subspace W_1 let $\operatorname{col}(\mathsf{W}_1) = \sum_{u \in \mathsf{W}_1} \operatorname{col}(u)$. Since $S \cap b(\Gamma)$ is dense in S (Lemma 6), (10) and (11) give $\operatorname{Lie}(S) = \operatorname{col}(\mathsf{N}^{-1}\mathsf{W})$. Similarly, for a word w let n_w be the order of $\pi_{b(M_w)}$ and $v_w := c(b(M_w)^{n_w}) = \sum_{i<n_w} \pi^i_{b(M_w)} c(b(M_w))$; as S_w is the identity component of the closure of $\langle b(M_w)^{n_w e}\rangle$ and $c(b(M_w)^{n_w e}) = e v_w$, we get $\operatorname{Lie}(S_w) = \operatorname{col}(\mathsf{N}^{-1} v_w)$. Its spanning vectors have entries of absolute value at most $n_w|w|\beta_0$ with $n_w \le d!$, hence polynomial size. Moreover, for words x, w, using $c(g^{-1}) = -\pi_g^{-1} c(g)$ and $\pi_{b(M_w)^{n_w}} = 1$,

$$v_{xwx^{-1}} = c\big(b(M_x)\, b(M_w)^{n_w}\, b(M_x)^{-1}\big) = \pi_{b(M_x)} v_w. \tag{12}$$

Let W' be the span of the $\pi_{b(M_x)} v_w$ with $|w| \le 2d-1$ and x a positive word with $|x| \le d^2$. Then $\mathsf{W}' \subseteq \mathsf{W}$, because $M_w^{n_w} \in K_\pi$. For a fixed u, let O_i be the span of the $\pi_{b(M_x)} u$ with x a positive word of length $\le i$ in the letters M_a. Then $O_i \subseteq O_{i+1}$, and $O_i = O_{i+1}$ implies that O_i is stable under all π_{b_a}, hence under the finite group they generate, which contains all π_g ($g \in b(\Gamma)$). Since O_i lies in the image of the linear map $\operatorname{span}\{\pi_g\} \to \mathcal{Q}$, $A \mapsto Au$, its dimension is at most $\dim \operatorname{span}\{\pi_g\} \le |X_S|^2 \le d^2$; so O_{d^2} is the span of all $\pi_g u$, and W' is stable under all π_g. We claim $\mathsf{W}' = \mathsf{W}$. Otherwise let $\mathcal{Q}_1$ be a π-stable complement of W' in $\mathcal{Q}$ (the π_g form a finite group) and q the equivariant projection onto $\mathcal{Q}_1$ along W'. Then $\bar c = q \circ c$ is again a cocycle with values in the permutation module $\mathcal{Q}$, and $q(\mathsf{W}) \ne 0$. Unwinding the cocycle identity, for $g_i = b(M_{a_i}^{\varepsilon_i})$,

$$\bar c(g_1 \cdots g_s)_\chi = \sum_{i=1}^{s} \bar c(g_i)_{\chi_{i-1}}, \qquad \chi_0 = \chi, \quad \chi_i = \pi_{g_i}^{-1} \chi_{i-1}.$$

Let Ω be the graph with vertex set X_S and, for each letter a and each χ, an edge from χ to $\pi_{b_a}^{-1}\chi$ labelled $\bar c(b_a)_\chi \in \mathbb{Q}^{\mathcal{P}}$. Since $\bar c(b_a^{-1})_\chi = -\bar c(b_a)_{\pi_{b_a}\chi}$, the letter a^{-1} traverses an edge backwards with the opposite label. Thus words correspond to walks, $\bar c(b(M_w))_\chi$ is the label of the walk of w from χ, and $w \in K_\pi$ iff all these walks are closed. As $q(\mathsf{W}) \ne 0$, some $g \in K_\pi$ and some χ give a closed walk with nonzero label. The label of a walk is the sum of the labels of the edges it traverses, counted with sign, so it depends only on the vector in $\mathbb{Z}^{E(\Omega)}$ of signed traversal counts. For a closed walk this vector lies in the cycle space of Ω, which has a $\mathbb{Z}$-basis given by the fundamental cycles of a spanning forest (the coefficient of a fundamental cycle is the signed count of its non-forest edge). Hence some fundamental cycle has nonzero label. Read from the root r of its tree, it is a closed walk of length at most $2(|X_S|-1)+1 \le 2d-1$, i.e. the walk from r of a word w with $|w| \le 2d-1$ and $\pi_{b(M_w)} r = r$. Then $q(v_w)_r = \sum_{i<n_w} \bar c(b(M_w))_{\pi^{-i}_{b(M_w)} r} = n_w\, \bar c(b(M_w))_r \ne 0$, so $v_w \notin \mathsf{W}'$, a contradiction.

Hence $\operatorname{Lie}(S) = \operatorname{col}(\mathsf{N}^{-1}\mathsf{W}') = \sum_{x,w} \operatorname{col}(\mathsf{N}^{-1} v_{xwx^{-1}}) = \sum_{x,w} \operatorname{Lie}(S_{xwx^{-1}})$ by (12), where $|xwx^{-1}| \le 2d^2 + 2d - 1$. □

Lemma 43 (A maximal torus of polynomial size). *There are a regular $Y \in \operatorname{Lie}(S)$ and $v_0 \in U_P(\mathbb{Q})$, both of polynomial size, such that $v_0 S v_0^{-1} \subseteq \mathcal{G}^\circ$.*

Proof. Let w be a word with $|w| \le 2d^2 + 2d$. Lemma 41(1), applied to $h = b(M_w)$ and $g = M_w$, gives $u_w \in U_P(\mathbb{Q})$ of polynomial size with $(M_w)_s = u_w b(M_w) u_w^{-1}$. Hence the torus Θ_w, the identity component of the closure of $\langle (M_w)_s\rangle$, equals $u_w S_w u_w^{-1}$; it lies in $\mathcal{G}^\circ$, because $(M_w)_s \in \mathcal{G}$ (A2) and Θ_w is connected. So $\operatorname{Ad}(u_w)\operatorname{Lie}(S_w) \subseteq \operatorname{Lie}(\mathcal{G}^\circ)$. Let $Y_{w,1}, Y_{w,2}, \ldots$ be spanning vectors of $\operatorname{Lie}(S_w)$ of polynomial size (Lemma 42). Altogether they span $\operatorname{Lie}(S)$, so we can choose among them a basis $Y_1, \ldots, Y_n$ of $\operatorname{Lie}(S)$, with $n = \dim S \le d$,

where Y_i is a spanning vector of $\mathrm{Lie}(S_{w_i})$; only these n words are used below. For each of the at most d^2 pairs (χ,ψ) in the definition of regularity, $c \mapsto (y_\chi - y_\psi)\big(\sum_i c_i Y_i\big)$ is a nonzero linear form on $\mathbb{Q}^n$, which vanishes on at most $(d^2+1)^{n-1}$ points of the box $\{0,\dots,d^2\}^n$. So some c in the box avoids all of them. Put $Y = \sum_i c_i Y_i$, which is regular, and $X = \sum_i c_i \,\mathrm{Ad}(u_{w_i}) Y_i \in \mathrm{Lie}(\mathcal{G}^\circ)$. Both have polynomial size, being sums of $n \le d$ terms of polynomial size, and $b(X) = Y$, because $\mathrm{Ad}(u)B - B \in \mathfrak{u}_P$ for $u \in U_P$ and block diagonal B. Lemma 41(2) gives $v_0 \in U_P(\mathbb{Q})$ of polynomial size with $X_s = \mathrm{Ad}(v_0)Y$.

It remains to show $v_0 S v_0^{-1} \subseteq \mathcal{G}^\circ$. Let T' be a maximal torus of $\mathcal{G}^\circ$. Then $b(T') = b(\mathcal{G}^\circ) = H^\circ = S$ (A2), since $\mathcal{G}^\circ = T'U$ and $b(U) = 1$. The tori T' and S are maximal tori of the connected solvable group $b^{-1}(S) = S \ltimes U_P$, so $T' = vSv^{-1}$ for some $v \in U_P$ [Bor91, 10.6]. Put $\mathfrak{u}' = \mathrm{Ad}(v^{-1})\mathfrak{u}$ and $X' = \mathrm{Ad}(v^{-1})X$. As $\mathrm{Lie}(\mathcal{G}^\circ) = \mathrm{Lie}(T') \oplus \mathfrak{u}$, we have $X' \in \mathrm{Lie}(S) \oplus \mathfrak{u}'$, and comparing images under b gives $X' = Y + N$ with $N \in \mathfrak{u}'$. Now $\mathfrak{u}'$ is $\mathrm{Ad}(S)$-stable, so it is the sum of its intersections with $\mathfrak{z}$ and with $\mathfrak{u}_P^{\neq 0}$, and $\mathrm{ad}(Y)$ maps $\mathfrak{u}' \cap \mathfrak{u}_P^{\neq 0}$ bijectively onto itself. If the component $N^{\neq 0}$ of N in $\mathfrak{u}_P^{\neq 0}$ lies in $\mathfrak{u}_P^{(t)}$, let $\Xi \in \mathfrak{u}' \cap \mathfrak{u}_P^{\neq 0}$ with $[\Xi, Y] = N^{\neq 0}$; then $\Xi \in \mathfrak{u}_P^{(t)}$, since $\mathrm{ad}(Y)$ preserves distances, and $\mathrm{Ad}(e^{-\Xi})(Y+N) = Y + N - N^{\neq 0} + R$ with $R = -[\Xi, N] + \frac{1}{2}[\Xi,[\Xi, Y+N]] - \cdots$, which lies in $\mathfrak{u}' \cap \mathfrak{u}_P^{(t+1)}$ (every term is a bracket of $\Xi \in \mathfrak{u}_P^{(t)}$ with an element of $\mathfrak{u}'$, using $[\Xi, Y] = N^{\neq 0} \in \mathfrak{u}'$). Replacing N by $N - N^{\neq 0} + R$, the new component in $\mathfrak{u}_P^{\neq 0}$ lies in $\mathfrak{u}_P^{(t+1)}$. After fewer than m such steps we obtain $v = e^{\Xi_1}\cdots e^{\Xi_r} \in \exp(\mathfrak{u}') = v^{-1}Uv$ with $\mathrm{Ad}(v^{-1})X' \in Y + \mathfrak{z}$, so $X'_s = \mathrm{Ad}(v)Y$ as in Lemma 41(2). Thus $v'' = vv \in U_P$ satisfies $\mathrm{Ad}(v'')Y = X_s$, and $v''Sv''^{-1} = (vvv^{-1})T'(vvv^{-1})^{-1} \subseteq \mathcal{G}^\circ$, as $vvv^{-1} \in U$. Finally $\mathrm{Ad}(v_0)Y = \mathrm{Ad}(v'')Y$, so $v''^{-1}v_0 \in C_{U_P}(Y) = Z$ by (9), and Z centralizes S; hence $v_0 S v_0^{-1} = v''Sv''^{-1} \subseteq \mathcal{G}^\circ$. □

From now on we conjugate by v_0: we replace M_a by $v_0^{-1}M_a v_0$, and $\mathcal{G}$, U, $\mathfrak{u}$ by their conjugates. This changes neither the b_a nor polynomial size, and now $S \subseteq \mathcal{G}^\circ$.

Lemma 44 (Moving the generators into the normalizer of S). *For each a there is $\hat{x}_a \in U(\mathbb{Q})$ of polynomial size such that $m_a := \hat{x}_a^{-1}M_a$ normalizes S. Moreover $m_a \in \mathcal{G}$, m_a has polynomial size, and $b(m_a) = b_a$.*

Proof. As $S \subseteq \mathcal{G}$ normalizes U, the Lie algebra $\mathfrak{u}$ is $\mathrm{Ad}(S)$-stable, so $\mathfrak{u} = \mathfrak{u}^0 \oplus \mathfrak{u}^{\neq 0}$ with $\mathfrak{u}^0 = \mathfrak{u} \cap \mathfrak{z}$ and $\mathfrak{u}^{\neq 0} = \mathfrak{u} \cap \mathfrak{u}_P^{\neq 0}$, and $C_U(S) = \exp(\mathfrak{u}^0)$, since e^X commutes with $s \in S$ iff $\mathrm{Ad}(s)X = X$. We claim $U = \mu\big(\exp(\mathfrak{u}^{\neq 0}) \times C_U(S)\big)$. The set $\mu^{-1}(U)$ is closed and irreducible of dimension $\dim U$, because μ is an isomorphism (Lemma 41(3)). It contains $\exp(\mathfrak{u}^{\neq 0}) \times \exp(\mathfrak{u}^0)$, which is closed and irreducible of dimension $\dim \mathfrak{u}^{\neq 0} + \dim \mathfrak{u}^0 = \dim U$; so the two sets are equal.

The torus S is a maximal torus of $\mathcal{G}^\circ$, since the maximal tori of $\mathcal{G}^\circ$ map isomorphically onto $b(\mathcal{G}^\circ) = S$. As $\mathcal{G}^\circ$ is normal in $\mathcal{G}$, $M_a S M_a^{-1}$ is a maximal torus of $\mathcal{G}^\circ$ too, so $M_a S M_a^{-1} = xSx^{-1}$ for some $x \in U$ [Bor91, 10.6]. Write $x = e^y z$ with $y \in \mathfrak{u}^{\neq 0}$ and $z \in C_U(S)$, and put $\hat{x}_a = e^y \in U$. Then $\hat{x}_a S \hat{x}_a^{-1} = M_a S M_a^{-1}$, so $m_a = \hat{x}_a^{-1}M_a \in \mathcal{G}$ normalizes S. Let Y be as in Lemma 43 and $Y_a = \mathrm{Ad}(b_a)Y$, which is regular. The element $\mathrm{Ad}(m_a)Y$ lies in $\mathrm{Lie}(S)$, and elements of $\mathrm{Lie}(S)$ are determined by their images under b; so $\mathrm{Ad}(m_a)Y = Y_a$, i.e. $\mathrm{Ad}(\hat{x}_a)Y_a = \mathrm{Ad}(M_a)Y$. By (9) and the injectivity of μ, $\hat{x}_a$ is the only element of $\exp(\mathfrak{u}_P^{\neq 0})$ with this property. The linear system $(I+N)Y_a = \mathrm{Ad}(M_a)Y\,(I+N)$ in $N \in \mathfrak{u}_P$ has rational coefficients of polynomial size and the solution $\hat{x}_a - I$, a priori over $\overline{\mathbb{Q}}$; so it is consistent over $\mathbb{Q}$. Let w_a be I plus its reduced echelon solution (free variables set to zero), which is rational of polynomial size by Cramer's rule and Hadamard's bound. Then $w_a \in \hat{x}_a C_{U_P}(Y_a) = \hat{x}_a Z$ by (9), so $\hat{x}_a = e^y$ with y the component of w_a given by Lemma 41(3). Hence $\hat{x}_a$ is rational of polynomial size, and so is m_a. Finally $b(m_a) = b(M_a) = b_a$, because $\hat{x}_a \in U_P$. □

Lemma 45 (The part of U centralizing S). *Let N' be the closure of the group generated by S and $m_1,\dots,m_k$, and $U'' = N' \cap U_P$. Then $U'' \subseteq U$, and $\mathrm{Lie}(U'')$ is spanned by matrices of polynomial size.*

Proof. $N' \subseteq \mathcal{G}$ normalizes S, and $U'' \subseteq \mathcal{G} \cap U_P = U$. An element $u \in U_P$ normalizing S centralizes it, because $usu^{-1} \in S$ and $b(usu^{-1}) = s$; so $U_P \cap N_{\mathrm{GL}_d}(S) = Z$.

The algebra $C(S) = \bigoplus_\chi \mathrm{End}(V_\chi)$ (the centralizer of S in $M_d(\mathbb{Q})$) is stable under conjugation by $b(\Gamma)$, and S acts trivially on it. Let Ψ be the group of these conjugation operators. It is finite, as its kernel contains $S \cap b(\Gamma)$, which has finite index in $b(\Gamma)$. By Lemma 31(2), applied to the generators $\mathrm{Ad}(b_a)|_{C(S)}$ in dimension $\dim C(S) = \sum_\chi n_\chi^2 \le d^2$, $|\Psi| \le (2d^2+1)^{d^4}$, and the elements of Ψ have entries of polynomial bit length with a common denominator of polynomial bit length. The centralizer of $C(S)$ in GL_d is D, so $\mathrm{Ad}(b(g))$ is trivial on $C(S)$ iff $b(g) \in D$.

Let $Y_N = \{g \in N' : b(g) \in D\}$ and $\varphi(g) = b(g)^{-1}g$ for $g \in Y_N$. For $g \in Y_N$ the element $n = b(g)^{-1}g \in U_P$ normalizes S, since g and $b(g)$ do; so $n \in Z \subseteq C(S)$, and n commutes with $b(g)$, which is central in $C(S)$. As $b(g)$ is semisimple and n unipotent, $g_s = b(g)$ and $g_u = n$, and both lie in N' (A2). Hence $\varphi(g) \in N' \cap U_P = U''$. Moreover φ is a homomorphism: $\varphi(gh) = b(h)^{-1}\varphi(g)b(h)\varphi(h) = \varphi(g)\varphi(h)$, as $\varphi(g) \in Z \subseteq C(S)$ commutes with $b(h) \in D$. Since $U'' \subseteq Y_N$ and φ is the identity on U'', $U'' = \varphi(Y_N)$.

Let $\Gamma'' = \langle m_1, \dots, m_k \rangle$, let N_0' be its closure, and let $K'' = \Gamma'' \cap Y_N$, the kernel of $\Gamma'' \to \Psi$, $g \mapsto \mathrm{Ad}(b(g))|_{C(S)}$; it has index at most $|\Psi|$. The product $N_0'S$ is a closed subgroup containing Γ'' and S, so $N' = N_0'S$, and $Y_N = (N_0' \cap Y_N)S$ as $S \subseteq Y_N$. Since $\varphi(S) = 1$, $U'' = \varphi(N_0' \cap Y_N)$. The closure of K'' is a closed subgroup of finite index in $N_0' \cap Y_N$, so its image under φ is a closed subgroup of finite index of the connected group U'', hence equal to U''. Thus U'' is the closure of the group generated by $\varphi(K'')$, and so by the $\varphi(\varsigma)$ for the Schreier generators $\varsigma = r_\psi m_a r_{\psi'}^{-1}$ of K'', where r_ψ is a shortest positive word in the m_a mapping to $\psi \in \Psi$. These are words of length $s < 2|\Psi|$ in the $m_a^{\pm 1}$. As in the proof of Lemma 37, a word $\varsigma = m_{a_1}^{\varepsilon_1} \cdots m_{a_s}^{\varepsilon_s}$ factors as $\varsigma = b(\varsigma) \prod_{i=1}^{s}(1 + x_i)$ with $1 + x_i = \beta_i^{-1} b(m_{a_i}^{\varepsilon_i})^{-1} m_{a_i}^{\varepsilon_i} \beta_i$ and $\beta_i = b(m_{a_{i+1}}^{\varepsilon_{i+1}} \cdots m_{a_s}^{\varepsilon_s}) \in b(\Gamma)$. Here $b(m_a)^{-1}m_a = b_a^{-1}m_a$ and $b(m_a^{-1})^{-1}m_a^{-1} = b_a m_a^{-1}$ lie in U_P and normalize S, hence lie in Z; so $n_i := b(m_{a_i}^{\varepsilon_i})^{-1} m_{a_i}^{\varepsilon_i} - I \in \mathfrak{z} \subseteq C(S)$ has polynomial size, and $x_i = \beta_i^{-1} n_i \beta_i = \psi_i(n_i)$ with $\psi_i = \mathrm{Ad}(\beta_i^{-1})|_{C(S)} \in \Psi$; as $\beta_i \in L_P$ normalizes S, $x_i \in \mathfrak{z}$. Hence the x_i are strictly block upper triangular with entries in $\varrho^{-1}\mathbb{Z}$ of absolute value at most 2^{p_0}, for some ϱ and p_0 of polynomial bit length. Products of m of them vanish, so $\varphi(\varsigma) = \prod_i (1 + x_i)$ is a sum of at most ms^{m-1} products of fewer than m factors x_i, and has bit length $O(m \log s + mp_0 + m \log \varrho)$, which is polynomial since $\log |\Psi|$ is. The same holds for $\log \varphi(\varsigma)$. Finally, the closure of the group generated by unipotent elements $u_i \in U_P$ is $\exp \mathfrak{l}$, where $\mathfrak{l}$ is the Lie algebra generated by the $\log u_i$: $\exp \mathfrak{l}$ is a closed subgroup (Baker–Campbell–Hausdorff) containing the u_i, and the closure contains $\exp(\tau \log u_i)$ for all τ (A3), so its Lie algebra contains $\mathfrak{l}$. By (F3), $\mathrm{Lie}(U'')$ is spanned by matrices of polynomial size. □

Proposition 46. *Let V be the smallest subspace of $\mathfrak{u}_P$ that contains $\log \hat{x}_1, \dots, \log \hat{x}_k$ and $\mathrm{Lie}(U'')$ and is stable under $\mathrm{Ad}(m_a)^{\pm 1}$ $(1 \le a \le k)$ and under $\mathrm{ad}(Y')$ for $Y' \in \mathrm{Lie}(S)$. Then $\mathfrak{u}$ is the Lie algebra generated by V.*

Proof. Let $\mathfrak{n}$ be the Lie algebra generated by V. By (F3) it is stable under the automorphisms $\mathrm{Ad}(m_a)^{\pm 1}$ and the derivations $\mathrm{ad}(Y')$, hence under $\mathrm{Ad}(S)$ (S is connected) and under $\mathrm{Ad}(N')$, since stabilizing a subspace is a closed condition. Since $\hat{x}_a \in U$, $U'' \subseteq U$, and $N' \subseteq \mathcal{G}$ normalizes U, we have $\mathfrak{n} \subseteq \mathfrak{u}$. Conversely, $\mathcal{N} = \exp \mathfrak{n}$ is a closed connected unipotent subgroup normalized by N', so $N'\mathcal{N}$ is a closed subgroup. It contains $M_a = \hat{x}_a m_a$, hence $\mathcal{G}$. If $g = n'n \in U = \mathcal{G} \cap U_P$ with $n' \in N'$ and $n \in \mathcal{N} \subseteq U_P$, then $n' \in N' \cap U_P = U'' = \exp \mathrm{Lie}(U'') \subseteq \mathcal{N}$, so $g \in \mathcal{N}$. Thus $U \subseteq \mathcal{N}$ and $\mathfrak{u} \subseteq \mathfrak{n}$. □

Proof of Theorem 40. By Lemmas 42, 44 and 45 ($\log \hat{x}_a = \sum_{i<m} (-1)^{i+1} (\hat{x}_a - I)^i / i$ has polynomial size), the generators of V in Proposition 46 are spanned by matrices of polynomial size, and the operators $\mathrm{Ad}(m_a)^{\pm 1}$ and $\mathrm{ad}(Y')$ (Y' in the basis $Y_1, \dots, Y_n$ of Lemma 43; stability under these is stability under all of $\mathrm{ad}(\mathrm{Lie}\, S)$) have polynomial size. By (F2) and (F3), $\mathfrak{u}$ is spanned by matrices of polynomial size, in the

weight basis conjugated by v_0. Undoing the conjugation by v_0 and the block diagonal change of basis, both of polynomial size, gives the first claim in the adapted basis. For every j, $U_j = \pi_j(U)$ (Lemma 29) and π_j is linear, so in characteristic zero $\mathfrak{u}_j = \mathrm{Lie}(U_j) = \pi_j(\mathfrak{u})$; these are exactly the subspaces computed in step 3 (Proposition 35(2)) and entering Definition 36 and Lemma 38. Projecting the spanning matrices of $\mathfrak{u}$ onto the distances $\le j$ gives spanning sets of $\mathfrak{u}_j$ of polynomial size, so $\mathrm{ht}(\mathfrak{u}_j)$ is polynomial by (F1) for every j, and so is B_f. By Theorem 28(2) the algorithm of Section 9.3 then runs in deterministic polynomial time on every invertible series. □

Remark 47. (i) The algorithm is unchanged; only its analysis is new. The proof is constructive except for the spanning set of $\mathrm{Lie}(U'')$, which uses Schreier generators of a subgroup of possibly exponential index. That set is used only to bound heights, in the same way as in Lemma 37, which is the case $S = 1$ of the argument (then $Y = 0$, $v_0 = I$, $\hat{x}_a = I$, $m_a = M_a$ and $U'' = U$).

(ii) Remark 39(ii) shows that unipotent parts of words of length $\le n$ may all be trivial in dimension $n + 2$. Lemma 42 shows that the torus behaves better: $\mathrm{Lie}(H^\circ)$ is always spanned by the Lie algebras of the closures of the $b(M_w)$ with $|w| \le 2d^2 + 2d$. For the affine Weyl group of type $\tilde{A}_n$ in its multiplicative form of Remark 39(ii) (a homomorphic image of the Coxeter group, so words of length $\le n$ still have finite order), words of length at most n have finite order, so length $n + 1$ is necessary; in our exact experiments ($n \le 3$) it was also sufficient.

(iii) In particular no randomness is needed in Theorem 28(3).

10 Beyond the invertible sector

This section studies series with singular minimal letter matrices. It opens with a local necessary condition derived from Theorem 1 (Corollary 48). Theorem 50 characterizes CCRA-computability exactly when every singular minimal letter matrix has rank at most one. Theorem 56 is a new necessary condition of arithmetic nature, derived from an inverse-weight lemma for compressed machines (Proposition 52); it implies that CCRA over $\mathbb{Q}$ are not closed under reversal (Corollary 57), which proves Conjecture 51 of [MPS26b], and that the candidate of [MPS26b, Example 52] is a witness. A companion lifting statement (Proposition 53) controls translations as well and yields a second obstruction (Theorem 60), which shows that a block-symmetric elementary symmetric function, equal to its own reversal, is not CCRA-computable although all local groups are commutative. Theorem 63 shows that these necessary conditions are also sufficient for series over $\{a, \#\}$ with M_a invertible and all words containing # of rank two. Proposition 65 shows that the normal form (2) of Theorem 1 fails beyond the invertible sector: registers must interact. Throughout, f has minimal representation $(\lambda, (M_a)_a, \gamma)$ of dimension d, words act on row vectors from the right, and for a subspace $X \subseteq \mathbb{Q}^{1\times d}$ we write $X^\perp = \{v \in \mathbb{Q}^d : Xv = 0\}$.

10.1 A local necessary condition

Theorem 1 constrains singular series locally.

Corollary 48 (Local necessary condition). *Let f be computed by a CCRA, with minimal representation $(\lambda, (M_a)_a, \gamma)$ of dimension d. Let e be a word with $\mathrm{rank}\, M_e^2 = \mathrm{rank}\, M_e$, let $R = \mathbb{Q}^{1\times d}M_e$ be the row space of M_e, and let Z be a finite set of words with $\mathrm{rank}(M_eM_zM_e) = \mathrm{rank}\, M_e$ for all $z \in Z$. Then the maps $v_z : R \to R$, $r \mapsto rM_eM_zM_e$, are bijective, and the Zariski closure of the group they generate satisfies condition (3) of Theorem 1: its identity component is triangularizable over $\mathbb{Q}$ and its unipotent elements commute.*

Proof. Write $N_z = M_eM_zM_e$. The rank condition on e gives $\mathbb{Q}^{1\times d} = R \oplus \{r : rM_e = 0\}$ and $RM_e = R$, so $RN_z = RM_zM_e = \mathbb{Q}^{1\times d}N_z$ has dimension $\mathrm{rank}\, M_e = \dim R$; hence v_z is a bijection of R. Let $\mathcal{H} \subseteq \mathrm{GL}(R)$ be the closure of the group generated by the v_z.

For words u, v consider $g_{u,v}(z_1 \cdots z_n) = f(ue \cdot ez_1e \cdots ez_ne \cdot ev) = c_v(\rho_u N_{z_1} \cdots N_{z_n})$, where $\rho_u = \lambda M_u M_e \in R$ and $c_v : r \mapsto rM_eM_v\gamma$. Reading ue, then the blocks eze, and finally ev before the output shows that $g_{u,v}$ is computed by a CCRA over the alphabet Z. By the row version of Section 2, its minimal representation is $X_u/Y_{u,v}$ with $X_u = \mathrm{span}\{\rho_u N_w : w \in Z^*\}$ and $Y_{u,v} = \{r \in X_u : c_v(rN_w) = 0 \ \forall w \in Z^*\}$, where $N_{z_1 \cdots z_n} = N_{z_1} \cdots N_{z_n}$; its letter operators are induced by the bijections ν_z, so $g_{u,v}$ is invertible. By Theorem 1, the closure $\mathcal{H}_{u,v}$ of the induced group on $X_u/Y_{u,v}$ satisfies (3), and restriction maps $\mathcal{H}$ onto $\mathcal{H}_{u,v}$ by (A2).

If $h_1, h_2 \in \mathcal{H}$ are unipotent, their images in $\mathcal{H}_{u,v}$ commute, so $[h_1, h_2]$ maps X_u into $Y_{u,v}$; with $w = \varepsilon$ this gives $c_v(\rho_u[h_1, h_2]) = 0$ for all u, v. The rows ρ_u span R (as the λM_u span the dual space), and the functionals c_v separate R (if $rM_eM_v\gamma = 0$ for all v then $rM_e = 0$, so $r = 0$). Hence $[h_1, h_2] = 0$. For triangularizability, $\bigcap_v Y_{u,v} = 0$ by the same separation, so X_u embeds $\mathcal{H}^\circ$-equivariantly in a finite sum $\bigoplus_v X_u/Y_{u,v}$, each of which carries an $\mathcal{H}^\circ$-invariant complete rational flag; thus each X_u has one, and so does $R = \sum_u X_u$, a quotient of $\bigoplus_u X_u$. □

For $e = \varepsilon$ the corollary says that any finite set of words whose minimal matrices are invertible generates a group satisfying (3), even if other letters are singular. For general e it constrains the action of the words eze on the row space of every word matrix M_e with $\operatorname{rank} M_e^2 = \operatorname{rank} M_e$.

Lemma 49. *If f is computed by a CCRA, $v, v' \in \Sigma^*$ and $\Sigma' \subseteq \Sigma$, then $u \mapsto f(vuv')$ on Σ'^* is computed by a CCRA.*

Proof. Start in $\delta(s_0, v)$ with valuation $P_{s_0,v}(x^{(0)})$, keep the transitions on Σ', and in state s output $\omega_{\delta(s,v')} \circ P_{s,v'}$, which is copyless as a composite of copyless maps. □

10.2 Rank-one resets

Theorem 50. *Suppose that every minimal letter matrix of f is invertible or has rank at most one. Let Σ_0 be the set of letters with invertible matrix and $\mathcal{G}_0 \subseteq \mathrm{GL}_d(\overline{\mathbb{Q}})$ the Zariski closure of the group generated by $\{M_a : a \in \Sigma_0\}$. Then f is CCRA-computable if and only if $\mathcal{G}_0^\circ$ is triangularizable over $\mathbb{Q}$ and any two unipotent elements of $\mathcal{G}_0$ commute. Consequently CCRA-computability is decidable in this class, and the CCRA series in this class are closed under reversal.*

Proof. Let $\Sigma_1 = \Sigma \setminus \Sigma_0$. For $\sigma \in \Sigma_1$ write $M_\sigma = c_\sigma r_\sigma$ with a column c_σ and a row r_σ (both zero if $M_\sigma = 0$), and put $r_\triangleright = \lambda$, $c_\triangleleft = \gamma$. For $i \in \Sigma_1 \cup \{\triangleright\}$ and $j \in \Sigma_1 \cup \{\triangleleft\}$ define $g_{ij} : \Sigma_0^* \to \mathbb{Q}$, $g_{ij}(u) = r_i M_u c_j$. Every word has a unique factorization $w = u_0\sigma_1u_1 \cdots \sigma_k u_k$ with $u_m \in \Sigma_0^*$ and $\sigma_m \in \Sigma_1$, and

$$f(w) = g_{\triangleright\sigma_1}(u_0)\, g_{\sigma_1\sigma_2}(u_1) \cdots g_{\sigma_k\triangleleft}(u_k). \tag{13}$$

The series g_{ij} is computed by $(r_i, (M_a)_{a\in\Sigma_0}, c_j)$, so its minimal representation is an invariant subquotient X_i/Y_{ij} of the row space, with $X_i = \mathrm{span}\{r_iM_u : u \in \Sigma_0^*\}$ and $Y_{ij} = \{x \in X_i : xM_uc_j = 0 \ \forall u \in \Sigma_0^*\}$ (row version of Section 2). Its letter operators are induced by invertible matrices, so g_{ij} is invertible, and by (A2) the Zariski closure $\mathcal{G}_{ij}$ of its group is the image of $\mathcal{G}_0$ under the induced-action morphism.

Sufficiency. Condition (3) passes from $\mathcal{G}_0$ to $\mathcal{G}_{ij}$: identity components map onto identity components (A2), the traces of a complete $\mathcal{G}_0^\circ$-invariant rational flag form a chain of invariant rational subspaces of X_i/Y_{ij} with quotients of dimension at most one, and unipotent elements of $\mathcal{G}_{ij}$ are images of unipotent elements of $\mathcal{G}_0$ (A2). By Theorem 1, each g_{ij} is computed by a CCRA C_{ij} over Σ_0. The machine for f keeps in its control the last letter $i \in \Sigma_1 \cup \{\triangleright\}$ read and the control states of the machines C_{ij}, $j \in \Sigma_1 \cup \{\triangleleft\}$; it has an accumulator ζ, initially 1, and for each j a block $\mathcal{B}_j$ of registers, large enough for every C_{ij}, holding the registers of C_{ij} for the current i. A letter $a \in \Sigma_0$ updates each block $\mathcal{B}_j$ by C_{ij}. A letter $\sigma \in \Sigma_1$ performs $\zeta := \zeta \cdot \omega_{i\sigma}$, where $\omega_{i\sigma}$ is the current output polynomial of $C_{i\sigma}$ applied to $\mathcal{B}_\sigma$, sets $i := \sigma$, and reinitializes every block $\mathcal{B}_j$ and its control component to the initial data of $C_{\sigma j}$; all other old register values are discarded. In state i

the output is $\zeta \cdot \omega_{i\triangleleft}$. Every update uses each old register at most once, and (13) shows that the machine computes f.

Necessity. Let f be CCRA-computable. Since the rows λM_v span $\mathbb{Q}^{1\times d}$ and the columns $M_{v'}\gamma$ span $\mathbb{Q}^d$, for nonzero c_i and r_j (otherwise $g_{ij}=0$) there are words v, v' with $\lambda M_v c_i \neq 0 \neq r_j M_{v'}\gamma$, and $f(v\,i\,u\,j\,v') = (\lambda M_v c_i)\, g_{ij}(u)\, (r_j M_{v'}\gamma)$; similarly for $i=\triangleright$ or $j=\triangleleft$. By Lemma 49 every g_{ij} is CCRA-computable, so $\mathcal{G}_{ij}$ satisfies (3) by Theorem 1. Let $h_1, h_2 \in \mathcal{G}_0$ be unipotent. Their images in $\mathcal{G}_{ij}$ commute, so $[h_1,h_2]$ maps X_i into $\bigcap_j Y_{ij}$. If $x \in \bigcap_j Y_{ij}$, then $xM_w\gamma = 0$ for every word w: either $w \in \Sigma_0^*$, or $w = u\sigma w'$ with $u \in \Sigma_0^*$ and $xM_w\gamma = (xM_u c_\sigma)(r_\sigma M_{w'}\gamma) = 0$. Hence $x=0$. As every row λM_w lies in $X_\triangleright$ or, if σ is the last letter of w in Σ_1, in X_σ, we have $\mathbb{Q}^{1\times d} = \sum_i X_i$, and $[h_1,h_2]=0$. For triangularizability, $\mathcal{G}_0^\circ$ preserves each X_i and each Y_{ij}, and its image in $\mathrm{GL}(X_i/Y_{ij})$ is connected, hence contained in $\mathcal{G}_{ij}^\circ$, which fixes a complete rational flag. Since $\bigcap_j Y_{ij} = 0$, X_i embeds $\mathcal{G}_0^\circ$-equivariantly into $\bigoplus_j X_i/Y_{ij}$; intersecting the concatenated flags with the image gives a complete invariant rational flag of X_i, and $\mathbb{Q}^{1\times d} = \sum_i X_i$ is a quotient of $\bigoplus_i X_i$. So $\mathcal{G}_0^\circ$ is triangularizable over $\mathbb{Q}$ on rows, hence on columns.

Consequences. Decidability follows from the procedure of Section 8 applied to $\{M_a : a \in \Sigma_0\}$. The reversal has minimal representation $(\gamma^{\mathsf{T}}, (M_a^{\mathsf{T}})_a, \lambda^{\mathsf{T}})$, transposition preserves ranks, and $g \mapsto (g^{\mathsf{T}})^{-1}$ preserves condition (3) as in the proof of Corollary 4. □

10.3 Inverse weights and translations

We return to the setting of Section 3: a monoid $\mathcal{Y}$ acts on $\mathbb{Q}^r$ by copyless maps P_y, subspaces $N \subseteq W \subseteq V_r$ are invariant under all B_y, $D = W/N$, and every induced operator $\rho(y)$ is invertible. Let e, k, L, A, Q be as chosen before Lemma 9, let $R_y = Q \circ P_y|_L$ be the affine coordinate permutations of the proof of Proposition 10, and identify L with $\mathbb{Q}^k$ through its free coordinates. Put $\mathcal{Z} = \{eye : y \in \mathcal{Y}\}$.

Lemma 51. *For $z \in \mathcal{Z}$ we have $R_z = P_z|_L$, and $R_{zz'} = R_{z'} \circ R_z$ for $z, z' \in \mathcal{Z}$. Hence $z \mapsto R_z$ is an antihomomorphism from $\mathcal{Z}$ to the group of affine coordinate permutations of $\mathbb{Q}^k$, and $\mathcal{Z}_0 = \{z \in \mathcal{Z} : R_z$ has trivial permutation part$\}$ is a subsemigroup containing $z^{k!}$ for every $z \in \mathcal{Z}$.*

Proof. $P_{eye} = P_e \circ P_y \circ P_e$ takes values in L, where Q is the identity; so $R_z = P_z|_L$ and $P_z(L) \subseteq L$, whence $R_{zz'} = (P_{z'} \circ P_z)|_L = R_{z'} \circ R_z$. □

For $z \in \mathcal{Z}_0$ write $R_z(t) = (\alpha_i(z) t_i + \beta_i(z))_{i=1}^k$ with $\alpha_i(z) \in \mathbb{Q}^\times$. The multiplier α_i is visible to the machine: if $t \in L$ and $t' = t + \epsilon_i$ differs from t by 1 in the free coordinate i, then

$$\alpha_i(z) = P_z(t')_i - P_z(t)_i \qquad (z \in \mathcal{Z}_0). \tag{14}$$

Proposition 52 (Inverse weights). *Let $D_0 \subset D_1 \subset D_2 \subseteq D$ be subspaces invariant under $\rho(\mathcal{Z}_0)$ with $\dim D_1/D_0 = \dim D_2/D_1 = 1$, and let $\kappa_1(z), \kappa_2(z)$ be the scalars by which $\rho(z)$ acts on D_1/D_0 and on D_2/D_1. Let $\mathcal{H}$ be the Zariski closure of the group generated by $\rho(\mathcal{Z}_0)$, and suppose that some unipotent element of $\mathcal{H}$ induces a non-identity map on D_2/D_0. Then there is a free coordinate i with*

$$\alpha_i(z) = \kappa_2(z)/\kappa_1(z) \qquad \text{for all } z \in \mathcal{Z}_0.$$

Proof. Let $\Theta : W''/N'' \to D$ be the intertwining isomorphism of Proposition 10, and let $N'' \subseteq W_0'' \subseteq W_1'' \subseteq W_2'' \subseteq W''$ be the preimages of D_0, D_1, D_2; all are invariant under B_{R_z}, $z \in \mathcal{Z}_0$. Let $\mathcal{S}$ be the closed subgroup of $\mathrm{GL}(V_k \otimes \overline{\mathbb{Q}})$ stabilizing these five subspaces, $\pi : \mathcal{S} \to \mathrm{GL}(D)$ and $\pi_2 : \mathcal{S} \to \mathrm{GL}(D_2/D_0)$ the induced-action morphisms, and $\mathcal{G}_L \subseteq \mathcal{S}$ the Zariski closure of the group generated by $\{B_{R_z} : z \in \mathcal{Z}_0\}$. As in Corollary 12, $\pi(\mathcal{G}_L) = \mathcal{H}$. Fix $\bar{e}_1 \in D_1/D_0$ and $\bar{e}_2 \in (D_2/D_0) \setminus (D_1/D_0)$. Every element of $\pi_2(\mathcal{S})$ preserves the line D_1/D_0, so the unipotent elements of $\pi_2(\mathcal{G}_L)$ are of the form $u_s : \bar{e}_1 \mapsto \bar{e}_1,\ \bar{e}_2 \mapsto \bar{e}_2 + s\bar{e}_1$.

For $b \in \overline{\mathbb{Q}}^k$ let $\tau_b(t) = t + b$, and let $U' = \{b : B_{\tau_b} \in \mathcal{G}_L\}$, a closed subgroup of $\overline{\mathbb{Q}}^k$. If $b \in U'$, then $U' \cap \overline{\mathbb{Q}}b$ contains $\mathbb{Z}b$, whose Zariski closure is the line $\overline{\mathbb{Q}}b$; so U' is a linear subspace. Define $\psi : U' \to \overline{\mathbb{Q}}$ by $\pi_2(B_{\tau_b}) = u_{\psi(b)}$. It is additive, and $c \mapsto \psi(cb) - c\psi(b)$ is a polynomial function of $c \in \overline{\mathbb{Q}}$ vanishing on $\mathbb{Z}$; so ψ is linear.

$\psi \neq 0$. Let $v \in \mathcal{H}$ be unipotent and non-identity on D_2/D_0. Write $v = \pi(g)$ with $g \in \mathcal{G}_L$; then $v = v_u = \pi(g_u)$ by (A2), and $g_u \in \mathcal{G}_L \subseteq \mathcal{W}_k$ is unipotent, hence $g_u = B_{\tau_b}$ for some b by Lemma 11. Then $b \in U'$, and $\pi_2(g_u)$ is the map induced by v on D_2/D_0, so $\psi(b) \neq 0$.

Equivariance. Let $z \in \mathcal{Z}_0$ and $\Delta_z = \mathrm{diag}(\alpha_1(z), \dots, \alpha_k(z))$. A direct computation gives $R_z \circ \tau_b \circ R_z^{-1} = \tau_{\Delta_z b}$, hence $B_{\tau_{\Delta_z b}} = B_{R_z}^{-1} B_{\tau_b} B_{R_z} \in \mathcal{G}_L$ for $b \in U'$, so $\Delta_z U' = U'$. In the basis $(\bar{e}_1, \bar{e}_2)$, $\pi_2(B_{R_z})$ is the upper triangular matrix with diagonal $(\kappa_1(z), \kappa_2(z))$, and for such a matrix h one computes $h^{-1} u_s h = u_{s\kappa_2(z)/\kappa_1(z)}$. Applying π_2,

$$\psi(\Delta_z b) = \frac{\kappa_2(z)}{\kappa_1(z)}\, \psi(b) \qquad (b \in U',\ z \in \mathcal{Z}_0).$$

Weight decomposition. Call free coordinates i, i' equivalent if $\alpha_i = \alpha_{i'}$ on $\mathcal{Z}_0$. For an equivalence class C let P_C be the coordinate projection onto $\overline{\mathbb{Q}}^C$. For each class $C' \neq C$ choose $z_{C'} \in \mathcal{Z}_0$ with $\alpha_C(z_{C'}) \neq \alpha_{C'}(z_{C'})$; then $P_C = \prod_{C' \neq C} (\Delta_{z_{C'}} - \alpha_{C'}(z_{C'}))/(\alpha_C(z_{C'}) - \alpha_{C'}(z_{C'}))$ is a polynomial in the operators Δ_z, so $P_C U' \subseteq U'$ and $U' = \bigoplus_C P_C U'$. As $\psi \neq 0$, some $b \in P_C U'$ has $\psi(b) \neq 0$. Then $\Delta_z b = \alpha_C(z) b$, and the displayed identity gives $\alpha_C(z) = \kappa_2(z)/\kappa_1(z)$ for all $z \in \mathcal{Z}_0$. Any $i \in C$ is as required. □

In the invertible sector κ_2/κ_1 is a rational character and the conclusion is harmless. In general it says that some register, whose multiplier is computed by the machine through (14), must realize the *inverse* of the weight by which the unipotent part translates. This is the source of the obstruction in Theorem 56 below.

The multipliers are only half of the data of the compressed registers; the translations are also rational series, which gives a second constraint. Let $\mathcal{G}'_L \subseteq \mathrm{GL}(V_k \otimes \overline{\mathbb{Q}})$ be the Zariski closure of the group generated by $\{B_{R_z} : z \in \mathcal{Z}\}$, let $F'_L = \mathcal{G}'_L/\mathcal{G}'^{\circ}_L$, let $\mathcal{Z}_1 = \{z \in \mathcal{Z} : B_{R_z} \in \mathcal{G}'^{\circ}_L\}$, and let $\mathcal{H}'$ be the Zariski closure of the group generated by $\rho(\mathcal{Z})$. By Lemma 51, $z \mapsto B_{R_z}$ is a homomorphism on $\mathcal{Z}$ ($B_{R_{zz'}} = B_{R_{z'} \circ R_z} = B_{R_z} B_{R_{z'}}$), so $\Psi(z) = B_{R_z} \mathcal{G}'^{\circ}_L$ is a homomorphism from $\mathcal{Z}$ to the finite group F'_L, and $\mathcal{Z}_1 = \Psi^{-1}(1)$ is a subsemigroup containing $z^{|F'_L|}$ for every $z \in \mathcal{Z}$. Since $\mathcal{G}'^{\circ}_L$ is connected, it lies in $\mathcal{W}^{\circ}_k$, so $\mathcal{Z}_1 \subseteq \mathcal{Z}_0$.

Proposition 53 (Rational lifting). *By the argument of Corollary 12, the unipotent elements of $\mathcal{H}'$ commute; let U_H be the unipotent radical of $\mathcal{H}'^{\circ}$. There are a maximal torus T_H of $\mathcal{H}'^{\circ}$, a linear map $\Lambda : \overline{\mathbb{Q}}^k \supseteq U'' \to \mathrm{Lie}(U_H)$, and for each free coordinate i a constant $c_i \in \overline{\mathbb{Q}}$ such that for every $z \in \mathcal{Z}_1$ the decomposition $\rho(z) = t(z)u(z)$ with $t(z) \in T_H$, $u(z) \in U_H$ satisfies*

$$\log u(z) = \Lambda(\gamma(z)), \qquad \gamma_i(z) = R_z(c)_i - c_i = \big(\alpha_i(z) - 1\big)c_i + R_z(0)_i,$$

and every character of T_H takes on $t(z)$ the value of a Laurent monomial in $\alpha_1(z), \dots, \alpha_k(z)$. In particular, extending Λ linearly to $\overline{\mathbb{Q}}^k$, each coordinate of $\log u(z)$ is a $\overline{\mathbb{Q}}$-linear combination of the constant 1 and the rational series $z \mapsto P_z(0)_i$ and $z \mapsto P_z(\epsilon_i)_i - P_z(0)_i$ (computed from fixed valuations in L as in (14)).

Proof. Let $\pi : \mathcal{S}' \to \mathrm{GL}(D)$ be the induced-action morphism on the stabilizer $\mathcal{S}'$ of W'' and N'', which contains $\mathcal{G}'_L$; as in Corollary 12, $\pi(\mathcal{G}'_L) = \mathcal{H}'$ and $\pi(\mathcal{G}'^{\circ}_L) = \mathcal{H}'^{\circ}$. The group $\mathcal{G}'^{\circ}_L$ is connected and solvable, being contained in $\mathcal{W}^{\circ}_k$, the group of pullbacks of maps $t \mapsto (a_i t_i + b_i)_i$. Choose a maximal torus T_L of $\mathcal{G}'^{\circ}_L$. It lies in a maximal torus of $\mathcal{W}^{\circ}_k$, and these are the conjugates of the group of pullbacks of diagonal scalings by pullbacks of translations [Bor91, 10.6]; so there is $c \in \overline{\mathbb{Q}}^k$ such that T_L consists of pullbacks of maps $\sigma_a(t) = (a_i(t_i - c_i) + c_i)_i$. The unipotent elements of $\mathcal{G}'^{\circ}_L$ form its unipotent radical U_L [Bor91, 10.6], and by

(A3) and Lemma 11 they are the pullbacks B_{τ_b} with b in the subspace $U'' = \{b : B_{\tau_b} \in \mathcal{G}'_L\}$ (a subspace by the argument in the proof of Proposition 52). For $z \in \mathcal{Z}_1$ write $R_z = \tau_{\gamma(z)} \circ \sigma_{\alpha(z)}$ with $\gamma(z) = R_z(c) - c$; then $B_{R_z} = B_{\sigma_{\alpha(z)}} B_{\tau_{\gamma(z)}}$. Since $B_{R_z} \in \mathcal{G}'^{\circ}_L = T_L \ltimes U_L$ and the decomposition of an element of $\mathcal{W}^{\circ}_k$ as a product of a pullback of some σ_a and a pullback of a translation is unique, $B_{\sigma_{\alpha(z)}} \in T_L$ and $B_{\tau_{\gamma(z)}} \in U_L$, so $\gamma(z) \in U''$. The image $T_H = \pi(T_L)$ is a maximal torus of $\mathcal{H}'^{\circ}$ [Bor91, 11.14], and $\pi(U_L) = U_H$ by (A2); by uniqueness of the decomposition in $\mathcal{H}'^{\circ} = T_H \ltimes U_H$, $t(z) = \pi(B_{\sigma_{\alpha(z)}})$ and $u(z) = \pi(B_{\tau_{\gamma(z)}})$. As U_H is commutative, log is an isomorphism of U_H onto the vector group $\mathrm{Lie}(U_H)$, and $b \mapsto \log \pi(B_{\tau_b})$ is a homomorphism of vector groups in characteristic zero, hence linear; call it Λ. Characters of T_H pull back to characters of T_L, and these extend to characters of the torus $\{B_{\sigma_a}\} \cong \mathbb{G}_m^k$ (restriction of characters to a subtorus is onto), which are Laurent monomials in a. Finally $R_z(t)_i = \alpha_i(z) t_i + R_z(0)_i$ gives the formula for $\gamma_i(z)$, and $R_z = P_z|_L$ by Lemma 51. □

10.4 Affine contexts and non-closure under reversal

An *exponential polynomial* is a function $E : \mathbb{N} \to \overline{\mathbb{Q}}$ of the form $E(\ell) = \sum_{\mu \in \mathcal{M}} p_\mu(\ell) \mu^\ell$ with $\mathcal{M} \subseteq \overline{\mathbb{Q}}^\times$ finite and $p_\mu \in \overline{\mathbb{Q}}[T]$. We use three standard facts: (E1) the functions $\ell \mapsto \ell^s \mu^\ell$ are linearly independent on every set $\{\ell \geq \ell_0\}$, and the functions $(\ell_1, \dots, \ell_k) \mapsto \prod_m \ell_m^{s_m} \mu_m^{\ell_m}$ are linearly independent on $\mathbb{N}^k$; (E2) sums and products of exponential polynomials are exponential polynomials; (E3) for a square matrix Y of size n, the entries of Y^ℓ agree with exponential polynomials for $\ell \geq n$, by the Jordan form. Consequently, if $(\lambda', (M'_a)_a, \gamma')$ represents a rational series φ and $w_0, \dots, w_m, y$ are words, then $\ell \mapsto \varphi(w_0 y^\ell w_1 y^\ell \cdots y^\ell w_m)$ agrees with an exponential polynomial for large ℓ.

Lemma 54. *Let $q \in \overline{\mathbb{Q}}[T]$ be nonconstant and E an exponential polynomial. Then $q(\ell)E(\ell) = 1$ fails for infinitely many ℓ.*

Proof. Otherwise $qE - 1 = \sum_\mu (qp_\mu)(\ell)\mu^\ell - 1$ vanishes for $\ell \geq \ell_0$, and by (E1) the coefficient of $\mu = 1$ gives $qp_1 = 1$ in $\overline{\mathbb{Q}}[T]$, which is impossible. □

Definition 55. An *affine context* for f consists of words u, x_1, w, x_2, linearly independent rows $r_1, r_2 \in \mathbb{Q}^{1\times d}$ and polynomials $\alpha, \beta \in \mathbb{Q}[T]$ such that, with $b_\ell = x_1 w^\ell x_2$:

(i) $\lambda M_u \in \mathbb{Q}r_1 + \mathbb{Q}r_2$ and $\lambda M_u \notin \mathbb{Q}r_1$;
(ii) $r_1 M_{b_\ell} = r_1$ and $r_2 M_{b_\ell} = \beta(\ell) r_1 + \alpha(\ell) r_2$ for all $\ell \geq 0$;
(iii) α is nonconstant, and the rational function $\beta/(\alpha - 1)$ is nonconstant.

Thus the blocks b_ℓ act on the plane $\mathbb{Q}r_1 + \mathbb{Q}r_2$ by $r_1 \mapsto r_1$, $r_2 \mapsto \beta(\ell) r_1 + \alpha(\ell) r_2$: an affine map $s \mapsto s + \beta(\ell)$ on the coefficient of r_1, scaled by $\alpha(\ell)$, whose fixed points move with ℓ. The proof below uses only that the columns $M_v \gamma$ span $\mathbb{Q}^d$, so affine contexts may be exhibited in any representation with this property.

Theorem 56. *If f has an affine context, then f is not computed by any CCRA over $\mathbb{Q}$.*

Proof. Suppose that a CCRA C with notation as in Section 2 computes f.

Step 1: control. Let $n = |S|$ and $p = n!$. For every state s and $j \geq n$ we have $\delta(s, w^{j+p}) = \delta(s, w^j)$. Choose $c \geq n$ such that $|\alpha(\ell)| > 1$ for all $\ell \geq c$, and let $I = c + p\mathbb{N}$. Then $\tau(s) = \delta(s, b_\ell)$ does not depend on $\ell \in I$, and τ^p is idempotent. Put $u' = u b_c^p$ and $q = \delta(s_0, u') = \tau^p(\delta(s_0, u))$, so $\tau^p(q) = q$. Let $\mathcal{Y} \subseteq \Sigma^*$ be the monoid generated by the words $b_{\ell_1} \cdots b_{\ell_p}$ with all $\ell_j \in I$. Every $y \in \mathcal{Y}$ fixes q, and $\mathcal{Y}$ acts on $\mathbb{Q}^r$ by $P_y = P_{q,y}$.

Step 2: the local quotient. Define W, N and $K_{u'}$ as in Lemma 15, with $\mathcal{Y}$ in place of L_R and u' in place of ue. The proof of Lemma 15 uses only that the words in question form a monoid fixing q, so W, N are invariant under every B_y, and $D = W/N \cong \mathbb{Q}^d/K_{u'}$ intertwines B_y with the operator $\rho(y)$ induced by M_y. Let $X = \mathbb{Q}r_1 + \mathbb{Q}r_2$, which is invariant under each M_{b_ℓ} by (ii). For $s, t \in \mathbb{Q}$ with $t \neq 0$, the line

$\mathbb{Q}(sr_1 + tr_2)$ is M_{b_ℓ}-invariant iff $s/t = \beta(\ell)/(\alpha(\ell) - 1)$, and for $\ell \in I$ the matrix of M_{b_ℓ} on X has the distinct eigenvalues 1 and $\alpha(\ell)$, so $M_{b_\ell}^p$ has the same invariant lines. By (i), $\lambda M_{u'} = \mu_1 r_1 + \mu_2 r_2$ with $\mu_2 \neq 0$. If $\operatorname{span}\{\lambda M_{u'} M_y : y \in \mathcal{Y}\}$ were one-dimensional, the line $\mathbb{Q}\lambda M_{u'}$ would be invariant under every $M_{b_\ell}^p$, $\ell \in I$, contradicting (iii). Hence this span is X, and $K_{u'} = X^\perp$. Choose $v_1, v_2 \in \mathbb{Q}^d$ with $r_i v_j = \delta_{ij}$, and let $\bar{e}_1, \bar{e}_2$ be their images in $D \cong \mathbb{Q}^d/X^\perp$. The subspace $X^\perp$ is invariant under each M_{b_ℓ}, and by (ii) the operator induced by M_{b_ℓ} on $\mathbb{Q}^d/X^\perp$ maps $\bar{e}_1 \mapsto \bar{e}_1 + \beta(\ell)\bar{e}_2$ and $\bar{e}_2 \mapsto \alpha(\ell)\bar{e}_2$. Hence for $y \in \mathcal{Y}$, $\rho(y)$ fixes the line $\mathbb{Q}\bar{e}_2$, acts on it by a scalar $\bar{\alpha}(y)$ equal to the product of the values $\alpha(\ell)$ over the blocks b_ℓ of y, and acts trivially on $D/\mathbb{Q}\bar{e}_2$. In particular every $\rho(y)$ is invertible, $\bar{\alpha}$ is multiplicative, and $|\bar{\alpha}(y)| > 1$ for $y \neq \varepsilon$.

Step 3: compression. Proposition 10 applies to $\mathcal{Y}$, W, N. Since $k(P_\varepsilon) = r$ is the largest possible value, we may and do choose the minimizing element $e \neq \varepsilon$. Let k, L, R, $\mathcal{Z}$, $\mathcal{Z}_0$ and α_i be as in Lemma 51; $k \geq 1$ because $\dim D = 2$.

Step 4: a unipotent element. We claim that $\rho(\mathcal{Z}_0)$ is not commutative. Otherwise, for $y \in \mathcal{Y}$ the matrices $\rho((eye)^{k!}) \in \rho(\mathcal{Z}_0)$ commute pairwise. The matrix $\rho(eye) = \rho(e)\rho(y)\rho(e)$ has the distinct eigenvalues 1 and $\bar{\alpha}(e)^2\bar{\alpha}(y)$, the latter of absolute value > 1, so its 1-eigenline equals that of its $k!$-th power. Commuting diagonalizable 2×2 matrices with distinct eigenvalues have the same eigenlines, and here the other eigenline is always $\mathbb{Q}\bar{e}_2$; hence all $\rho(eye)$ have a common 1-eigenvector v_0. Taking $y = \varepsilon$, $\rho(e)^2 v_0 = v_0$, and since $\bar{\alpha}(e)^2 \neq 1$ this gives $\rho(e)v_0 = v_0$. Then $\rho(e)\rho(y)\rho(e)v_0 = v_0$ yields $\rho(y)v_0 = v_0$ for all $y \in \mathcal{Y}$, in particular for $y = b_\ell^p$, $\ell \in I$. But the 1-eigenline of $\rho(b_\ell^p)$ is that of the operator induced by M_{b_ℓ}, spanned by $\bar{e}_1 + \frac{\beta(\ell)}{1-\alpha(\ell)}\bar{e}_2$, and by (iii) these lines are pairwise distinct for infinitely many $\ell \in I$, a contradiction. So there are $z, z' \in \mathcal{Z}_0$ with $\rho(z)\rho(z') \neq \rho(z')\rho(z)$. All elements of $\rho(\mathcal{Y})$ are triangular in the basis $(\bar{e}_2, \bar{e}_1)$ with diagonal $(\bar{\alpha}(\cdot), 1)$, so the commutator $\rho(z)\rho(z')\rho(z)^{-1}\rho(z')^{-1}$ is a unipotent element of $\mathcal{H}$, the closure of the group generated by $\rho(\mathcal{Z}_0)$, and it is not the identity.

Step 5: inverse weight. Apply Proposition 52 with $D_0 = 0$, $D_1 = \mathbb{Q}\bar{e}_2$, $D_2 = D$, so that $\kappa_1 = \bar{\alpha}$ and $\kappa_2 = 1$. There is a free coordinate i of L (a register index) with $\alpha_i(\zeta)\bar{\alpha}(\zeta) = 1$ for all $\zeta \in \mathcal{Z}_0$.

Step 6: pumping. For $\ell \geq 0$ let $y_\ell = b_{c+p\ell}\, b_c^{p-1} \in \mathcal{Y}$ and $\zeta_\ell = (ey_\ell e)^{k!} \in \mathcal{Z}_0$. Then $\bar{\alpha}(\zeta_\ell) = C\alpha(c + p\ell)^{k!}$ with $C = (\bar{\alpha}(e)^2\alpha(c)^{p-1})^{k!} \neq 0$. Fix $t \in L$. By (14), $\alpha_i(\zeta_\ell) = \varphi(\zeta_\ell)$, where $\varphi(v) = P_{q,v}(t + \epsilon_i)_i - P_{q,v}(t)_i$ is the difference of two functions computed by CCRAs (start in q with valuation $t + \epsilon_i$, respectively t, and output register i); so φ is a rational series (Section 2). Since $b_{c+p\ell} = x_1 w^c (w^p)^\ell x_2$, the word ζ_ℓ has the form $v_0(w^p)^\ell v_1 \cdots (w^p)^\ell v_{k!}$ with fixed words v_j, so $\ell \mapsto \alpha_i(\zeta_\ell)$ agrees with an exponential polynomial E for large ℓ. Thus $C\alpha(c + p\ell)^{k!}E(\ell) = 1$ for all large ℓ, where $\ell \mapsto C\alpha(c + p\ell)^{k!}$ is a nonconstant polynomial. This contradicts Lemma 54. □

Corollary 57. *Over the alphabet $\{a, \#\}$, write $w = u_0\#u_1\#\cdots\#u_k$ with $u_m \in a^*$, and let*

$$g(w) = \sum_{i=0}^{k} \prod_{j<i} |u_j|, \qquad h(w) = \sum_{i=0}^{k} \prod_{j<i} \big(|u_j| + 1\big).$$

The reversals $g^{\mathrm{r}}(w) = g(w^{\mathrm{rev}})$ and h^{r} are computed by CCRA, but g and h are not. Likewise, for the function $F(0^{m_1}1^{k_1}\cdots 0^{m_t}1^{k_t}) = (\cdots((m_1 + k_1)m_2 + k_2)\cdots)m_t + k_t$ of [MPS26b, Example 52]*, where $m_1, k_t \geq 0$ and all other exponents are positive (so the decomposition is unique and $F(\varepsilon) = 0$), F is computed by a CCRA and its reversal is not. In particular, the class of functions computed by CCRA over $\mathbb{Q}$ is not closed under reversal, for every alphabet with at least two letters.*

Proof. *Machines.* Since $g^{\mathrm{r}}(u_0\#\cdots\#u_k) = \sum_{i=0}^{k}\prod_{j>i}|u_j| =: G_k$ satisfies $G_k = G_{k-1}|u_k| + 1$ and $G_0 = 1$, the CCRA with one state, registers x, A initialized to 0, updates $A := A + 1$ on a and $(x, A) := (xA + 1, 0)$ on $\#$, and output $xA + 1$ computes g^{r}: after reading $u_0\#\cdots\#u'_m$ with u'_m a prefix of u_m, $x = G_{m-1}$ (with

$G_{-1} = 0$) and $A = |u'_m|$. For h^{r} initialize and reset A to 1 instead. For F, a two-state control records whether the last letter was 0 (initially yes); registers $v = 1$, $c = 0$; on 0, $c := c + 1$ after a 0 and $c := 1$ after a 1; on 1, $(v, c) := (vc + 1, 0)$ after a 0 and $v := v + 1$ after a 1; output vc after a 0 and v after a 1. All updates are copyless.

Affine contexts. On row coordinates (S, Q, P) let

$$M_a = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad M_{\#} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad M'_{\#} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \quad \lambda = (1, 0, 1),\ \gamma = (1, 0, 0)^{\mathsf{T}},$$

that is, $a : (S, Q, P) \mapsto (S, Q + P, P)$, $\# : (S, Q, P) \mapsto (S + Q, 0, Q)$ and $\#' : (S, Q, P) \mapsto (S + Q + P, 0, Q + P)$. Induction on the number of blocks shows that $(\lambda, M_a, M_{\#}, \gamma)$ represents g and $(\lambda, M_a, M'_{\#}, \gamma)$ represents h: after $u_0\# \cdots \#u'_m$ the state is $(S, |u'_m|P, P)$ with S the sum of the terms $i \le m$ and P the product of the factors contributed by $u_0, \ldots, u_{m-1}$. Both are minimal, since $\lambda, \lambda M_a, \lambda M_{\#}$ and $\gamma, M_{\#}\gamma, M_a M_{\#}\gamma$ are bases, and likewise $\lambda, \lambda M_a, \lambda M'_{\#}$ and $\gamma, M'_{\#}\gamma, M_a M'_{\#}\gamma$. Take $u = x_1 = \varepsilon$, $w = a$, $x_2 = \#$, $r_1 = (1, 0, 0)$, $r_2 = (0, 0, 1)$. Then $r_1 M_{a^\ell \#} = r_1$ and $r_2 M_{a^\ell \#} = \ell r_1 + \ell r_2$, while $r_2 M_{a^\ell \#'} = (\ell + 1)(r_1 + r_2)$; and $\lambda = r_1 + r_2$. So g has an affine context with $\alpha = \beta = T$ and h with $\alpha = \beta = T + 1$; in both cases $\beta/(\alpha - 1)$ is nonconstant. For $F^{\mathrm{r}}(w) = F(w^{\mathrm{rev}})$ use row coordinates (S, R, Q, P) with

$$1 : (S, R, Q, P) \mapsto (S + R + Q,\ R + Q,\ 0,\ R + Q), \qquad 0 : (S, R, Q, P) \mapsto (S,\ 0,\ Q + P,\ P),$$

$\lambda = (0, 1, 0, 1)$ and output $S + Q$; here R is the weight added by each 1 of the current run of ones, P the product of the completed runs of zeros, and $Q = P$ times the length of the current run of zeros. This representation of F^{r} is minimal of dimension 4. With $u = \varepsilon$, $x_1 = w = 0$, $x_2 = 1$, $r_1 = (1, 0, 0, 0)$, $r_2 = (0, 1, 0, 1) = \lambda$ we get $r_1 M_{0^{\ell+1}1} = r_1$ and $r_2 M_{0^{\ell+1}1} = (\ell + 1)r_1 + (\ell + 1)r_2$, an affine context with $\alpha = \beta = T + 1$. Theorem 56 applies in all three cases. For a larger alphabet $\Sigma \supseteq \{a, \#\}$, extend g by 0 outside $\{a, \#\}^*$: the reversal of the extension is computed by adding a sink state, and the extension is not CCRA-computable by Lemma 49 with $\Sigma' = \{a, \#\}$. □

Polynomiality of α enters the proof of Theorem 56 only in Step 6. The following general form shows that what matters is whether the block scalar is a pure exponential along pumping.

Lemma 58. *Let E, E' be exponential polynomials with $E(\ell)E'(\ell) = 1$ for all large ℓ. Then there is $p' \ge 1$ such that for every residue s there are $C_s, \mu_s \in \overline{\mathbb{Q}}^{\times}$ with $E(s + p'\ell) = C_s \mu_s^{\ell}$ for all large ℓ.*

Proof. Let $\Lambda \subseteq \overline{\mathbb{Q}}^{\times}$ be the finitely generated group generated by the bases of E and E', and p' the order of its torsion subgroup. On the residue class $s + p'\mathbb{N}$, both functions are exponential polynomials in ℓ with bases in $\Lambda^{p'}$, which is free abelian, say of rank m. By (E1), exponential polynomials with bases in $\Lambda^{p'}$, taken as functions of large ℓ, form a ring isomorphic to the Laurent polynomial ring $\overline{\mathbb{Q}}[T][X_1^{\pm 1}, \ldots, X_m^{\pm 1}]$, whose units are the nonzero constant multiples of Laurent monomials. Hence $E(s + p'\ell)$ is a unit of this ring, that is, of the form $C_s \mu_s^{\ell}$. □

Theorem 59 (General form). *Let u, x_1, w, x_2 and r_1, r_2 satisfy (i) and (ii) of Definition 55 for some functions $\alpha, \beta : \mathbb{N} \to \mathbb{Q}$; these agree with exponential polynomials for large ℓ. Suppose that $\alpha(\ell) \ne 0$ for all large ℓ, and that $|\alpha(\ell)| > 1$ for all large ℓ or $|\alpha(\ell)| < 1$ for all large ℓ; that $\beta/(\alpha - 1)$ is not eventually constant on any infinite arithmetic progression; and that on no infinite arithmetic progression does α agree, for large ℓ, with a function $C\mu^{\ell}$, $C, \mu \in \overline{\mathbb{Q}}^{\times}$. Then f is not computed by any CCRA over $\mathbb{Q}$.*

Proof. With v_1, v_2 as in Step 2, $\alpha(\ell) = r_2 M_{x_1} M_w^{\ell} M_{x_2} v_2$ and $\beta(\ell) = r_2 M_{x_1} M_w^{\ell} M_{x_2} v_1$, which agree with exponential polynomials by (E3). The proof of Theorem 56 goes through verbatim up to Step 6, with c

chosen so that α has no zeros on I, and with $|\cdot| < 1$ in place of $|\cdot| > 1$ in the second case, the hypothesis on $\beta/(\alpha-1)$ being used in Steps 2 and 4. Step 6 yields an exponential polynomial E with $C\alpha(c+p\ell)^{k!}E(\ell) = 1$ for large ℓ. Let Λ be the group generated by the bases of E and of the exponential polynomial $\ell \mapsto \alpha(c+p\ell)$, and p' the order of its torsion subgroup. On a residue class $s + p'\mathbb{N}$ both functions have bases in the free abelian group $\Lambda^{p'}$, so, as in the proof of Lemma 58, they are elements of a Laurent polynomial ring over $\overline{\mathbb{Q}}[T]$ in which $\alpha^{k!}$ is a unit. A divisor of a unit is a unit, so $\alpha(c+p(s+p'\ell)) = C''\mu''^{\ell}$ for large ℓ, contrary to the hypothesis. □

For instance, block scalars $\alpha(\ell) = 2^\ell - 1$ or $\alpha(\ell) = \ell 2^\ell$ are excluded just like polynomial ones, whereas $\alpha(\ell) = 2^\ell$ is allowed; see Remark 62(2).

Proposition 53 gives a second, independent obstruction, which applies when the two weights coincide and the inverse-weight condition is void.

Theorem 60 (Unipotent contexts). *Let u, x_1, w, x_2 be words, $b_\ell = x_1 w^\ell x_2$, and $r_1, r_2 \in \mathbb{Q}^{1\times d}$ linearly independent rows with $\lambda M_u \in \mathbb{Q}r_1 + \mathbb{Q}r_2$, $\lambda M_u \notin \mathbb{Q}r_1$, such that for all $\ell \geq 0$*

$$r_1 M_{b_\ell} = \kappa(\ell)\, r_1, \qquad r_2 M_{b_\ell} = \kappa(\ell)\big(s(\ell)\, r_1 + r_2\big)$$

with $\kappa(\ell) \neq 0$ for all large ℓ. If on no infinite arithmetic progression the function s agrees, for large ℓ, with an exponential polynomial, then f is not computed by any CCRA over $\mathbb{Q}$.

Proof. Suppose a CCRA computes f. Step 1 of the proof of Theorem 56 applies with $c \geq n$ chosen so that $\kappa(\ell) \neq 0$ for $\ell \geq c$, and gives $I = c + p\mathbb{N}$, u', q and the monoid $\mathcal{Y}$ of products of p blocks b_ℓ, $\ell \in I$. Let $X = \mathbb{Q}r_1 + \mathbb{Q}r_2$, invariant under each M_{b_ℓ}. By hypothesis $s(\ell) \neq 0$ for some $\ell \in I$; writing $\lambda M_{u'} = \mu_1 r_1 + \mu_2 r_2$ with $\mu_2 \neq 0$, the row $\lambda M_{u'} M_{b_\ell}^p$ has r_1-coefficient $\kappa(\ell)^p(\mu_1 + p\, s(\ell)\mu_2)$ and r_2-coefficient $\kappa(\ell)^p\mu_2$, so it is independent of $\lambda M_{u'}$. Hence the rows $\lambda M_{u'} M_y$, $y \in \mathcal{Y}$, span X, $K_{u'} = X^\perp$, and $D = W/N \cong \mathbb{Q}^d/X^\perp$ is two-dimensional. With v_1, v_2 and $\bar{e}_1, \bar{e}_2$ as in Step 2, the operator induced by M_{b_ℓ} on $\mathbb{Q}^d/X^\perp$ is $\kappa(\ell)(I + s(\ell)\mathcal{N})$, where $\mathcal{N}\bar{e}_1 = \bar{e}_2$ and $\mathcal{N}\bar{e}_2 = 0$. These operators lie in the closed commutative group $\mathcal{G} = \{x(I + y\mathcal{N}) : x \neq 0\}$, so for $y \in \mathcal{Y}$ we have $\rho(y) = \bar{\kappa}(y)(I + S(y)\mathcal{N})$ with $\bar{\kappa}$ multiplicative and S additive, $S(b_{\ell_1}\cdots b_{\ell_p}) = \sum_j s(\ell_j)$; every $\rho(y)$ is invertible.

Apply Proposition 10 with a minimizing $e \neq \varepsilon$, and let $\mathcal{Z}_1$, F_L' and $\mathcal{H}'$ be as above. The group $\mathcal{H}'$ lies in $\mathcal{G}$, so its unipotent elements commute and its maximal tori are contained in the scalars. Hence in Proposition 53, $t(z) = \bar{\kappa}(z)I$ and $\log u(z) = S(z)\mathcal{N}$, and $S(z)$ is a $\overline{\mathbb{Q}}$-linear combination of finitely many rational series in z, for $z \in \mathcal{Z}_1$. Let $y_\ell = b_{c+p\ell} b_c^{p-1}$ and $\zeta_\ell = (e y_\ell e)^m$ with $m = |F_L'|$, so that $\zeta_\ell \in \mathcal{Z}_1$. Then

$$S(\zeta_\ell) = m\big(2S(e) + (p-1)s(c) + s(c+p\ell)\big),$$

and, as in Step 6 of the proof of Theorem 56, the left-hand side agrees with an exponential polynomial for large ℓ. So does $\ell \mapsto s(c+p\ell)$, contrary to the hypothesis. □

Example 61 (A palindromic obstruction)**.** Over $\{a, \#\}$ let $E(u_0\# \cdots \# u_k) = \sum_{i=0}^{k} \prod_{j\neq i}(|u_j| + 1)$, the elementary symmetric function of degree k in the numbers $|u_j| + 1$. It is invariant under permutations of the blocks, so $E(w^{\mathrm{rev}}) = E(w)$. On row coordinates (S, P, Q_S, Q_P) it is represented by $a : (S, P, Q_S, Q_P) \mapsto (S, P, Q_S + S, Q_P + P)$, $\# : (S, P, Q_S, Q_P) \mapsto (Q_S + S + P,\ Q_P + P,\ 0,\ 0)$, $\lambda = (0, 1, 0, 0)$ and output $Q_S + S + P$; this representation is minimal, and every word matrix containing # has rank two. With $u = x_1 = \varepsilon$, $w = a$, $x_2 = \#$, $r_1 = (1, 0, 0, 0)$ and $r_2 = (0, 1, 0, 0) = \lambda$ we get $r_1 M_{a^\ell\#} = (\ell+1)r_1$ and $r_2 M_{a^\ell\#} = r_1 + (\ell+1)r_2$, so $\kappa(\ell) = \ell + 1$ and $s(\ell) = 1/(\ell+1)$, which is an exponential polynomial on no arithmetic progression by Lemma 54. By Theorem 60, E is not CCRA-computable. Here the block products generate a commutative group whose Zariski closure is $\{x(I + y\mathcal{N})\} \cong \mathbb{G}_m \times \mathbb{G}_a$, condition (3) holds for every local group, and the unipotent

weight is trivial, so Proposition 52 gives nothing: the obstruction is the translation $s(\ell) = 1/(\ell+1)$, which a register would have to accumulate.

The claims about representations and machines in Corollary 57 and Example 61 were also checked by exact computation (Section 10.7). We record what the earlier necessary conditions say about g and h.

Remark 62 (What the local condition must look like)**.** (1) *Maximal subgroups do not detect the obstruction.* Let $\bar{M}$ be the Zariski closure in $M_3(\overline{\mathbb{Q}})$ of the matrix monoid of g or of h. The word matrices are the powers M_a^n and matrices of rank at most two, and all of them fix the row r_1; hence $\bar{M} \subseteq \{M_a^t : t \in \overline{\mathbb{Q}}\} \cup \{X : \operatorname{rank} X \le 2\}$ and $r_1 X = r_1$ for all $X \in \bar{M}$. Let $\epsilon \in \bar{M}$ be idempotent and H_ϵ the group of units of $\epsilon\bar{M}\epsilon$, which acts faithfully on $\overline{\mathbb{Q}}^{1\times 3}\epsilon \ni r_1$ and fixes r_1. If $\epsilon = I$, $H_\epsilon = \{M_a^t\}$; if $\operatorname{rank}\epsilon = 2$, H_ϵ embeds in $\{r_1 \mapsto r_1,\ r \mapsto br_1 + ar\} \cong \mathrm{Aff}_1$; if $\operatorname{rank}\epsilon = 1$, H_ϵ is trivial. So every maximal subgroup of $\bar{M}$ stabilizes a complete flag through the rational row r_1 and has commuting unipotent elements. The groups of Corollary 48 lie in such an H_ϵ with ϵ rational, namely the idempotent ϵ with the same image and kernel as M_e, which exists because $\operatorname{rank} M_e^2 = \operatorname{rank} M_e$ and lies in $\bar{M}$, being the identity of the closure of $\{M_e^n : n \ge 1\}$ inside the group of matrices with the same image and kernel as M_e (isomorphic to $\mathrm{GL}(R)$), which is a group by the argument of Lemma 6; indeed $M_e M_z M_e \in \epsilon\bar{M}\epsilon$ has the same rank as ϵ; for them the flag $\mathbb{Q}r_1 \subset \mathbb{Q}^{1\times 3}\epsilon$ is rational, so the conclusion of Corollary 48 is true for g and h. For h the monoid is even more regular: every word matrix containing # has rank exactly two, so each of them, $M'_\#$ included, lies in a subgroup of $M_3(\mathbb{Q})$. Since transposition preserves condition (3) and maps the closure of the monoid of f to that of f^{r}, *no characterization of CCRA-computability can be invariant under transposition of the minimal representation*, and any combinatorial condition on resets must distinguish left from right.

(2) *The extra condition is arithmetic.* By Theorem 59, the obstruction is that the block scalar fails to be a pure exponential along pumping. By contrast, if the block scalar is 2^ℓ, as for $\sum_i \prod_{j<i} 2^{|u_j|}$, then the inverse weight $2^{-\ell}$ is realized by the register $x := x/2$, and that series is computed by the CCRA with registers T, D initialized to 1, updates $(T, D) := (T/2, 2D)$ on a and $T := T + 1$ on #, and output TD (indeed it is invertible). In the reversal g^{r} the roles of r_1 and r_2 are exchanged, the required multiplier is $\kappa_2/\kappa_1 = |u_j|$ itself, and it is realized by the counter A in $x := xA + 1$.

10.5 One invertible letter and one rank-two reset

The necessary conditions above are sharp in the following class, which contains h, h^{r} and E (but not g, whose empty block is singular). Let $\Sigma = \{a, \#\}$, let M_a be invertible, let $\operatorname{rank} M_\# = 2$, and assume that every word containing # has rank two. Factor $M_\# = CR$ with $C \in \mathbb{Q}^{d\times 2}$, $R \in \mathbb{Q}^{2\times d}$ and put $T(n) = RM_a^n C$. Since $M_\# M_a^n M_\# = CT(n)R$, the rank assumption says $T(n) \in \mathrm{GL}_2(\mathbb{Q})$ for all $n \ge 0$. For $y \in \mathbb{Q}^{1\times 2}$ we have $(yR)M_{a^n\#} = yT(n)R$, and for $v = a^{n_1}\# \cdots a^{n_m}\#$ we write $T(v) = T(n_1)\cdots T(n_m)$. Let $\mathcal{T} \subseteq \mathrm{GL}_2(\overline{\mathbb{Q}})$ be the Zariski closure of the group generated by $\{T(n) : n \ge 0\}$. Changing the factorization conjugates all $T(n)$ by one rational matrix.

Theorem 63. *In this class, f is CCRA-computable if and only if the following hold.*

(a) *Every eigenvalue of M_a is a root of a rational number.*
(b) *$\mathcal{T}^\circ$ is triangularizable over $\mathbb{Q}$.*
(c) *If (b) holds and $\mathcal{T}^\circ$ contains a unipotent element $\neq 1$, then in a rational basis (y_1, y_2) of $\mathbb{Q}^{1\times 2}$ such that $\mathbb{Q}y_2$ is $\mathcal{T}$-invariant, writing $T(n) = \left(\begin{smallmatrix} a(n) & b(n) \\ 0 & d(n) \end{smallmatrix}\right)$, the sequences $d(n)/a(n)$ and $b(n)/a(n)$ are linear recurrence sequences.*

Condition (c) does not depend on the choices: rescaling y_1, y_2 multiplies b/a by a constant, and replacing y_1 by $y_1 + cy_2$ changes b by $c(d-a)$, hence adds $c(d/a - 1)$ to b/a. For h (with $y_1 = r_2$, $y_2 = r_1$) one gets

$a = b = n + 1$, $d = 1$, and $d/a = 1/(n + 1)$ is not a linear recurrence sequence. For E one gets $a = d = n + 1$, $b = 1$, and b/a fails. For h^{r} one gets $a = 1$, $d = n + 1$ and a polynomial b (for instance $b = 1$ for a suitable factorization), so (c) holds. Conditions (a) and (c) can be checked effectively (we only sketch this): (a) is a test on the irreducible factors of the characteristic polynomial of M_a, and (c) amounts to divisibility of explicit exponential polynomials with algebraic bases on residue classes. We do not discuss an algorithm for (b).

Proof. *Preliminaries.* (P1) If (b) holds and $\mathcal{T}^\circ$ contains a unipotent element $\neq 1$, then $\mathcal{T}^\circ$ lies in the group of upper triangular matrices for some rational basis, its unipotent radical $U \cong \mathbb{G}_a$ is normalized by $\mathcal{T}$ and fixes a unique line, which is defined over $\mathbb{Q}$; so that line is $\mathcal{T}$-invariant, a basis as in (c) exists, and every element of $\mathcal{T}$ is upper triangular in it. Moreover a, b, d are linear in M_a^n, hence linear recurrence sequences whose characteristic roots are eigenvalues of M_a. (P2) Let $\mathcal{G}_a$ be the Zariski closure of $\{M_a^n : n \geq 0\}$, a group by Lemma 6, with component group of order N_a. For each residue s, $\{M_a^n : n \equiv s \bmod N_a\}$ is dense in $M_a^s\mathcal{G}_a^\circ$, so $\Phi(X) = RXC$ maps $M_a^s\mathcal{G}_a^\circ$ into the closure of $\{T(n) : n \equiv s \bmod N_a\}$; the intersection of the irreducible set $\Phi(M_a^s\mathcal{G}_a^\circ)$ with GL_2 is irreducible, contains $T(s)$, and lies in $\mathcal{T}$, hence in one coset of $\mathcal{T}^\circ$. Thus $n \mapsto T(n)\mathcal{T}^\circ$ has period N_a. (P3) By [BFLM22], a linear recurrence sequence over $\mathbb{Q}$ whose characteristic roots are roots of rational numbers is computed by a CCRA over a one-letter alphabet.

Sufficiency. Assume (a)–(c). For $w = a^{n_0}\#a^{n_1}\#\cdots\#a^{n_k}$ with $k \geq 1$,

$$f(w) = s(n_0)\,T(n_1)\cdots T(n_{k-1})\,t(n_k), \qquad s(n) = \lambda M_a^n C, \quad t(n) = R M_a^n \gamma, \tag{15}$$

and $f(a^n) = \lambda M_a^n \gamma$. All entries of s, t, T are linear in M_a^n, so by (a) and (P3) each is computed by a one-letter CCRA, and so is every product of such entries. The machine runs, inside each block a^n, fresh copies of one-letter CCRAs for the finitely many block quantities used below, and combines them when it reads #; it also keeps the values computed in the first block, and at the end it uses the copies for $t(n_k)$ and, if $k = 0$, for $\lambda M_a^n \gamma$. It remains to evaluate the middle product copylessly.

If $\mathcal{T}^\circ$ contains no unipotent element $\neq 1$, it is a torus, split over $\mathbb{Q}$ by (b); choose a rational basis in which it is diagonal. Let $F = \mathcal{T}/\mathcal{T}^\circ$ and $\varphi(n) = T(n)\mathcal{T}^\circ$, periodic by (P2), so that the control can track φ of the prefix. As in Section 5 choose products r_g of matrices $T(n)$ with $r_g\mathcal{T}^\circ = g$, $r_1 = I$, and put $c(g, n) = r_g T(n) r_{g\varphi(n)}^{-1} \in \mathcal{T}^\circ(\mathbb{Q})$, a diagonal matrix whose entries $c_1(g, n), c_2(g, n)$ are linear in M_a^n. Then $T(n_1)\cdots T(n_m) = c(g_0, n_1)\cdots c(g_{m-1}, n_m)\, r_{g_m}$, and the middle product contributes $\sum_{i=1,2} Y_i\, s(n_0) E_{ii} r_g\, t(n_k)$ with product registers $Y_i := Y_i\, c_i(g, n)$. Expanding the bilinear form, each monomial $Y_i s_p(n_0) t_q(n_k)$ uses its own copies, and the machine is copyless.

If $\mathcal{T}^\circ$ contains a unipotent element $\neq 1$, use the basis of (c). With $A = \prod_j a(n_j)$, $D = \prod_j d(n_j)$ and B the upper right entry of the product over $j = 1, \ldots, k-1$, we have $B = AV$, where V satisfies the Horner recursion $V := V \cdot \frac{d(n)}{a(n)} + \frac{b(n)}{a(n)}$ at each #, starting from $V = 0$. Hence

$$f(w) = s_1(n_0)t_1(n_k)A + s_1(n_0)t_2(n_k)AV + s_2(n_0)t_2(n_k)D.$$

The sequences a, d have characteristic roots among the eigenvalues of M_a, and $q = d/a$ is a linear recurrence sequence by (c). Here the characteristic roots of a linear recurrence sequence are the bases that occur with nonzero coefficient in its exponential-polynomial form for large n. Let $\Gamma \subseteq \overline{\mathbb{Q}}^\times$ be the group generated by the eigenvalues of M_a and the characteristic roots of q and of b/a, and let p' be a multiple of the order of the torsion subgroup of Γ, which is finite because Γ is a finitely generated abelian group. Fix a residue s and write $n = s + p'm$. As functions of m, the sequences a and d are exponential polynomials with bases in the group Γ_0 generated by the p'-th powers of the eigenvalues of M_a, and q is one with bases in $\Gamma^{p'} = \{\gamma^{p'} : \gamma \in \Gamma\} \supseteq \Gamma_0$. Both groups are torsion-free: if $(\gamma^{p'})^j = 1$ for some $j \geq 1$, then γ is a torsion element, so $\gamma^{p'} = 1$. With $K = \overline{\mathbb{Q}}[T]$, (E1) identifies exponential polynomials in m with bases in $\Gamma^{p'}$ with

elements of the group ring $K[\Gamma^{p'}]$. This ring is a free $K[\Gamma_0]$-module on the cosets of Γ_0, and multiplication by a, a nonzero element of $K[\Gamma_0]$ (all $a(n) \neq 0$ because $T(n)$ is invertible), preserves the coset components. Since $qa = d$ has all its mass on the trivial coset and $K[\Gamma_0]$ is a domain (the group ring of a torsion-free abelian group over a domain), all nontrivial coset components of q vanish. So on every residue class the bases of q, as a function of m, lie in Γ_0, which consists of roots of rationals by (a). A characteristic root μ of q cannot vanish from all residue classes: the characteristic roots with the same p'-th power as μ are among the $\mu\zeta$ with $\zeta^{p'} = 1$, and vanishing on every class would give $\sum_\zeta p_{\mu\zeta}(n)\zeta^n = 0$ for all large n, hence all $p_{\mu\zeta} = 0$ by (E1). So $\mu^{p'} \in \Gamma_0$, and the characteristic roots of q are roots of rationals. The same argument applies to b/a (whose characteristic roots were included in Γ), and a rational sequence satisfying a recurrence over $\overline{\mathbb{Q}}$ satisfies one over $\mathbb{Q}$. So all quantities are computed by one-letter CCRAs, and the displayed formula is evaluated with two copies of A, $s_1(n_0)$ and $t_2(n_k)$ and one copy of V and D; all updates are copyless.

Necessity. Assume that a CCRA computes f. Corollary 48 with $e = \varepsilon$ and $Z = \{a\}$ shows that $\mathcal{G}_a^\circ$ is triangularizable over $\mathbb{Q}$; as $M_a^{N_a} \in \mathcal{G}_a^\circ$, its eigenvalues are rational, which gives (a).

For (b) and (c) we run the construction of Section 4 on the block words $(a^*\#)^*$. Choose a nonempty block word t_0 for which $\delta(S, t_0)$ has minimum size among block words, let $e_0 = t_0^N$ with $N = |\delta(S, t_0)|!$, $R_0 = \delta(S, e_0)$, and let L_0 be the monoid of block words fixing R_0 pointwise (the word e_0 for the control is not to be confused with the element e chosen below by the compression); Lemma 13 holds for block words. By the argument of Lemma 14, with Lemma 6 applied to the dense monoid $\{T(v)\}$ of $\mathcal{T}$, the closure $\mathcal{K}_0$ of $\{T(v) : v \in L_0\}$ is a group containing $\mathcal{T}^\circ$. For every word u, $\delta(s_0, ue_0) \in R_0$, and the span X_u of the rows $\lambda M_{ue_0} M_v$, $v \in L_0$, lies in $\mathbb{Q}^{1\times 2}R$ (as e_0 ends with #) and is $\mathcal{K}_0$-invariant. Lemma 15 with L_0 in place of L_R gives a local quotient $D_u \cong \mathbb{Q}^d/X_u^\perp$, the dual of X_u, on which $v \in L_0$ acts invertibly, through $T(v)$.

Necessity of (b). By Proposition 10 and Corollary 12 applied to L_0 and D_u, the closure $\mathcal{H}_u$ of the group generated by the operators induced on D_u has a $\mathbb{Q}$-triangularizable identity component. The restriction morphism maps $\mathcal{K}_0$ onto $\mathcal{H}_u$ by (A2), so it maps the connected group $\mathcal{T}^\circ$ into $\mathcal{H}_u^\circ$; hence $\mathcal{T}^\circ$ stabilizes a complete rational flag of X_u (the dual of a flag of D_u). The rows λM_{ue_0} span $\mathbb{Q}^{1\times d}M_{e_0} = \mathbb{Q}^{1\times 2}R$, which is therefore the sum of finitely many of the subspaces X_u, an equivariant quotient of their direct sum; the image of a complete invariant rational flag of the direct sum under this surjection is an invariant rational chain with successive quotients of dimension at most one, which gives a complete $\mathcal{T}^\circ$-invariant rational flag of $\mathbb{Q}^{1\times 2}R \cong \mathbb{Q}^{1\times 2}$. This is (b).

Necessity of (c). Now assume (b) and $U \neq 1$. If $\dim X_u \leq 1$ for every u, each nonzero X_u would be the line $\mathbb{Q}y_2R$, the only line invariant under U, so all rows λM_{ue_0} would lie in it, contradicting $\dim \mathbb{Q}^{1\times d}M_{e_0} = 2$. Fix u with $X_u = \mathbb{Q}^{1\times 2}R$, so that $D = D_u$ is two-dimensional and $v \in L_0$ acts on it by the matrix $T(v)$ in the basis dual to (y_1R, y_2R); in this basis the column $\bar{e}_1$ spans an invariant line with weight $\kappa_1 = a(v)$ and the quotient has weight $\kappa_2 = d(v)$, where $a(v), d(v)$ are the diagonal entries of $T(v)$. Apply Proposition 10 to L_0 with a minimizing $e \neq \varepsilon$. The power-map argument once more (with $\Phi(h) = (T(e)hT(e))^{k!}$) shows that the closure of the group generated by $\rho(\mathcal{Z}_0)$ contains $\mathcal{T}^\circ \supseteq U$, whose nontrivial elements act nontrivially on D. Proposition 52 gives a free coordinate i with $\alpha_i(\zeta) = d(\zeta)/a(\zeta)$ on $\mathcal{Z}_0$, and Proposition 53 (unipotent elements of $\mathcal{T}$ commute) together with the computation $\left(\begin{smallmatrix} x & c'(w-x) \\ 0 & w \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & \sigma \\ 0 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right) \iff x = a,\ w = d,\ \sigma = \frac{b}{a} - c'\left(\frac{d}{a} - 1\right)$, for the maximal torus $T_H \subseteq \{\left(\begin{smallmatrix} x & c'(w-x) \\ 0 & w \end{smallmatrix}\right)\}$ of $\mathcal{H}'^\circ$ (every torus of the upper triangular group is of this form for some c'), shows that $b(\zeta)/a(\zeta)$ is, on $\mathcal{Z}_1$, a $\overline{\mathbb{Q}}$-linear combination of rational series in ζ. Here $b(\zeta)$ is the upper right entry of $T(\zeta)$.

Now we build words with a single varying block. Let $x_n = e_0a^n\#e_0$; it permutes R_0, and the permutations of R_0 induced by words e_0ve_0 form a finite group, so we may fix block words w_σ such that $y_n = x_n\, e_0w_\sigma e_0 \in L_0$ when σ is the permutation induced by x_n. The homomorphism $\Psi : \mathcal{Z} \to F_L'$ defined before Proposition 53

has as image a subsemigroup of a finite group, hence a subgroup; fix $Z_g \in \mathcal{Z}$ with $\Psi(Z_g) = g^{-1}$. Put $\zeta_n = ey_ne\,Z_{\Psi(ey_ne)} \in \mathcal{Z}_1$. Along each residue class of n modulo the control period of a (with $n \geq |S|$), the coefficients of R_{ey_ne} are exponential polynomials in n, so the condition that $B_{R_{ey_ne}}$ lies in a given component of $\mathcal{G}'_L$ is the vanishing of finitely many exponential polynomials; by the Skolem–Mahler–Lech theorem [EvdPSW03], $\Psi(ey_ne)$ and the permutation induced by x_n are eventually periodic in n. So there is a period P such that for n large in each residue class modulo P, ζ_n is a fixed word with the single factor $a^n\#$ in the middle, and $T(\zeta_n) = A_1T(n)A_2$ with fixed upper triangular A_1, A_2. Then

$$\frac{d}{a}(T(\zeta_n)) = \frac{d}{a}(A_1)\frac{d}{a}(A_2)\cdot\frac{d(n)}{a(n)}, \qquad \frac{b}{a}(T(\zeta_n)) = \Big(\frac{b}{a}(A_1)\frac{d(n)}{a(n)} + \frac{b(n)}{a(n)}\Big)\frac{d}{a}(A_2) + \frac{b}{a}(A_2),$$

by the Horner rule $\frac{b}{a}(XY) = \frac{b}{a}(X)\frac{d}{a}(Y) + \frac{b}{a}(Y)$ for upper triangular X, Y. The left-hand sides are values at ζ_n of $\overline{\mathbb{Q}}$-linear combinations of rational series, hence agree with exponential polynomials in n on each residue class for large n. Solving, so do $d(n)/a(n)$ and $b(n)/a(n)$; a rational sequence that agrees with exponential polynomials on all residue classes modulo P for large n is a linear recurrence sequence. This proves (c). □

Remark 64. For an alphabet $\Sigma_0 \cup \{\#\}$ with several invertible letters and the same rank assumption, $T(z) = RM_zC$ for $z \in \Sigma_0^*$, and the sufficiency proof of Theorem 63 goes through (with (P2) replaced by the same argument for the cosets of the identity component of the closure of the group generated by $\{M_a : a \in \Sigma_0\}$, which a permutation automaton tracks) if condition (3) holds for the group generated by $\{M_a : a \in \Sigma_0\}$ and for $\mathcal{T}$, and, when $\mathcal{T}^\circ$ has unipotent elements, $z \mapsto d(z)/a(z)$ and $z \mapsto b(z)/a(z)$ are CCRA-computable on Σ_0^*. The necessity argument uses the Skolem–Mahler–Lech theorem to find words with a single varying block, and we do not know whether it extends.

10.6 Registers must interact

Call a CCRA *affine* if every update has the form $x_i := \alpha_i x_{\sigma(i)} + \beta_i$ with $\alpha_i, \beta_i \in \mathbb{Q}$ and a permutation σ of the registers (both depending on the transition); $\alpha_i = 0$, that is, a reset, is allowed, and the outputs may be arbitrary polynomials. By Theorem 1, every invertible CCRA series is computed by an affine CCRA.

Proposition 65. *The CCRA series g^{r} of Corollary 57 is not computed by any affine CCRA.*

Proof. Suppose an affine CCRA with n states and r registers computes g^{r}, and let Δ bound the degrees of its output polynomials. Put $p = n!\,r!$ and $c = n$. For every state s, $\delta(s, a^n)$ lies on an a-cycle, so the composite update over a^p from it is $x_i \mapsto \mu_i x_i + \nu_i$ (its permutation part is the $r!$-th power of a permutation). Hence the composite update along $a^{c+p\ell}$ from s is $x_i \mapsto A_{s,i}(\ell)x_{\sigma_s(i)} + B_{s,i}(\ell)$ with σ_s independent of ℓ, $A_{s,i}(\ell) = C_{s,i}\mu_{s,i}^{\ell}$ for some constants $C_{s,i}, \mu_{s,i}$ (possibly 0), and $B_{s,i}$ agreeing with an exponential polynomial for $\ell \geq 1$ (a factor 0^ℓ coming from a reset on the cycle vanishes for $\ell \geq 1$); and $\delta(s, a^{c+p\ell})$ does not depend on ℓ. Below all $\ell_j \geq 1$, and (E1) is applied on $\{1, 2, \dots\}^k$.

Fix $k > \Delta$ and consider $w(\ell_1, \dots, \ell_k) = \#a^{c+p\ell_1}\#a^{c+p\ell_2}\cdots\#a^{c+p\ell_k}$; its control run does not depend on ℓ. Call a function on $\mathbb{N}^k$ *m-sparse* if it is a finite sum of products $\prod_{j=1}^k E_j(\ell_j)$ of exponential polynomials in which at most m factors are not of the form $C\mu^{\ell_j}$. By induction along the word, every register value after reading $w(\ell)$ is 1-sparse: an affine update $x \mapsto A(\ell_j)x' + B(\ell_j)$ multiplies each summand of a 1-sparse function by a function $C\mu^{\ell_j}$ of the current block and adds a function of ℓ_j alone, and the updates on # have constant coefficients. Hence $g^{\mathrm{r}}(w(\ell))$ is Δ-sparse, so by (E1) its expansion in the functions $\prod_j \ell_j^{s_j}\mu_j^{\ell_j}$ only involves functions with $s_j \neq 0$ for at most Δ indices j. But $g^{\mathrm{r}}(w(\ell)) = \sum_{i=0}^k \prod_{j=i+1}^k (c + p\ell_j)$ is a polynomial in which the monomial $\ell_1\cdots\ell_k$ has coefficient $p^k \neq 0$, a contradiction by (E1). □

So outside the invertible sector the normal form (2) of Theorem 1 fails, even with resets: the product xA of two registers in the machine for g^{r} is unavoidable. The same proof applies to h^{r} and to F.

10.7 Discussion

Compression with resets. For the machine computing g^{r} in Corollary 57, the word ## maps every valuation to $(x, A) = (1, 0)$, so a minimizing word in Proposition 10 has $k = 0$; the operator $\bar{E}$ then has rank one on the three-dimensional minimal representation, and global compression yields nothing. The correct replacement, used in Corollary 48 and in the proof of Theorem 56, is to compress only monoids of return words that act invertibly on a local quotient, and to record, through Proposition 52, which inverse weights the compressed registers must carry.

A conjectural picture. Theorems 1, 50, 59, 60 and 63 suggest the following shape for a characterization. For every monoid $\mathcal{Y}$ of return words that acts invertibly on a local quotient, the induced representation factors, on the semigroup $e\mathcal{Y}e$ for a word e depending on the machine and after passing to a fixed power of each element, through a morphism of algebraic groups from a group of diagonal affine maps, in such a way that the multipliers and translations of the lifted maps are rational series of the words; this machine-dependent form is what Propositions 52 and 53 prove. This condition implies condition (3) for the local groups, the inverse-weight condition and the translation condition. Condition (3) for local groups together with trivial inverse weights does not suffice, even for single-rank resets: the function E of Example 61 satisfies (3) for all its local groups and has $\kappa_2/\kappa_1 = 1$, yet is not CCRA-computable. Theorem 63 confirms this picture for one invertible letter and one rank-two reset, where conditions (a)–(c) are the intrinsic form of the lifting condition. We conjecture that an intrinsic version of the full lifting condition is sufficient in general, at least when all words containing a singular letter have the same rank: there the block products are invertible, and the cocycle construction of Section 5 should run at the level of blocks, with the torus coordinates, their required inverses, and the translation coordinates computed inside each block. The lifting condition as stated depends on the machine through e; an intrinsic formulation, and a decision procedure, are open.

Exact checks. Exact computations (Python, `fractions` and `sympy`) verified the machines of Corollary 57 on all words of length at most 12, and verified symbolically the representations, minimality claims and context identities used above. They also found affine contexts in the reversals of randomly generated CCRAs. For example, for the one-state CCRA over $\{a, b\}$ with registers (x_0, x_1) initialized to $(1, 1)$, updates $(x_0, x_1) := (x_1 - 1, x_0)$ on a and $(x_0, x_1) := (x_0x_1 - 1, \frac{1}{2})$ on b, and output $\frac{1}{2}x_1 + 1$, the minimal representation of the reversal has dimension 4 and an affine context with $u = ab$, $x_1 = \varepsilon$, $w = aa$, $x_2 = b$, $\alpha(\ell) = \frac{1}{2} - \ell$, β quadratic and $\beta/(\alpha - 1)$ nonconstant (checked symbolically), so non-closure under reversal is not confined to hand-made examples. These checks support, but are not part of, the proofs.

11 Limits and open problems

Remark 66. The proof of Theorem 1 uses only that $\mathbb{Q}$ has characteristic zero, and it extends, with routine modifications, to any field of characteristic zero, with constants and triangularizability taken over that field. Lemma 18, Theorem 3(2) and the algorithms use $\mathbb{Q}$ specifically.

Positive characteristic is excluded: finite unipotent roots and (A3), (A4) fail there.

Exact computational sanity checks. Exact rational computations (Python, `fractions`) confirmed the verdicts for all examples above and verified the normal-form machines of Theorem 1(2) by exact equivalence of weighted automata, including a CCRA with S_3 control whose minimal representation has dimension 22 and finite quotient of order 6, recovered from the minimal representation alone. These checks support, but do not replace, the proofs.

Open problems.

(1) *Singular minimal representations.* Characterize CCRA-computability for all rational series over $\mathbb{Q}$, and decide it. Resets break the identity $\bar{E} = I$ of Proposition 10, so the minimal representation is no longer a subquotient of one compressed system. Section 10 shows that condition (3) for maximal subgroups is not sufficient, that the missing conditions are arithmetic and not invariant under transposition (inverse weights and translations of compressed registers must be rational series), and that CCRA are not closed under reversal. An intrinsic formulation of the local lifting condition of Section 10.7, and its decidability, remain open.

(2) *Complexity.* Theorem 40 gives a deterministic polynomial-time algorithm for invertible series, but its analysis uses Schreier generators of a subgroup of possibly exponential index. Is there a polynomial bound on the length of words whose unipotent parts generate $\mathrm{Lie}(R_u(\mathcal{G}_f))$ (Remark 39(ii); Lemma 42 gives such a bound for the torus), and what is the parallel complexity of the problem?

(3) *Succinctness.* The normal form may use exponentially many registers in the minimal dimension, and its control is the possibly large component group. What are the optimal state and register counts of CCRA for invertible series, and how do they compare with unrestricted CCRA?

(4) *Constants and semirings.* For a subsemiring $R \subseteq \mathbb{Q}$, which series have CCRA with constants in R? The normal form requires the inverses $\chi(t)^{-1}$ of torus characters, as in the register $h := h/2 + b$ of Example 20. Analogous structural questions remain open for copyless models over $\mathbb{N}$ and the tropical semirings.

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